

Lecture 11
Quantum Information for Newcomers
Quantum Information Theory and Physical Qubits

Deep Learning in Classical and Quantum Physics

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Plan of the Lecture

The lecture separates the rules that every qubit obeys, on any platform, from the physical systems that carry qubits.

Part I Quantum Information Theory

- one qubit and the Bloch sphere
- gates as rotations, bases and shots
- registers, many qubits, multi-qubit gates
- entanglement and Hamiltonians
- density matrices, measurement, noise
- fidelity and distances between states
- what quantum information allows and forbids

Part II Physical Qubits

- requirements for a physical qubit
- an atom in laser light, pulses
- transmon, spin, photon, comparison
- couplings and native two-qubit gates
- a register of interacting qubits
- appendix, derivations for the atom and the Mølmer–Sørensen gate

Part I

Introduction to Quantum Information Theory

The rules of qubits, gates, measurement and noise, the same for every physical system.



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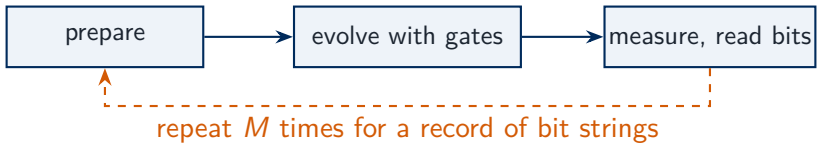
Prepare, Evolve, Measure



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Physical Systems as Carriers of Information

A classical bit is a two-state system that we prepare, transform and read out. A qubit is its quantum counterpart.



One run returns one bit per qubit. Theory predicts probabilities, which we compare with records of many repeated runs, called *shots*.

Running example, a two-qubit register

Qubits 0 and 1 are prepared, interact and are read out one bit each. We describe the register in a computer and predict its records.

Prerequisites and a Dictionary

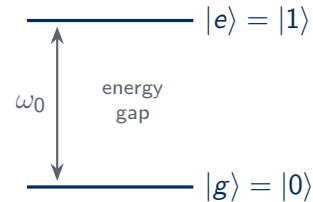
We assume complex numbers, vectors and matrices, eigenvalues, basic probability and a little Python, and no prior course in quantum mechanics.

physics	mathematics	computer science
state, ket $\|\psi\rangle$	unit column vector in $\mathbb{C}^{2^N}$	array of 2^N complex numbers
bra $\langle\psi\|$, $\langle\phi\|\psi\rangle$	conjugate row, inner product	`psi.conj()`, `vdot`
observable	Hermitian matrix	quantity averaged over shots
Hamiltonian $\hat{\mathcal{H}}$ (energy)	Hermitian generator of time evolution	cost function of variational algorithms
gate, time evolution	unitary matrix	reversible instruction
measurement in a basis	projectors, Born rule	sampling a bit string
two systems	tensor (Kronecker) product	`np.kron`, binary indexing
density matrix	positive matrix of trace one	probability-weighted states
partial trace	linear map $\mathbb{C}^{4\times4} \to \mathbb{C}^{2\times2}$	marginalising a qubit

The Qubit as a Two-Level System

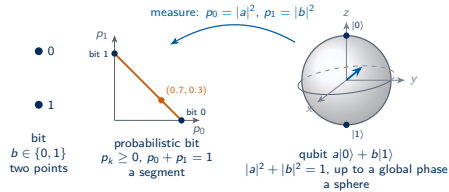
A **qubit** is any quantum system with two perfectly distinguishable states. Its states form a two-dimensional complex vector space, so it can be in a superposition of both, and a measurement finds one of them.

Often the two states are two energy levels, the lower *ground* state and the higher *excited* state, separated by the energy $\hbar\omega_0$. We set $\hbar = 1$, so energies are frequencies. A standard example is a *spin-1/2*, the intrinsic angular momentum of an electron, whose component along z is either up or down. We label the states as shown on the right. Part II shows atoms, circuits, spins and photons as qubits.



$$|0\rangle = |\uparrow\rangle \equiv |g\rangle,$$
$$|1\rangle = |\downarrow\rangle \equiv |e\rangle$$

Bit, Probabilistic Bit and Qubit



	bit	probabilistic bit	qubit
state	0 or 1	(p_0, p_1), $p_k \geq 0$, $\sum p_k = 1$	$(a, b) \in \mathbb{C}^2$, $\lvert a\rvert^2 + \lvert b\rvert^2 = 1$
operations	logic gates	stochastic matrices	unitary matrices (reversible)
readout	the bit	sample a bit with p_k	sample a bit with $\lvert a\rvert^2, \lvert b\rvert^2$
N units	N bits	2^N probabilities	2^N complex amplitudes

A qubit still outputs a bit. Its complex amplitudes can cancel each other in interference, and a classical description of N qubits stores 2^N complex amplitudes.

Four Rules of Quantum Mechanics

The description of N qubits rests on four postulates motivated by experiments.

1. The **state** is a unit vector $|\psi\rangle \in \mathbb{C}^{2^N}$, defined up to a global phase.
2. An isolated system **evolves** by a unitary matrix, $|\psi\rangle \to \hat{U}|\psi\rangle$, and $\hat{U} = e^{-it\hat{\mathcal{H}}}$ with the energy matrix $\hat{\mathcal{H}}$ of the system.
3. A **measurement** in an orthonormal basis $\{|k\rangle\}$ gives the outcome k with probability $|\langle k|\psi\rangle|^2$ (the Born rule) and leaves the state $|k\rangle$.
4. Two systems **together** live in the tensor product $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$.

Density matrices and noise extend the first three rules to systems that we know only statistically or only in part.

One Qubit

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The State of One Qubit

The two basis states, spin up and spin down along z, form the **computational basis**,

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A general state is a **superposition** with complex *amplitudes* a and b,

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle = \begin{pmatrix} a \\ b \end{pmatrix}, \qquad \underbrace{|a|^2 + |b|^2 = 1}_{\text{normalisation}}.$$

In a computer one qubit is an array of two complex numbers.

Running example

$$|\psi_{\text{ex}}\rangle = \frac{\sqrt{3}}{2}\,|0\rangle + \frac{e^{i\pi/4}}{2}\,|1\rangle, \qquad \tfrac{3}{4} + \tfrac{1}{4} = 1.$$

Dirac Notation

symbol	object	example
ket $\lvert\psi\rangle$	column vector	$\lvert\psi\rangle = (a, b)^\top$
bra $\langle\psi\rvert = (\lvert\psi\rangle)^\dagger$	row vector, conjugated	$\langle\psi\rvert = (a^*, b^*)$
$\langle\phi\rvert\psi\rangle$	inner product, a number	$\sum_k \phi_k^* \psi_k$, $\langle 0\rvert 1\rangle = 0$
$\lvert a\rangle\langle b\rvert$	outer product, a matrix	$\lvert 0\rangle\langle 0\rvert = \mathrm{diag}(1, 0)$

- The dagger $\dagger$ is complex conjugation followed by transposition.
- $\lvert 0\rangle\langle 0\rvert$ and $\lvert 1\rangle\langle 1\rvert$ are **projectors** with $\hat{P}^2 = \hat{P}$, and $\lvert 0\rangle\langle 0\rvert\psi\rangle = a\lvert 0\rangle$ keeps one component.
- A matrix with $\hat{O}^\dagger = \hat{O}$ is **Hermitian**, and its eigenvalues are real.

The Born Rule

A measurement in the computational basis returns one bit, with the probabilities

$$p(0) = |\langle 0|\psi\rangle|^2 = |a|^2, \qquad p(1) = |\langle 1|\psi\rangle|^2 = |b|^2, \qquad p(0) + p(1) = 1.$$

The measurement leaves the qubit in the basis state it found, and repeating the measurement gives that bit again.

Running example

$p(0) = |\sqrt{3}/2|^2 = 0.750$ and $p(1) = |e^{i\pi/4}/2|^2 = 1/4$. The phase $e^{i\pi/4}$ does not enter these probabilities.

Global and Relative Phase

Multiplying the whole state by a **global phase**, $|\psi\rangle \to e^{i\alpha}|\psi\rangle$, changes no prediction, because every prediction contains amplitudes in pairs,

$$(e^{i\alpha}a)^*(e^{i\alpha}b) = \underbrace{e^{-i\alpha}e^{i\alpha}}_{=1}a^*b.$$

The global phase can be chosen to make a real and non-negative.

A **relative phase** between the two amplitudes is physical. The two states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

have $|a|^2 = |b|^2 = \frac{1}{2}$. Both give a fair coin in the computational basis, although they are orthogonal, $\langle +|-\rangle = 0$.

Which measurement tells $|+\rangle$ from $|-\rangle$ with certainty?

Pauli Matrices and Expectation Values

Recording $+1$ for bit 0 and -1 for bit 1 measures the spin along z, whose mean is

$$\langle \hat{Z} \rangle = (+1)\, p(0) + (-1)\, p(1) = p(0) - p(1).$$

The spin components along x, y, z are the Pauli matrices, $\hat{\sigma}_x \equiv \hat{X}$, $\hat{\sigma}_y \equiv \hat{Y}$, $\hat{\sigma}_z \equiv \hat{Z}$,

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

$\hat{X}^2 = \hat{Y}^2 = \hat{Z}^2 = \hat{I}$, with eigenvalues ± 1 and eigenvectors $|0\rangle, |1\rangle$ for $\hat{Z}$, $|\pm\rangle$ for $\hat{X}$ and $|\pm i\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$ for $\hat{Y}$.

The **expectation value** of an observable $\hat{O}$ with eigenvalues λ_k is $\langle \hat{O} \rangle = \langle \psi | \hat{O} | \psi \rangle = \sum_k \lambda_k\, p(\lambda_k)$. For the running example $\langle \hat{Z} \rangle = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$.

Pauli Algebra

Multiplying the matrices out gives one rule for all products, with $a, b, c \in \{x, y, z\}$,

$$\hat{\sigma}_a \hat{\sigma}_b = \delta_{ab}\,\hat{I} + i \sum_c \varepsilon_{abc}\,\hat{\sigma}_c,$$

where $\varepsilon_{abc} = +1$ for the cyclic orders xyz, yzx, zxy, -1 for the reversed orders and 0 when an index repeats. Hence $\hat{\sigma}_a^2 = \hat{I}$, $\hat{X}\hat{Y} = i\hat{Z}$, $\hat{Y}\hat{Z} = i\hat{X}$ and $\hat{Z}\hat{X} = i\hat{Y}$.

- Different Pauli matrices *anticommute*, $\{\hat{\sigma}_a, \hat{\sigma}_b\} = 2\delta_{ab}\hat{I}$, and their commutator is $[\hat{\sigma}_a, \hat{\sigma}_b] = 2i \sum_c \varepsilon_{abc}\,\hat{\sigma}_c$.
- $\mathrm{Tr}\hat{\sigma}_a = 0$ and $\mathrm{Tr}(\hat{\sigma}_a\hat{\sigma}_b) = 2\delta_{ab}$, so $\hat{I}, \hat{X}, \hat{Y}, \hat{Z}$ are an orthogonal basis, and any 2×2 matrix is $c_0\hat{I} + \mathbf{c} \cdot \hat{\boldsymbol{\sigma}}$ with $c_0 = \frac{1}{2}\mathrm{Tr}\hat{A}$, $c_a = \frac{1}{2}\mathrm{Tr}(\hat{\sigma}_a\hat{A})$.
- The spin operators are $\hat{S}_a = \hat{\sigma}_a/2$. The transition matrices $\hat{S}^+ = |0\rangle\langle 1|$ and $\hat{S}^- = |1\rangle\langle 0|$ move one basis state into the other, $\hat{S}^+|1\rangle = |0\rangle$, $\hat{S}^+|0\rangle = 0$, and many texts use $\hat{\sigma}^\pm = \hat{X} \pm i\hat{Y} = 2\hat{S}^\pm$.

The Bloch Sphere

Up to a global phase,

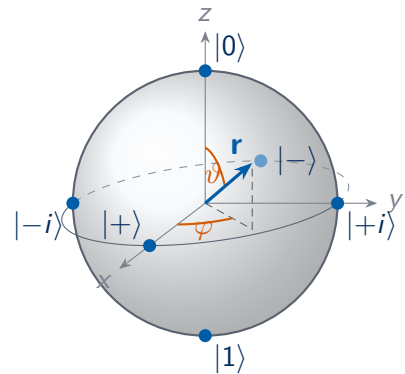
$$|\psi\rangle = \cos\tfrac{\vartheta}{2}\,|0\rangle + e^{i\varphi}\sin\tfrac{\vartheta}{2}\,|1\rangle.$$

The **Bloch vector** collects the three expectation values,

$$\mathbf{r} = (\langle\hat{X}\rangle, \langle\hat{Y}\rangle, \langle\hat{Z}\rangle)$$

$$= (\sin\vartheta\cos\varphi,\ \sin\vartheta\sin\varphi,\ \cos\vartheta).$$

The poles are $|0\rangle$ and $|1\rangle$, and the equator holds $|\pm\rangle$ and $|\pm i\rangle$. The relative phase is the azimuth φ, visible in $\langle\hat{X}\rangle$ and $\langle\hat{Y}\rangle$, and pure states have $|\mathbf{r}| = 1$.



The running example has $\vartheta = \pi/3$, $\varphi = \pi/4$ and $\mathbf{r}_{\mathrm{ex}} = (0.612,\ 0.612,\ 0.500)$.

Evolution and Gates as Rotations



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The Evolution Operator

The *Hamiltonian* $\hat{\mathcal{H}} = \hat{\mathcal{H}}^\dagger$ is the energy matrix, whose eigenvalues are the energy levels. The state obeys the Schrödinger equation ($\hbar = 1$)

$$i\frac{d}{dt}|\psi(t)\rangle = \hat{\mathcal{H}}|\psi(t)\rangle.$$

For a **time-independent** $\hat{\mathcal{H}}$ we define $\hat{U}(t) = e^{-it\hat{\mathcal{H}}} = \sum_{k\geq 0}(-it\hat{\mathcal{H}})^k/k!$ by its power series and differentiate each term,

$$\frac{d}{dt}\hat{U}(t) = \sum_{k\geq 1}\frac{k\,t^{k-1}(-i\hat{\mathcal{H}})^k}{k!} = (-i\hat{\mathcal{H}})\underbrace{\sum_{k\geq 1}\frac{(-it\hat{\mathcal{H}})^{k-1}}{(k-1)!}}_{=\hat{U}(t)} = -i\hat{\mathcal{H}}\,\hat{U}(t), \qquad \underbrace{\hat{U}(0) = \hat{I}}_{\text{start}},$$

so $|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle$ solves the equation with the right initial state.

Properties of the Evolution Operator

- $\hat{U}^\dagger = e^{+it\hat{\mathcal{H}}}$ because $\hat{\mathcal{H}}^\dagger = \hat{\mathcal{H}}$. The two exponentials commute, and $\hat{U}^\dagger \hat{U} = \hat{I}$ preserves the total probability.
- Evolving for $t_1 + t_2$ is evolving twice, $\hat{U}(t_1 + t_2) = \hat{U}(t_1)\,\hat{U}(t_2)$.

With $\hat{\mathcal{H}}|E_k\rangle = E_k|E_k\rangle$, every power is $\hat{\mathcal{H}}^n = \sum_k E_k^n |E_k\rangle\langle E_k|$, and the series sums to the spectral form

$$\hat{U}(t) = \sum_k e^{-iE_k t}\,|E_k\rangle\langle E_k|.$$

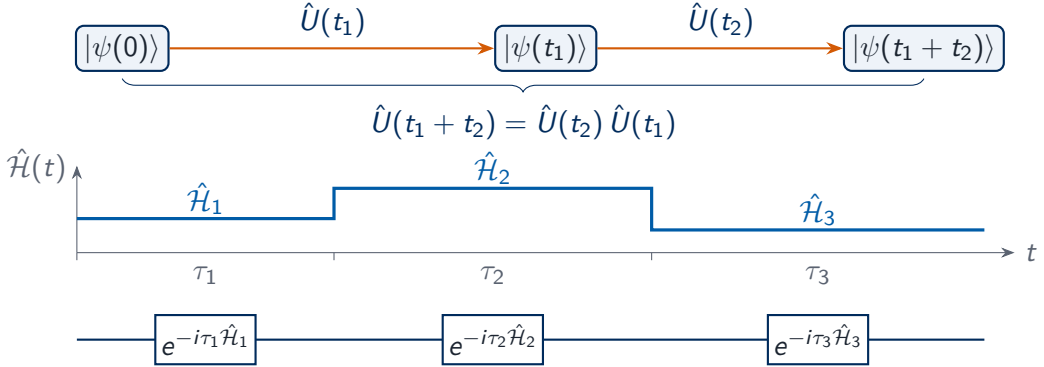
An energy eigenstate acquires only a phase, and none of its predictions changes in time.

Precession, $\hat{\mathcal{H}} = \frac{\omega}{2}\hat{Z}$ with an angular frequency ω

The eigenvalues are $\pm\frac{\omega}{2}$ on $|0\rangle$ and $|1\rangle$, so $\hat{U}(t) = \mathrm{diag}(e^{-i\omega t/2}, e^{i\omega t/2})$. The relative phase $e^{+i\omega t}$ on $|1\rangle$ turns the Bloch vector about z, $\varphi \to \varphi + \omega t$.

From a Hamiltonian to a Circuit

A gate is a Hamiltonian switched on for a chosen time. Successive evolutions multiply, the later one on the left.



Pulses $\hat{\mathcal{H}}_k$ of durations τ_k give $\hat{U} = e^{-i\tau_3\hat{\mathcal{H}}_3} e^{-i\tau_2\hat{\mathcal{H}}_2} e^{-i\tau_1\hat{\mathcal{H}}_1}$, with the first pulse rightmost.

Exponential of a Pauli Matrix

Every Pauli matrix squares to the identity. For $\hat{P}^2 = \hat{I}$ we have $\hat{P}^{2m} = \hat{I}$ and $\hat{P}^{2m+1} = \hat{P}$, and the series splits into even and odd powers,

$$e^{-ia\hat{P}} = \underbrace{\sum_m \frac{(-ia)^{2m}}{(2m)!}}_{=\cos a}\,\hat{I} + \underbrace{\sum_m \frac{(-ia)^{2m+1}}{(2m+1)!}}_{=-i\sin a}\,\hat{P} = \cos a\,\hat{I} - i\sin a\,\hat{P}.$$

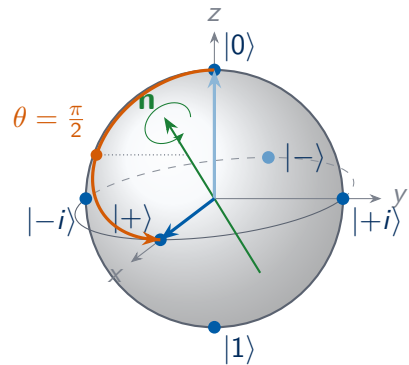
This matrix Euler formula holds for $\hat{X}$, $\hat{Y}$ and $\hat{Z}$, for $\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ with $|\mathbf{n}| = 1$, where anticommutation gives $(\mathbf{n}\cdot\hat{\boldsymbol{\sigma}})^2 = \hat{I}$, and for $\hat{X}\otimes\hat{X}$ and $\hat{Z}\otimes\hat{Z}$.

Rotations About Any Axis

The **rotation** about the unit axis $\mathbf{n}$ by the angle θ is

$$\begin{aligned}\hat{R}_{\mathbf{n}}(\theta) &= e^{-i\theta\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}/2} \\ &= \cos\tfrac{\theta}{2}\,\hat{I} - i\sin\tfrac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}, \\ \hat{R}_z(\theta) &= \mathrm{diag}(e^{-i\theta/2}, e^{i\theta/2}).\end{aligned}$$

$\hat{R}_{\mathbf{n}}(\theta)$ turns the Bloch vector by θ about $\mathbf{n}$, and $\hat{R}_{\mathbf{n}}(2\pi) = -\hat{I}$ is a global phase. For example $\hat{R}_x(\theta)|0\rangle = \cos\frac{\theta}{2}|0\rangle - i\sin\frac{\theta}{2}|1\rangle$ has the Bloch vector $(0, -\sin\theta, \cos\theta)$. The Hamiltonian $\frac{\omega}{2}\hat{Z}$ generates the precession $\hat{R}_z(\omega t)$, and $\frac{\Omega}{2}\hat{X}$ generates $\hat{R}_x(\Omega t)$.



The Hadamard gate $\hat{H} = i\hat{R}_{\mathbf{n}}(\pi)$, $\mathbf{n} = (\mathbf{e}_x + \mathbf{e}_z)/\sqrt{2}$, turns $|0\rangle$ into $|+\rangle$ along the orange arc.

Single-Qubit Gates as Rotations

Every single-qubit gate is $\hat{G} = e^{i\chi}\hat{R}_{\mathbf{n}}(\theta)$ with an axis $\mathbf{n}$, an angle θ and an unobservable global phase $e^{i\chi}$. As an evolution for unit time, $\hat{G} = e^{-i\hat{\mathcal{H}}_G}$ with $\hat{\mathcal{H}}_G = \frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}} - \chi\hat{I}$.

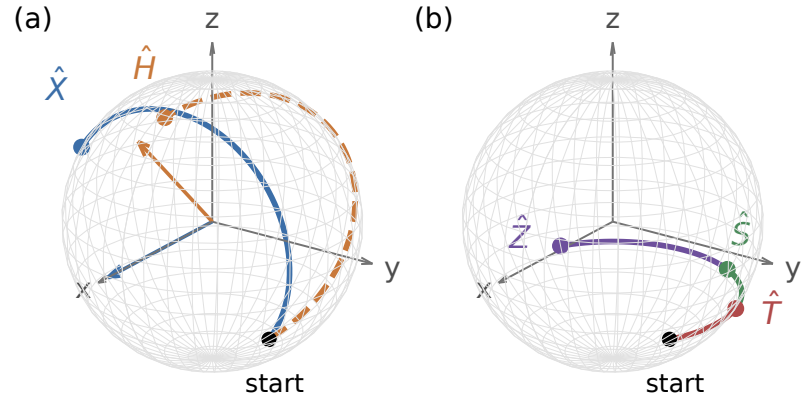
- The named gates are the bit flip $\hat{X}$, the phase flip $\hat{Z}$, the Hadamard gate $\hat{H}$ and the phase gates $\hat{S}$ and $\hat{T}$.
- For these five gates $\chi = \theta/2$, so the term $-\frac{\theta}{2}\hat{I}$ in $\hat{\mathcal{H}}_G$ produces only the global phase $e^{i\theta/2}$.
- The phase gates obey $\hat{T}^2 = \hat{S}$ and $\hat{S}^2 = \hat{Z}$.

Single-Qubit Gate Table

gate	matrix	rotation	axis $\mathbf{n}$	angle θ	generator $\hat{\mathcal{H}}_G$
$\hat{X}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$i\,\hat{R}_x(\pi)$	x	π	$\frac{\pi}{2}(\hat{X}-\hat{I})$
$\hat{Z}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$i\,\hat{R}_z(\pi)$	z	π	$\frac{\pi}{2}(\hat{Z}-\hat{I})$
$\hat{H}$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$i\,\hat{R}_{\mathbf{n}}(\pi)$	$(x+z)/\sqrt{2}$	π	$\frac{\pi}{2}(\hat{H}-\hat{I})$
$\hat{S}$	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$e^{i\pi/4}\hat{R}_z(\pi/2)$	z	$\pi/2$	$\frac{\pi}{4}(\hat{Z}-\hat{I})$
$\hat{T}$	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	$e^{i\pi/8}\hat{R}_z(\pi/4)$	z	$\pi/4$	$\frac{\pi}{8}(\hat{Z}-\hat{I})$

Gate Trajectories on the Bloch Sphere

We switch on $\hat{\mathcal{H}} = \frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ for $0 \le t \le 1$, so that $\hat{U}(t) = \hat{R}_{\mathbf{n}}(\theta t)$ ends at the gate. From the start $\vartheta = 2\pi/3$, $\varphi = \pi/3$, each path is an arc around the axis of its gate.



A One-Qubit Circuit

A **circuit** prepares a state, applies gates and measures. Each gate multiplies the state from the left, and the first gate stands rightmost in the matrix product. The box with a dial is a measurement in the computational basis.



matrix $\hat{S}\hat{H}$, $\hat{S}\hat{H}|0\rangle = \hat{S}|+\rangle = |+i\rangle$



matrix $\hat{H}\hat{S}$, $\hat{H}\hat{S}|0\rangle = \hat{H}|0\rangle = |+\rangle$

Interference, $\hat{H}\hat{Z}\hat{H} = \hat{X}$

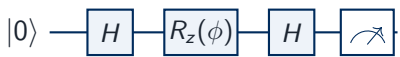
$\hat{H}$ exchanges the x and z axes, so a half turn about z between two Hadamard gates is a half turn about x,

$$|0\rangle \xrightarrow{\hat{H}} |+\rangle \xrightarrow{\hat{Z}} |-\rangle \xrightarrow{\hat{H}} |1\rangle.$$

$\hat{Z}$ changes only the relative phase, and the second $\hat{H}$ turns this phase into a different bit.

Ramsey Interference

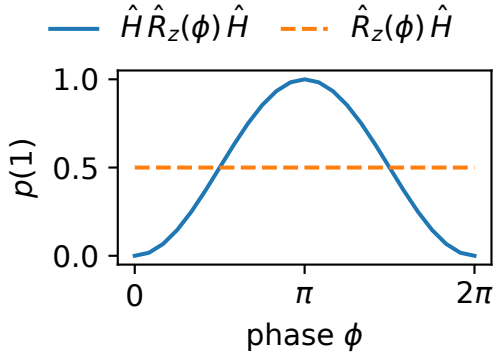
Replacing $\hat{Z}$ by $\hat{R}_z(\phi)$ gives the Ramsey circuit



with the amplitude of $|1\rangle$

$$\tfrac{1}{2}\left(e^{-i\phi/2} - e^{i\phi/2}\right) = -i \sin\tfrac{\phi}{2},$$

so $p(1) = \sin^2\frac{\phi}{2}$. Without the second $\hat{H}$, $p(1) = \frac{1}{2}$ for every ϕ, and the phase stays hidden.



With the closing $\hat{H}$ the probability oscillates between 0 and 1, and without it the record is a fair coin.

Other Bases and Shots



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Measurement Along x or y

A device reads out only the computational basis. To measure along another axis we first rotate that axis onto z.

$\hat{X}$ basis — H — $\hat{H}|+\rangle = |0\rangle,\ \hat{H}|-\rangle = |1\rangle$

$\hat{Y}$ basis — $S^\dagger$ — H — $\hat{S}^\dagger|\pm i\rangle = |\pm\rangle$, then $\hat{H}$

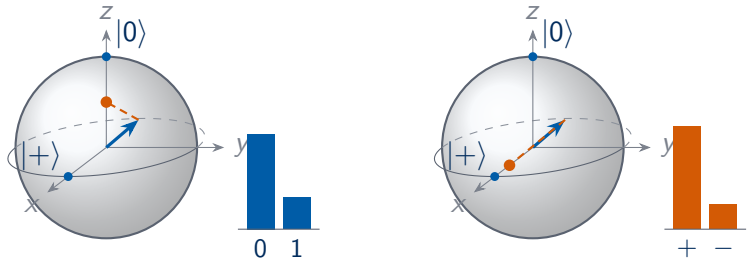
Bit 0 always stands for the eigenvalue $+1$ and bit 1 for -1, so in every basis

$$p_\alpha(0) = \frac{1 + \langle\hat{\sigma}_\alpha\rangle}{2}, \qquad p_\alpha(1) = \frac{1 - \langle\hat{\sigma}_\alpha\rangle}{2}, \qquad \alpha = x, y, z.$$

A measurement along an axis therefore reads the component of the Bloch vector along that axis.

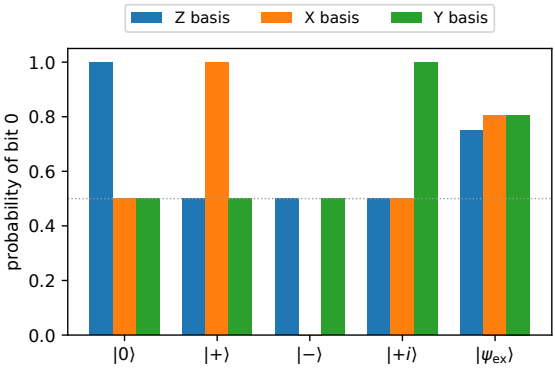
The Running Example in the $\hat{Z}$ and $\hat{X}$ Bases

The outcome probabilities follow from the projection of the Bloch vector $\mathbf{r}_{\mathrm{ex}}$ onto the measured axis (orange dot), and the bars show $p(0)$ and $p(1)$.



With $\langle \hat{Z} \rangle = \frac{1}{2}$ and $\langle \hat{X} \rangle = \langle \hat{Y} \rangle = 0.612$ the running example gives $p_z(0) = 0.750$ and $p_x(0) = p_y(0) = 0.806$. The phase $e^{i\pi/4}$, invisible in the $\hat{Z}$ record, shows in the $\hat{X}$ and $\hat{Y}$ records.

Records in Different Bases



The bars show the probability of bit 0. A bar at 1 or 0 is a certain outcome, and a bar at 0.5 is a fair coin.

- No state gives a certain outcome in two bases at once (**complementarity**).
- $|+\rangle$ and $|-\rangle$ give identical $\hat{Z}$ records and opposite $\hat{X}$ records.
- ψ_{ex} leans to bit 0 in all three bases, since its Bloch vector lies in the octant $x, y, z > 0$.

Estimating an Expectation Value from Shots

M repetitions return $z_j = \pm 1$, and the sample mean estimates $\langle \hat{Z} \rangle$,

$$\bar{z} = \frac{1}{M} \sum_{j=1}^{M} z_j, \qquad \mathrm{Var}(z_j) = \underbrace{\langle \hat{Z}^2 \rangle}_{=1 \text{ since } z_j^2 = 1} - \langle \hat{Z} \rangle^2 .$$

For independent shots the standard error of the mean is

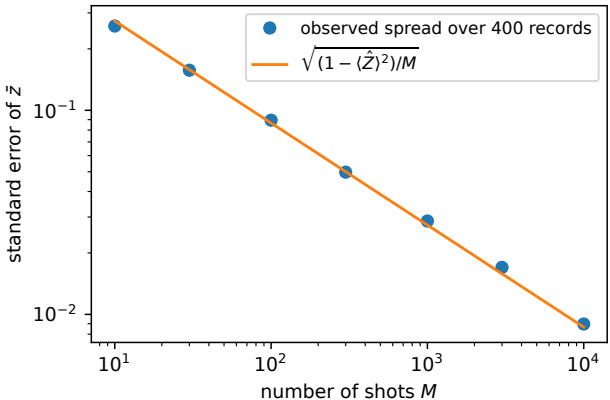
$$\mathrm{std}(\bar{z}) = \sqrt{\frac{1 - \langle \hat{Z} \rangle^2}{M}} .$$

Since $\langle \hat{Z} \rangle$ is unknown in practice, we divide the sample standard deviation of the record by $\sqrt{M}$. Doubling the precision costs four times as many shots.

One record of 1000 shots of ψ_{ex}, exact $\langle \hat{Z} \rangle = 0.5$

The record gives $\bar{z} = 0.484 \pm 0.028$, and the formula gives $\sqrt{0.75/1000} = 0.027$.

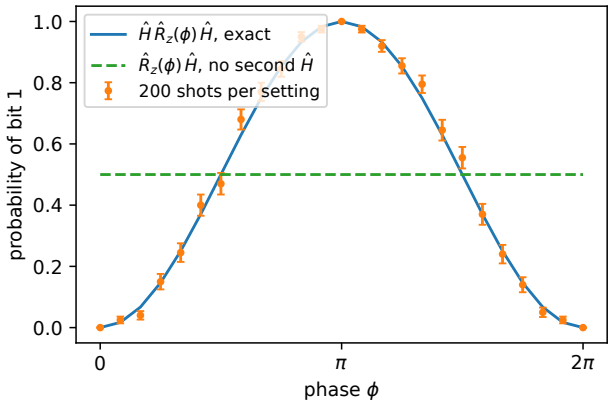
Shot Noise



The observed spread of the estimate over 400 repeated records falls with the number of shots M along a line of slope $-1/2$ on log–log axes, as $\sqrt{(1-\langle\hat{Z}\rangle^2)/M}$ predicts.

Every expectation value measured on a device carries this error.

A Ramsey Experiment with Shots



The Ramsey circuit is run with 200 shots per phase ϕ. The recorded frequencies scatter around $p(1) = \sin^2 \frac{\phi}{2}$.

The error bars $\sqrt{p(1-p)/M}$ are largest near $p = \frac{1}{2}$ and vanish where the outcome is certain.

The Two-Spin Register in a Computer and Its Bit Strings



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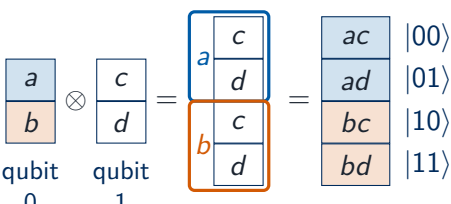
The Tensor Product

Qubit 0 in $|\psi^{(0)}\rangle$ and qubit 1 in $|\psi^{(1)}\rangle$ form the **product state** $|\psi^{(0)}\rangle \otimes |\psi^{(1)}\rangle$, with

$$|\psi^{(0)}\rangle = \begin{pmatrix} a \\ b \end{pmatrix} = a|0\rangle + b|1\rangle, \qquad |\psi^{(1)}\rangle = \begin{pmatrix} c \\ d \end{pmatrix} = c|0\rangle + d|1\rangle.$$

The product $\otimes$ is linear in each factor, so

$$|\psi^{(0)}\rangle|\psi^{(1)}\rangle = ac\,\underbrace{|0\rangle \otimes |0\rangle}_{|00\rangle} + ad\,\underbrace{|0\rangle \otimes |1\rangle}_{|01\rangle} + bc\,\underbrace{|1\rangle \otimes |0\rangle}_{|10\rangle} + bd\,\underbrace{|1\rangle \otimes |1\rangle}_{|11\rangle}.$$



Each entry of the first vector multiplies the whole second vector. In code the product is `jnp.kron(psi0, psi1)`. The first factor, qubit 0, gives the left bit of the label. For $a = 1, b = 0$ and $c = d = \frac{1}{\sqrt{2}}$ we obtain $|0\rangle|+\rangle = \frac{1}{\sqrt{2}}(1, 1, 0, 0)^\top$.

Basis and Order of the 2^N Amplitudes

The four products of basis states span $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$, and a general state is a normalised superposition $|\psi\rangle = \sum_{b_0,b_1} \psi_{b_0 b_1} |b_0 b_1\rangle$ with $\sum |\psi_{b_0 b_1}|^2 = 1$. Most such states are not products. The label $b_0 b_1$ is a *bit string*, and the states $|b_0 b_1\rangle$ form the **computational basis**, the common eigenvectors of $\hat{Z}$ on each qubit with $+1$ for $b_q = 0$.

Stored order, the index of a bit string

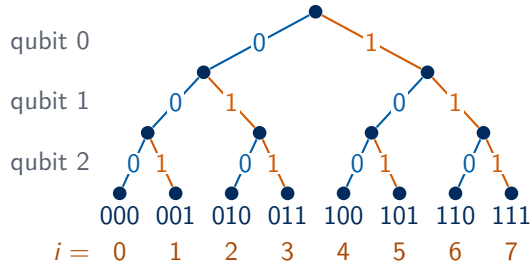
A computer stores N qubits as 2^N complex amplitudes in this basis, in the order of the Kronecker product,

$$i = \sum_{q=0}^{N-1} b_q \, 2^{N-1-q} \qquad \text{(qubit 0 is the most significant bit)},$$

so qubit 0 changes slowest, as in counting in binary.

Ordering Tables and Memory

$N = 2$		
bit string	i	$2b_0 + b_1$
00	0	$0 + 0$
01	1	$0 + 1$
10	2	$2 + 0$
11	3	$2 + 1$



- Rows and columns of operators use the same order.
- Some libraries, Qiskit among them, take qubit 0 as the least significant bit. Their bit strings read right to left and must be reordered before a comparison.
- At 16 bytes per complex amplitude the state takes $16 \cdot 2^N$ bytes, 16 GiB at $N = 30$, 16 TiB at $N = 40$ and 16 PiB at $N = 50$.

Kronecker Products of Operators

Operators on the register are Kronecker products as well. For 2×2 matrices the product is a 4×4 block matrix,

$$\hat{A} \otimes \hat{B} = \begin{pmatrix} A_{00}\hat{B} & A_{01}\hat{B} \\ A_{10}\hat{B} & A_{11}\hat{B} \end{pmatrix}, \qquad (\hat{A} \otimes \hat{B})_{2a+b,\, 2a'+b'} = A_{aa'} B_{bb'},$$

which acts factor by factor, $(\hat{A} \otimes \hat{B})(|u\rangle \otimes |v\rangle) = \hat{A}|u\rangle \otimes \hat{B}|v\rangle$. A gate $\hat{U}$ on qubit 0 alone is $\hat{U} \otimes \hat{I}$, and on qubit 1 alone $\hat{I} \otimes \hat{U}$, for example

$$\hat{X} \otimes \hat{I} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{matrix} 00 \leftrightarrow 10 \\ 01 \leftrightarrow 11 \end{matrix} \qquad \hat{I} \otimes \hat{X} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{matrix} 00 \leftrightarrow 01 \\ 10 \leftrightarrow 11 \end{matrix}$$

$\hat{X} \otimes \hat{I}$ flips the left bit and $\hat{I} \otimes \hat{X}$ the right bit. Diagonal operators read bits, $\hat{Z} \otimes \hat{I} = \mathrm{diag}(1, 1, -1, -1)$ and $\hat{I} \otimes \hat{Z} = \mathrm{diag}(1, -1, 1, -1)$, with $+1$ where $b_q = 0$. In code, `np.kron(X, I2)`.

Bit Strings of Two Qubits

For the register the Born rule gives the bit string $b_0 b_1$ with probability $p(b_0 b_1) = |\psi_{b_0 b_1}|^2$. A record of M shots is a list of M bit strings, summarised by three quantities.

- The **marginal** of qubit 0, $p_0(b_0) = \sum_{b_1} p(b_0 b_1)$, is what an observer of qubit 0 alone sees.
- The **single-qubit expectation** is $\langle \hat{Z}_q \rangle = p_q(0) - p_q(1)$.
- The **correlator** $\langle \hat{Z}_0 \hat{Z}_1 \rangle = \sum_{b_0 b_1} (-1)^{b_0 + b_1} p(b_0 b_1)$ equals $+1$ when the bits always agree, -1 when they always differ, and 0 when agreement and disagreement are equally likely.

Correlation and Independence

The bits are independent when the joint probabilities factorise, $p(b_0 b_1) = p_0(b_0)\, p_1(b_1)$. Independent bits have $\langle \hat{Z}_0 \hat{Z}_1 \rangle = \langle \hat{Z}_0 \rangle \langle \hat{Z}_1 \rangle$, which vanishes only when one of the two means is zero.

Product state $\hat{R}_y(2\pi/3)|0\rangle \otimes |0\rangle$ with qubit 0 tilted

Exact values are $\langle \hat{Z}_0 \rangle = -0.5$, $\langle \hat{Z}_1 \rangle = 1$ and $\langle \hat{Z}_0 \hat{Z}_1 \rangle = -0.5$, and 2000 shots give $\langle \hat{Z}_0 \hat{Z}_1 \rangle = -0.489 \pm 0.020$.

Bit Strings in Different Bases

A **measurement setting** assigns a basis to each qubit and is written as a pair of letters. In the setting $\hat{Z}\hat{X}$ qubit 0 is measured along z and qubit 1 along x, implemented as $\hat{H}$ on qubit 1 and then $\hat{Z}$ on both.
The state $|0\rangle \otimes |+\rangle$ has qubit 0 up along z and qubit 1 along $+x$. Counts in 1000 shots are

setting	00	01	10	11	
$\hat{Z}\hat{Z}$	508	492	0	0	qubit 0 fixed, qubit 1 random
$\hat{Z}\hat{X}$	1000	0	0	0	both fixed
$\hat{X}\hat{X}$	501	0	499	0	qubit 0 random, qubit 1 fixed

Only the setting that measures each qubit along its own axis gives a deterministic record. Since one state gives three different records, the setting is part of the experiment.

Two-Qubit Gates as Rotations



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Two-Qubit Gates from Interactions

Single-qubit gates map product states to product states. Correlations that no product state carries require an *interaction*, a term of the Hamiltonian that acts on both qubits. An Ising-type coupling, with energy proportional to the product of the same spin component $\alpha \in \{x, y, z\}$ of both qubits, gives

$$\hat{\mathcal{H}}_{\alpha\alpha} = \frac{J}{2}\,\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha \quad \Longrightarrow \quad \hat{R}_{\alpha\alpha}(\theta) = e^{-i\hat{\mathcal{H}}_{\alpha\alpha}t} = e^{-i\theta\,\hat{\sigma}_\alpha\otimes\hat{\sigma}_\alpha/2}, \qquad \theta = Jt.$$

Since $(\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha)^2 = \hat{\sigma}_\alpha^2 \otimes \hat{\sigma}_\alpha^2 = \hat{I}$, the identity $e^{-ia\hat{P}} = \cos a\,\hat{I} - i \sin a\,\hat{P}$ applies unchanged,

$$\hat{R}_{\alpha\alpha}(\theta) = \cos\tfrac{\theta}{2}\,\hat{I} - i \sin\tfrac{\theta}{2}\,\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha.$$

The angle θ is a rotation angle in two-dimensional subspaces, for $\hat{R}_{xx}$ the pair $\{|00\rangle, |11\rangle\}$ treated as a Bloch sphere.

CNOT, CZ and SWAP

With the projectors $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ on qubit 0,

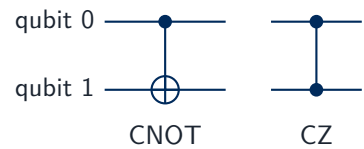
$$\widehat{\mathrm{CNOT}} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{X},$$

$$\widehat{\mathrm{CZ}} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{Z} = \mathrm{diag}(1, 1, 1, -1).$$

In the basis $00, 01, 10, 11$

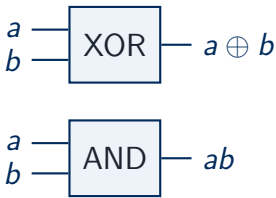
$$\widehat{\mathrm{CNOT}} = \begin{pmatrix} 1&0&0&0 \\ 0&1&0&0 \\ 0&0&0&1 \\ 0&0&1&0 \end{pmatrix}, \quad \widehat{\mathrm{SWAP}} = \begin{pmatrix} 1&0&0&0 \\ 0&0&1&0 \\ 0&1&0&0 \\ 0&0&0&1 \end{pmatrix}.$$

CNOT flips qubit 1 if and only if qubit 0 is 1, the reversible XOR $b_1 \to b_1 \oplus b_0$. CZ multiplies $|11\rangle$ by -1, and SWAP exchanges the qubits, $|01\rangle \leftrightarrow |10\rangle$.



The top wire is qubit 0, and time runs from left to right. The dot marks the **control**, and $\oplus$ the **target** that is flipped. CZ is symmetric in the two qubits and has a dot at both ends.

Reversible Classical Logic



two bits in,
one bit out



$00 \to 00$
$01 \to 01$
$10 \to 11$
$11 \to 10$

CNOT

XOR and AND map two bits to one, so the inputs cannot be recovered from the output. CNOT keeps a and computes $a \oplus b$ reversibly, so as a permutation of bit strings it is unitary.

The three-qubit Toffoli gate computes AND reversibly in the same way (section on N-qubit registers). With both, a quantum computer can run any classical program, and superposition and interference are additional resources.

$\hat{R}_{xx}$ as a Rotation of the Pair $|00\rangle, |11\rangle$

$\hat{X} \otimes \hat{X}$ flips both bits. With $c = \cos\frac{\theta}{2}$ and $s = \sin\frac{\theta}{2}$,

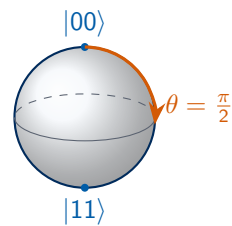
$$\hat{R}_{xx}(\theta) = c\hat{I} - is\,\hat{X} \otimes \hat{X} = \begin{pmatrix} c & 0 & 0 & -is \\ 0 & c & -is & 0 \\ 0 & -is & c & 0 \\ -is & 0 & 0 & c \end{pmatrix},$$

which acts as $\hat{R}_x(\theta) = \begin{pmatrix} c & -is \\ -is & c \end{pmatrix}$ on each pair $\{|00\rangle, |11\rangle\}$ and $\{|01\rangle, |10\rangle\}$.

The pair $|00\rangle, |11\rangle$ spans a Bloch sphere with north pole $|00\rangle$ and south pole $|11\rangle$. At $\theta = \pi/2$ the rotation takes $|00\rangle$ to the equator,

$$\hat{R}_{xx}(\tfrac{\pi}{2})|00\rangle = \frac{|00\rangle - i|11\rangle}{\sqrt{2}},$$

which is no product state.



$\hat{R}_{zz}$ as a Conditional Rotation

$\hat{Z} \otimes \hat{Z}$ is diagonal. Writing $\hat{Z} = |0\rangle\langle 0| - |1\rangle\langle 1|$ for qubit 0 gives

$$\hat{Z} \otimes \hat{Z} = |0\rangle\langle 0| \otimes \hat{Z} - |1\rangle\langle 1| \otimes \hat{Z} \quad \Longrightarrow \quad \hat{R}_{zz}(\theta) = |0\rangle\langle 0| \otimes \hat{R}_z(\theta) + |1\rangle\langle 1| \otimes \hat{R}_z(-\theta),$$

$$\hat{R}_{zz}(\theta) = \mathrm{diag}\Big(\underbrace{e^{-i\theta/2},\, e^{i\theta/2}}_{\text{qubit 0} = |0\rangle,\ \hat{R}_z(\theta)} \ , \quad \underbrace{e^{i\theta/2},\, e^{-i\theta/2}}_{\text{qubit 0} = |1\rangle,\ \hat{R}_z(-\theta)} \Big).$$

- Qubit 1 turns about z by $+\theta$ or $-\theta$ depending on the state of qubit 0.
- The populations of the four bit strings never change, only their relative phases.
- On $|{+}{+}\rangle$ these conditional phases create a state that is not a product.
- It is the Ising coupling $\frac{J}{2}\hat{Z}_0\hat{Z}_1$ switched on for the time $t = \theta/J$.

CNOT and CZ as Conditional Half Turns

Since $\hat{X} = i\hat{R}_x(\pi)$, CNOT is a half turn about x of qubit 1, conditioned on qubit 0,

$$\widehat{\mathrm{CNOT}} = (\hat{S} \otimes \hat{I})\,[\,|0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_x(\pi)],$$

where $\hat{S}$ on qubit 0 supplies the phase i on the branch where qubit 0 is 1. As an evolution operator, let $\hat{P} = \underbrace{\frac{\hat{I}-\hat{Z}}{2}}_{|1\rangle\langle 1| \text{ on qubit 0}} \otimes \underbrace{\frac{\hat{I}-\hat{X}}{2}}_{|-\rangle\langle -| \text{ on qubit 1}}$, a projector with $\hat{P}^2 = \hat{P}$.

Summing the series with $\hat{P}^k = \hat{P}$ gives

$$e^{-i\pi\hat{P}} = \hat{I} + \underbrace{(e^{-i\pi} - 1)}_{-2}\hat{P} = \hat{I} - 2\hat{P},$$

which flips the sign of $|1\rangle|-\rangle$ alone, the action of CNOT. Hence

$$\widehat{\mathrm{CNOT}} = \exp[-i\pi(\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{X})/4], \qquad \widehat{\mathrm{CZ}} = \exp[-i\pi(\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{Z})/4].$$

CZ is generated by the projector on $|11\rangle$, and the sign in the exponent is irrelevant, since $e^{\pm i\pi} = -1$.

CZ and CNOT from the Ising Rotation

Expanding $(\hat{I} - \hat{Z}_0)(\hat{I} - \hat{Z}_1)$ in the exponent of CZ gives the Ising rotation and two z rotations,

$$\widehat{\mathrm{CZ}} = e^{-i\pi/4}\big[\hat{R}_z(-\tfrac{\pi}{2}) \otimes \hat{R}_z(-\tfrac{\pi}{2})\big]\,\hat{R}_{zz}(\tfrac{\pi}{2}), \qquad \widehat{\mathrm{CNOT}} = (\hat{I} \otimes \hat{H})\,\widehat{\mathrm{CZ}}\,(\hat{I} \otimes \hat{H}).$$

- CZ is the Ising coupling $\frac{J}{2}\hat{Z}_0\hat{Z}_1$ switched on for the time $t = \pi/(2J)$, followed by one z rotation on each qubit.
- The Hadamard pair on the target turns CZ into CNOT, since $\hat{H}\hat{Z}\hat{H} = \hat{X}$.

SWAP and the Heisenberg Exchange

The exchange coupling $\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z}=4\,\hat{\mathbf{S}}_0\cdot\hat{\mathbf{S}}_1$, with $\hat{\mathbf{S}}_q=\hat{\boldsymbol{\sigma}}_q/2$, is the Heisenberg exchange, a model Hamiltonian of two coupled spins whose physical origin on each platform is the subject of Part II.

eigenstate	exchange symmetry	eigenvalue of the coupling
$\lvert 00\rangle,\ \lvert 11\rangle,\ (\lvert 01\rangle+\lvert 10\rangle)/\sqrt{2}$	symmetric (triplet)	$+1$
$(\lvert 01\rangle-\lvert 10\rangle)/\sqrt{2}$	antisymmetric (singlet)	-3

SWAP is $+1$ on the symmetric states and -1 on the singlet, which gives both forms

$$\begin{aligned}\widehat{\mathrm{SWAP}} &= \tfrac{1}{2}\big(\hat{I}+\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z}\big)\\ &= e^{i\pi/4}\exp\big[-i\tfrac{\pi}{4}(\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z})\big].\end{aligned}$$

In the exponential form the symmetric states acquire the factor $e^{i\pi/4}e^{-i\pi/4}=1$, and the singlet acquires $e^{i\pi/4}e^{3i\pi/4}=-1$.

iSWAP and $\sqrt{\mathrm{SWAP}}$ from Exchange

Both gates act only on the pair $|01\rangle, |10\rangle$,

$$\widehat{\mathrm{iSWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \sqrt{\widehat{\mathrm{SWAP}}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1+i}{2} & \frac{1-i}{2} & 0 \\ 0 & \frac{1-i}{2} & \frac{1+i}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- The excitation exchange $|01\rangle\langle 10| + |10\rangle\langle 01| = \frac{1}{2}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1)$, switched on as $\frac{J}{2}(\hat{X}\hat{X} + \hat{Y}\hat{Y})$ for $Jt = \pi/2$, gives $\widehat{\mathrm{iSWAP}}^\dagger$, and $\widehat{\mathrm{iSWAP}}$ for the opposite sign of J.
- The Heisenberg exchange $J_{\mathrm{ex}}\hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1 = \frac{J_{\mathrm{ex}}}{4}(\hat{X}\hat{X} + \hat{Y}\hat{Y} + \hat{Z}\hat{Z})$ gives $\sqrt{\widehat{\mathrm{SWAP}}}$ for $\int J_{\mathrm{ex}}dt = \pi/2$ and SWAP for $\int J_{\mathrm{ex}}dt = \pi$. The register Hamiltonian later uses $J = J_{\mathrm{ex}}/4$.

Parametrised and Fixed Gates

A **parametrised gate** $\hat{U}(\theta)$ is a family labelled by a continuous angle, and a **fixed gate** is one member at a fixed angle. A coupling $\frac{J(t)}{2}\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha$ commutes with itself at all times, so

$$\hat{U} = \hat{R}_{\alpha\alpha}(\theta), \qquad \theta = \int_0^T J(t')\,dt' \quad (= JT \text{ for a constant coupling}).$$

- Any angle is reached by choosing the duration.
- A fixed gate such as CZ corresponds to one fixed duration of its coupling.

Gates in an N-Qubit Register



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Pauli Strings

A **Pauli string** is a Kronecker product of one Pauli matrix or identity per qubit,

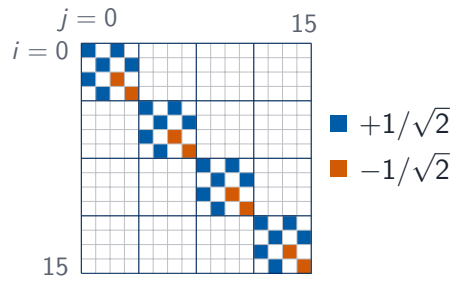
$$\hat{P}_0 \otimes \hat{P}_1 \otimes \cdots \otimes \hat{P}_{N-1}, \quad \hat{P}_q \in \{\hat{I}, \hat{X}, \hat{Y}, \hat{Z}\}, \qquad \text{for example } \hat{X}_0\hat{Z}_2 = \hat{X} \otimes \hat{I} \otimes \hat{Z} \otimes \hat{I} \ (N = 4).$$

- There are 4^N strings, each Hermitian with $\hat{P}^2 = \hat{I}$. Their eigenvalues are ± 1, and $e^{-ia\hat{P}} = \cos a\, \hat{I} - i \sin a\, \hat{P}$ applies.
- Two strings commute when the number of positions where both are non-identity and different is even, and anticommute otherwise.
- They form an operator basis with $\mathrm{Tr}(\hat{P}\hat{P}') = 2^N \delta_{PP'}$. Every Hamiltonian of N qubits is a real linear combination of strings, with the coefficient $\mathrm{Tr}(\hat{P}\hat{\mathcal{H}})/2^N$.
- Strings of $\hat{Z}$ and $\hat{I}$ alone are diagonal. $\hat{Z}_0\hat{Z}_2$ has the entry $(-1)^{b_0+b_2}$ at the bit string $b_0 b_1 b_2 b_3$, the parity of the selected bits.

A One-Qubit Gate on Qubit q of N

In Kronecker form the gate $\hat{U}$ stands at position q, with identities elsewhere. For $N = 4$ the Hadamard gate on qubit 2 is $\hat{H}_2 = \hat{I} \otimes \hat{I} \otimes \hat{H} \otimes \hat{I}$, a 16×16 matrix built by `reduce(np.kron, [I, I, H, I])`.

- It changes only bit 2, worth $2^{N-1-q} = 2$ in the index, so an entry (i, j) is non-zero only for $j = i$ or $j = i \oplus 2$.
- It mixes each pair i, $i \oplus 2$, for example $\psi'_{0000} = (\psi_{0000} + \psi_{0010})/\sqrt{2}$.
- Each row has two non-zero entries. The cost is $2 \cdot 2^N$ products, against 4^N for a dense multiplication.



CNOT on a Non-Neighbouring Pair

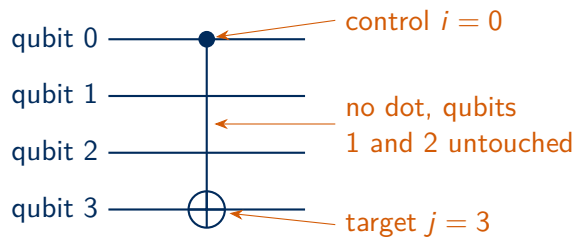
With control i, target j and the identity on every other qubit, the gate is a sum of two Kronecker products,

$$\widehat{\mathrm{CNOT}}_{i\to j} = |0\rangle\langle 0|_i \otimes \hat{I} + |1\rangle\langle 1|_i \otimes \hat{X}_j, \qquad \widehat{\mathrm{CNOT}}_{0\to 3} = |0\rangle\langle 0| \otimes \hat{I} \otimes \hat{I} \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{I} \otimes \hat{I} \otimes \hat{X}.$$

It permutes bit strings, $b_j \to b_j \oplus b_i$. For $N = 4$, $\widehat{\mathrm{CNOT}}_{0\to 3}$ acts as

before	after
1000 (8)	1001 (9)
1010 (10)	1011 (11)
1100 (12)	1101 (13)
1110 (14)	1111 (15)

and back, while strings with $b_0 = 0$ are unchanged.

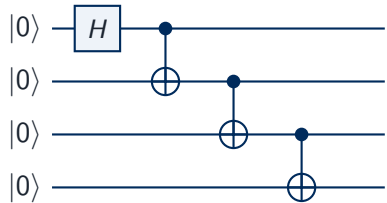


The GHZ State of N Qubits

The Greenberger–Horne–Zeilinger (GHZ) state of N qubits is

$$|\mathrm{GHZ}_N\rangle = \frac{|0\cdots0\rangle + |1\cdots1\rangle}{\sqrt{2}},$$

non-zero only at the indices 0 and $2^N - 1$. All bits agree in every shot of a $\hat{Z}$ measurement, while each bit alone is a fair coin.



A Hadamard gate on qubit 0 and the chain $\widehat{\mathrm{CNOT}}_{q-1\to q}$, $q = 1, \ldots, N-1$, prepare it. The fan-out $\widehat{\mathrm{CNOT}}_{0\to1}, \ldots, \widehat{\mathrm{CNOT}}_{0\to N-1}$ prepares the same state.

Sparse Operators

Most entries of $\hat{I} \otimes \cdots \otimes \hat{G} \otimes \cdots \otimes \hat{I}$ are zero. Code stores only the non-zero ones in compressed sparse rows (CSR), `scipy.sparse.kron(A, B, format="csr")`.

operator on N qubits	non-zero entries	dense entries
one-qubit gate $\hat{G}$ on qubit q	$2 \cdot 2^N$	4^N
Pauli string	2^N, one per row	4^N
$\widehat{\mathrm{CNOT}}_{i \to j}$ on any pair	2^N, a permutation	4^N
Hamiltonian with $\hat{L}$ Pauli strings	$\leq \hat{L} \cdot 2^N$	4^N

A matrix–vector product costs a time proportional to the number of non-zero entries, and at $N = 20$ it runs on a laptop. `scipy.sparse.linalg.expm_multiply` applies $e^{-it\hat{\mathcal{H}}}$ to $|\psi\rangle$ without forming the exponential. Sparse eigensolvers (Lanczos, `eigsh`) find ground states from products $\hat{\mathcal{H}}|\psi\rangle$ alone.

The State Tensor and `einsum`

A reshape of the 2^N vector into N axes of length 2 gives the **state tensor**, with one index per qubit, $\psi[b_0, \ldots, b_{N-1}] = \psi_i$ and $i = \sum_q b_q 2^{N-1-q}$. NumPy and JAX store arrays in C order, last axis fastest, so `psi.reshape((2,)*N)` moves no number.

Reshape and transpose

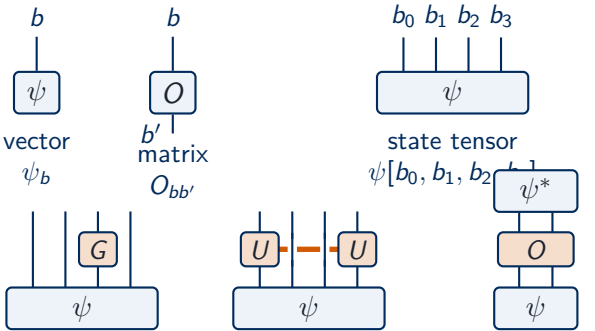
A reshape regroups, $(0,1,2,3) \to \psi[b_0, b_1] = \left(\begin{smallmatrix} 0 & 1 \\ 2 & 3 \end{smallmatrix}\right)$. A transpose exchanges, and flattening gives $(0,2,1,3)$, a different state with the qubits swapped.

`einsum` writes a sum over indices as letters. A letter in two inputs and absent from the output is summed, an output letter is kept, and output axes follow the order after `->`.

operation	`einsum`
$w_i = \sum_j A_{ij} v_j$	`"ij,j->i"`
$\langle a \vert b \rangle$	`"i,i->"` on `(a.conj(), b)`
$\mathrm{Tr} A$	`"ii->"`

Tensor Contractions

A box is a tensor, a line an index, a joined line a summed index. The results equal the Kronecker-product matrices at a cost proportional to 2^N per gate.



- one-qubit gate on qubit 2 of 4, `"Cc,abcd->abCd"`
- two-qubit gate on qubits 0 and 3, `"ADad,abcd->AbcD"`
- $\langle\psi|\hat{O}|\psi\rangle$, two-qubit $\hat{O}$, `"AB,ABab,ab->"`

Upper-case letters are row indices of operators.

The Toffoli Gate and Circuit Cost

Toffoli gate as an evolution operator

The Toffoli gate flips qubit 2 when qubits 0 and 1 are both 1, $b_2 \to b_2 \oplus (b_0 \wedge b_1)$, the reversible AND of classical computing. Its generator is the projector on $|1\rangle|1\rangle|-\rangle$, and as for CNOT $e^{-i\pi\hat{P}} = \hat{I} - 2\hat{P}$, so

$$\widehat{\text{Toffoli}} = \exp\Big[- i\tfrac{\pi}{8} (\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{X}) \Big].$$

- The *depth* of a circuit, the number of layers of gates acting in parallel, and its gate count measure its cost.
- Useful circuits are shallow, since each gate on a device adds noise.

Entanglement



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The Bell State and Its Predictions

The **Bell state** of two qubits is

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Each qubit alone

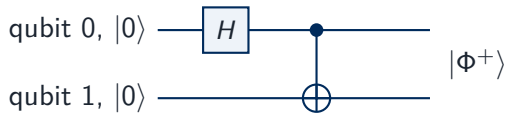
A measurement of one qubit in $\hat{Z}$, $\hat{X}$ or $\hat{Y}$ gives a fair coin, $\langle \hat{X}_q \rangle = \langle \hat{Y}_q \rangle = \langle \hat{Z}_q \rangle = 0$.

Both qubits together

The two bits agree in every shot in $\hat{Z}\hat{Z}$ and $\hat{X}\hat{X}$ and disagree in every shot in $\hat{Y}\hat{Y}$, $\langle \hat{Z}\hat{Z} \rangle = \langle \hat{X}\hat{X} \rangle = 1$, $\langle \hat{Y}\hat{Y} \rangle = -1$.

No product state $|u\rangle \otimes |v\rangle$ gives these records. Its probabilities factorise, $p(b_0 b_1) = p_0(b_0) p_1(b_1)$, and bits that always agree in $\hat{Z}$ would then require certain outcomes in p_0 and p_1, contradicting the fair coin of each qubit. The Bell state is therefore **entangled**, i.e. not a product of single-qubit states.

Preparing a Bell State



$$|00\rangle \xrightarrow{\hat{H}\otimes\hat{I}} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\widehat{\text{CNOT}}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

CNOT flips qubit 1 only in the branch where qubit 0 is 1. The interaction gives the same kind of state, $\hat{R}_{xx}(\frac{\pi}{2})|00\rangle = (|00\rangle - i|11\rangle)/\sqrt{2}$, which $\hat{S}$ on qubit 0 maps to $|\Phi^+\rangle$.

The Product Test

We arrange the four amplitudes as a 2×2 matrix $\Psi_{b_0 b_1} = \psi_{b_0 b_1}$, with rows labelled by qubit 0. A product state has $\psi_{b_0 b_1} = a_{b_0} c_{b_1}$, so

$$\text{product state} \iff \Psi = a\, c^{\top} \iff \operatorname{rank} \Psi = 1.$$

$|0\rangle \otimes |+\rangle$

$$\Psi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \operatorname{rank} \Psi = 1,$$

a product state.

$|\Phi^+\rangle$

$$\Psi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \operatorname{rank} \Psi = 2,$$

an entangled state.

The Schmidt Decomposition

The singular value decomposition of Ψ is the **Schmidt decomposition** of $|\psi\rangle$,

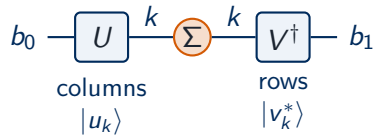
$$\Psi = U\,\Sigma\,V^\dagger \quad\Longrightarrow\quad |\psi\rangle = \sum_{k=1}^{2} s_k\,|u_k\rangle \otimes |v_k^*\rangle,$$

with $\Sigma = \mathrm{diag}(s_1, s_2)$, $s_1 \geq s_2 \geq 0$ and $s_1^2 + s_2^2 = 1$. The orthonormal Schmidt vectors $|u_k\rangle$ of qubit 0 are the columns of U, and the components of $|v_k^*\rangle$ of qubit 1 form the k-th row of $V^\dagger$ (`Vh` in NumPy).

Measuring qubit 0 in the basis $|u_k\rangle$ gives k with probability s_k^2. The number of non-zero **Schmidt coefficients** s_k is the **Schmidt rank**, 1 for a product, 2 for an entangled state.

	$b_1{=}0$	$b_1{=}1$
$b_0{=}0$	ψ_{00}	ψ_{01}
$b_0{=}1$	ψ_{10}	ψ_{11}

$= U\,\Sigma\,V^\dagger$



Schmidt Decompositions of the Two Examples

$|0\rangle \otimes |+\rangle$

$$\Psi = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \underbrace{1}_{s_1} \cdot \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{\text{column of } U} \underbrace{\tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \end{pmatrix}}_{\text{row of } V^\dagger,\ |v_1\rangle = |+\rangle}, \qquad s_2 = 0, \text{ a product.}$$

$|\Phi^+\rangle$

$$\Psi = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \tfrac{1}{\sqrt{2}}\,|0\rangle\langle 0| + \tfrac{1}{\sqrt{2}}\,|1\rangle\langle 1|, \qquad s_1 = s_2 = 0.7071, \text{ the largest spread.}$$

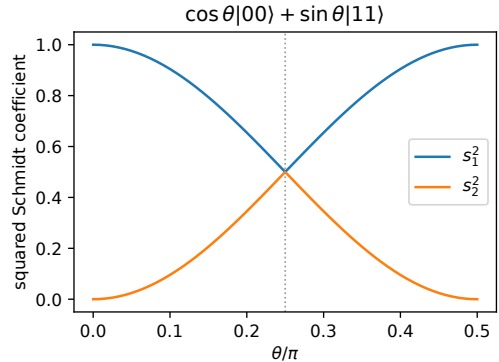
Gates $\hat{A} \otimes \hat{B}$ on single qubits act as $\Psi \to A\,\Psi B^\top$ with the matrices A, B of the gates and keep the s_k, so $\hat{R}_{xx}(\frac{\pi}{2})|00\rangle$ and $|\Phi^+\rangle$ are equally entangled.

A Family of Product and Entangled States

$$|\psi_\theta\rangle = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle$$

has $\Psi = \mathrm{diag}(\cos\theta, \sin\theta)$ and the Schmidt coefficients $|\cos\theta|$ and $|\sin\theta|$.

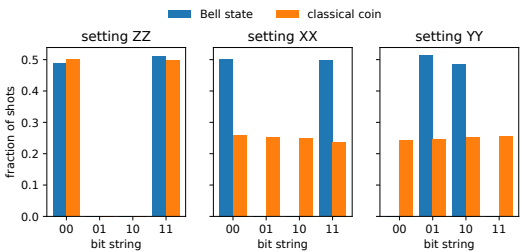
- At $\theta = 0$ the state is the product $|00\rangle$.
- At $\theta = \pi/4$ (dotted) it is the Bell state, with both squares equal to $\frac{1}{2}$.
- At $\theta = \pi/2$ it is the product $|11\rangle$, and beyond $\pi/4$ the two terms exchange roles.



The sorted squares $s_1^2 = \max(\cos^2\theta, \sin^2\theta)$ and $s_2^2 = \min(\cos^2\theta, \sin^2\theta)$.

Bell Correlations and a Classical Coin

A **classical correlated coin** prepares $|00\rangle$ or $|11\rangle$ with probability $\frac{1}{2}$ each and reproduces the $\hat{Z}\hat{Z}$ record of $|\Phi^+\rangle$.



Frequencies of the four bit strings in each setting, Bell state (blue) and coin (orange).

	$\langle \hat{Z}\hat{Z} \rangle$	$\langle \hat{X}\hat{X} \rangle$	$\langle \hat{Y}\hat{Y} \rangle$
Bell, exact	$+1$	$+1$	-1
Bell, sampled	$+1.000$	$+1.000$	-1.000
coin, sampled	$+1.000$	-0.006	-0.002

Since the coin holds $|00\rangle$ or $|11\rangle$ in every run, $\hat{X}$ and $\hat{Y}$ give fair, independent bits with $\langle \hat{X}\hat{X} \rangle = \langle \hat{Y}\hat{Y} \rangle = 0$. The samples use 4000 shots per setting.

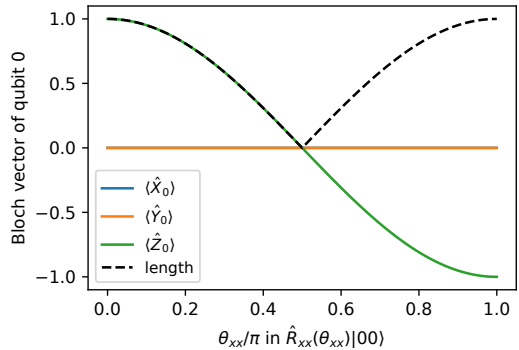
Entanglement Shrinks the Bloch Vector of Each Qubit

The Bloch vector of qubit 0 inside the register collects the expectation values of $\hat{X} \otimes \hat{I}$, $\hat{Y} \otimes \hat{I}$ and $\hat{Z} \otimes \hat{I}$,

$$\hat{R}_{xx}(\theta_{xx})|00\rangle = \cos\tfrac{\theta_{xx}}{2}|00\rangle - i \sin\tfrac{\theta_{xx}}{2}|11\rangle$$

$$\Longrightarrow \quad \mathbf{r}_0 = (0, 0, \cos\theta_{xx}).$$

Its length is one only for the products $|00\rangle$ and $|11\rangle$, and 0.000 at the Bell point $\theta_{xx} = \pi/2$, reached at $Jt = \pi/2$ for $\hat{\mathcal{H}} = \frac{J}{2}\hat{X} \otimes \hat{X}$. A vector inside the sphere describes no pure state of one qubit.



Components of $\mathbf{r}_0$ (solid) and its length $|\cos\theta_{xx}|$ (dashed) against the rotation angle of $\hat{R}_{xx}$. The x and y components stay zero.

The Register's Hamiltonian



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Hamiltonian of the Register

As a model Hamiltonian we take the exchange coupling J of the two qubits and a field h along z on both. Part II shows how physical devices provide such terms.

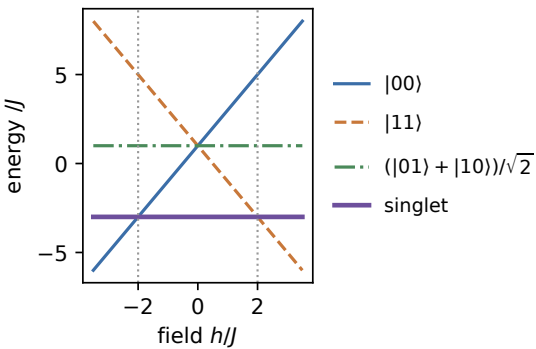
$$\hat{\mathcal{H}} = J\,(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1 + \hat{Z}_0\hat{Z}_1) + h\,(\hat{Z}_0 + \hat{Z}_1), \qquad \hat{X}_0\hat{X}_1 \equiv \hat{X}\otimes\hat{X}, \quad \hat{Z}_0 \equiv \hat{Z}\otimes\hat{I}.$$

Each term is a Pauli string with a real coefficient. *Exact diagonalisation* of the 4×4 matrix gives all energies and stationary states.

eigenstate	exchange	$\hat{Z}_0 + \hat{Z}_1$	energy
$\lvert 00\rangle$	J	$+2$	$J + 2h$
$\lvert 11\rangle$	J	-2	$J - 2h$
$(\lvert 01\rangle + \lvert 10\rangle)/\sqrt{2}$	J	0	J
singlet $(\lvert 01\rangle - \lvert 10\rangle)/\sqrt{2}$	$-3J$	0	$-3J$

For $J > 0$ and a weak field the ground state is the singlet, an entangled state with $s_1 = s_2 = 1/\sqrt{2}$.

Ground State against the Field



Energies against h for $J > 0$, the sign that favours anti-aligned spins.

- $(\hat{Z}_0 + \hat{Z}_1)$ gives zero on the singlet, whose level stays flat at $-3J$.
- $|11\rangle$ falls and crosses it at $h = 2J$, and $|00\rangle$ crosses at $h = -2J$ (dotted lines). A field $h > 0$ lowers $|1\rangle$, on which $\hat{Z} = -1$.
- For $|h| < 2J$ the ground state is the entangled singlet, and for $|h| > 2J$ a product state.

Hopping of an Excitation

Exchange without the $\hat{Z}\hat{Z}$ part,

$$\hat{\mathcal{H}}_{XX} = \tfrac{J}{2}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1) = J(\hat{S}_0^+\hat{S}_1^- + \hat{S}_0^-\hat{S}_1^+)$$

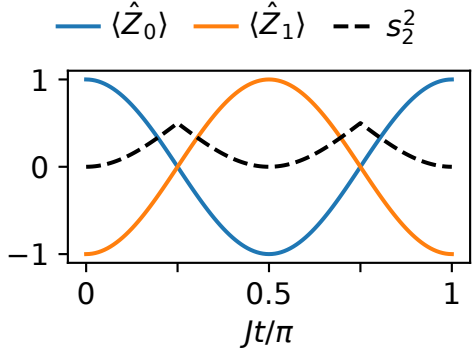
$$= J(|01\rangle\langle 10| + |10\rangle\langle 01|),$$

moves the excitation $|1\rangle$ between the qubits. From $|01\rangle$

$$|\psi(t)\rangle = \cos(Jt)\,|01\rangle - i\sin(Jt)\,|10\rangle,$$

$$\langle \hat{Z}_0(t)\rangle = \cos 2Jt = -\langle \hat{Z}_1(t)\rangle.$$

The Schmidt coefficients $|\cos Jt|$ and $|\sin Jt|$ give maximal entanglement, $s_2^2 = 0.500$, at $Jt = \pi/4$ and $3\pi/4$.



$\langle \hat{Z}_0\rangle$ and $\langle \hat{Z}_1\rangle$ oscillate in opposite phase. The dashed curve is the smaller squared Schmidt coefficient, maximal where they cross zero.

Three Couplings and Their Rotation Angles

The three model couplings carry different factors in front of their Pauli strings, so the same time gives different rotation angles.

coupling	Hamiltonian	reached at
Ising $\hat{X}\hat{X}$	$\frac{J}{2}\,\hat{X}\otimes\hat{X}$, $\hat{R}_{xx}(\theta_{xx})$ with $\theta_{xx}=Jt$	Bell type from $\lvert 00\rangle$ at $Jt=\pi/2$
hopping	$\frac{J}{2}(\hat{X}\hat{X}+\hat{Y}\hat{Y})$	maximal entanglement from $\lvert 01\rangle$ at $Jt=\pi/4$
Heisenberg	$J(\hat{X}\hat{X}+\hat{Y}\hat{Y}+\hat{Z}\hat{Z})=4J\,\hat{\mathbf{S}}_0\cdot\hat{\mathbf{S}}_1$	SWAP up to a phase at $Jt=\pi/4$

A rotation angle belongs to its Hamiltonian. Part II shows which platform provides which coupling.

Pauli Strings from Bit Strings

Bit 0 means the eigenvalue $+1$, so one shot gives the values

$$\hat{Z}_0 \to (-1)^{b_0}, \qquad \hat{Z}_1 \to (-1)^{b_1}, \qquad \hat{Z}_0\hat{Z}_1 \to \underbrace{(-1)^{b_0+b_1}}_{\text{parity of the bit string}} .$$

The mean over M shots estimates the expectation value. For a string with $\hat{X}$ or $\hat{Y}$ factors we first rotate those qubits with $\hat{H}$ or $\hat{H}\hat{S}^\dagger$ and then take the same parity.

One setting, several strings

The setting $\hat{Z}\hat{Z}$ gives $\hat{Z}_0$, $\hat{Z}_1$ and $\hat{Z}_0\hat{Z}_1$ from the same shots, all strings with $\hat{I}$ or the Pauli of the setting on each qubit. These strings commute.

setting	terms of $\hat{\mathcal{H}}$
$\hat{X}\hat{X}$	$\hat{X}_0\hat{X}_1$
$\hat{Y}\hat{Y}$	$\hat{Y}_0\hat{Y}_1$
$\hat{Z}\hat{Z}$	$\hat{Z}_0\hat{Z}_1,\ \hat{Z}_0,\ \hat{Z}_1$

Energy from Three Measurement Settings

We estimate the energy with $J = 1$, $h = 0.5$ and 1000 shots in each of the three settings. The three parts come from independent records, so their standard errors add in quadrature.

state	estimate from 3×1000 shots	exact
Bell $\lvert\Phi^+\rangle$	$+1.030$	$+1.000$
product $\lvert 0\rangle\lvert+\rangle$	$+0.590 \pm 0.065$	$+0.500$

- In the Bell state the three correlators have definite values, so exchange contributes J in every shot without error. Only the field part, $\pm 2h$ with mean zero, fluctuates, and the standard error is about half that of the product state.
- $\lvert 0\rangle\lvert+\rangle$ has random bits in more settings and the larger error.
- Both estimates agree with the exact energies within two standard errors.

Density Matrices



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Density Matrix of a Pure State

The correlated coin and a qubit of the Bell state have no state vector. Every prediction for $|\psi\rangle$ is written with one matrix, $\hat{\rho} = |\psi\rangle\langle\psi|$, since the trace is cyclic, $\langle\hat{O}\rangle = \langle\psi|\hat{O}|\psi\rangle = \mathrm{Tr}(\hat{O}\,|\psi\rangle\langle\psi|) = \mathrm{Tr}(\hat{O}\hat{\rho})$. For one qubit, $|\psi\rangle = (a, b)^\top$,

$$\hat{\rho} = \begin{pmatrix} a \\ b \end{pmatrix} \begin{pmatrix} a^* & b^* \end{pmatrix} = \begin{pmatrix} |a|^2 & ab^* \\ a^*b & |b|^2 \end{pmatrix} \qquad \begin{matrix} \text{diagonal, populations } p(0), p(1) \\ \text{off-diagonal, coherences (relative phase)} \end{matrix}$$

The rank is 1, as every column is a multiple of $|\psi\rangle$, and $\hat{\rho}^2 = |\psi\rangle \underbrace{\langle\psi|\psi\rangle}_{=1}\langle\psi| = \hat{\rho}$, so

$\mathrm{Tr}\hat{\rho}^2 = 1$.

A product register has $\hat{\rho} = (|\psi^{(0)}\rangle \otimes |\psi^{(1)}\rangle)(\langle\psi^{(0)}| \otimes \langle\psi^{(1)}|) = \hat{\rho}_0 \otimes \hat{\rho}_1$, and for $|0\rangle|+\rangle$

$$\hat{\rho} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \otimes \tfrac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \tfrac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \text{rank 1,} \quad \hat{\rho}^2 = \hat{\rho}.$$

Mixtures

A source that emits $|\psi_i\rangle$ with probability p_i, without telling us which, is known only statistically. Its predictions average the Born predictions, $\langle \hat{O} \rangle = \sum_i p_i \langle \psi_i | \hat{O} | \psi_i \rangle = \mathrm{Tr}(\hat{O}\hat{\rho})$, with

$$\hat{\rho} = \sum_i p_i \, |\psi_i\rangle\langle\psi_i|, \qquad p_k = \langle k|\hat{\rho}|k\rangle .$$

- $\hat{\rho}$ is Hermitian with $\mathrm{Tr}\hat{\rho} = 1$ and positive, $\langle v|\hat{\rho}|v\rangle = \sum_i p_i |\langle v|\psi_i\rangle|^2 \geq 0$, so its eigenvalues λ_k are probabilities.
- The **purity** $\mathrm{Tr}\hat{\rho}^2 = \sum_k \lambda_k^2 \in [1/d, 1]$ equals 1 only for rank 1, a pure state. A mixture of distinct states has rank > 1 and purity < 1.

The Coin and the Bell State as Density Matrices

For the register, the classical coin and the Bell state have

$$\hat{\rho}_{\mathrm{Bell}} = \tfrac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \text{ (rank 1)}, \qquad \hat{\rho}_{\mathrm{coin}} = \tfrac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{ (rank 2, } \mathrm{Tr}\hat{\rho}^2 = \tfrac{1}{2}\text{)}.$$

The two matrices share the diagonal and give identical $\hat{Z}\hat{Z}$ records. The coherences between $|00\rangle$ and $|11\rangle$ (orange) produce $\langle \hat{X}\hat{X} \rangle = 1$ and $\langle \hat{Y}\hat{Y} \rangle = -1$ for the Bell state and 0 for the coin.

Superposition and Mixture

We compare $|+\rangle$ with a source that emits $|0\rangle$ or $|1\rangle$ with probability $\frac{1}{2}$ each,

$$\hat{\rho}_{|+\rangle} = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \hat{\rho}_{\mathrm{mix}} = \tfrac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \tfrac{\hat{I}}{2}.$$

state	basis	exact $p(0), p(1)$	fraction of 0 in 2000 shots
$\|+\rangle$	$\hat{Z}$	0.5, 0.5	0.518
$\|+\rangle$	$\hat{X}$	1, 0	1.000
mixture	$\hat{Z}$	0.5, 0.5	0.495
mixture	$\hat{X}$	0.5, 0.5	0.523

The two states share the diagonal and hence the $\hat{Z}$ probabilities. The **coherences** (off-diagonal $\frac{1}{2}$) make the $\hat{X}$ outcome certain for $|+\rangle$, and the $\hat{X}$ basis tells the two states apart. One fixed basis sees only the diagonal in that basis and cannot determine an arbitrary density matrix.

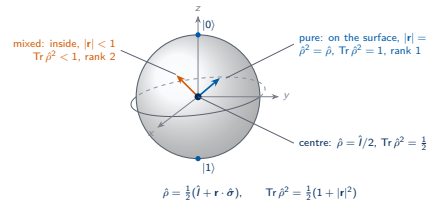
The Bloch Ball

Since $\hat{I}, \hat{X}, \hat{Y}, \hat{Z}$ are a basis of 2×2 matrices and $\mathrm{Tr}\hat{\rho} = 1$, three expectation values fix every density matrix of one qubit.

One qubit, $r_\alpha = \langle \hat{\sigma}_\alpha \rangle$

$$\hat{\rho} = \tfrac{1}{2}(\hat{I} + r_x \hat{X} + r_y \hat{Y} + r_z \hat{Z}),$$
$$|\mathbf{r}| \le 1, \qquad \mathrm{Tr}\hat{\rho}^2 = \tfrac{1}{2}(1 + |\mathbf{r}|^2).$$

The eigenvalues are $\frac{1}{2}(1 \pm |\mathbf{r}|)$, so pure states lie on the sphere and mixed states inside.



$\hat{I}/2$ sits at the centre with purity 0.5. A mixture averages Bloch vectors with its weights.

The Partial Trace

Let $|\psi\rangle = \sum_{a,b} \psi_{ab}|a\rangle|b\rangle$, with a for qubit 0 and b for qubit 1. A measurement on qubit 0 alone measures $\hat{O}_0 \otimes \hat{I}$, and with primed indices on the bra

$$\begin{aligned}\langle\psi|\hat{O}_0 \otimes \hat{I}|\psi\rangle &= \sum_{a,a',b,b'} \psi^*_{a'b'} \, (\hat{O}_0)_{a'a} \underbrace{\delta_{b'b}}_{\text{identity on qubit 1}} \psi_{ab} \\ &= \sum_{a,a'} (\hat{O}_0)_{a'a} \underbrace{\sum_b \psi_{ab}\psi^*_{a'b}}_{(\hat{\rho}_0)_{aa'}} = \mathrm{Tr}(\hat{O}_0\hat{\rho}_0).\end{aligned}$$

The Reduced Density Matrix

Reduced density matrix of qubit 0, the partial trace over qubit 1

$$(\hat{\rho}_0)_{aa'} = \sum_b \psi_{ab}\,\psi^*_{a'b}, \qquad \hat{\rho}_0 = \mathrm{Tr}_1|\psi\rangle\langle\psi|, \qquad (\hat{\rho}_1)_{bb'} = \sum_a \psi_{ab}\psi^*_{ab'}.$$

The delta sets the index of qubit 1 equal in ψ and ψ^*, and the sum over it is the sum over the unobserved bit. $\hat{\rho}_0$ contains everything that measurements on qubit 0 alone can reveal.

The Partial Trace Element by Element

For two qubits the four entries of $\hat{\rho}_0$ are

$$\hat{\rho}_0 = \begin{pmatrix} |\psi_{00}|^2 + |\psi_{01}|^2 & \psi_{00}\psi_{10}^* + \psi_{01}\psi_{11}^* \\ \psi_{10}\psi_{00}^* + \psi_{11}\psi_{01}^* & |\psi_{10}|^2 + |\psi_{11}|^2 \end{pmatrix}.$$

- The diagonal holds the marginal probabilities of qubit 0, summed over both outcomes of qubit 1.
- The off-diagonal entries pair amplitudes that agree on qubit 1. If qubit 1 carries a record of qubit 0, these terms vanish.

The Partial Trace in Matrix Form

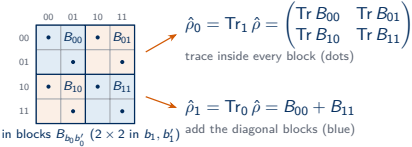
From the amplitudes, the reshape $\Psi_{ab} = \psi_{ab}$ with rows for qubit 0 turns the sum into a matrix product,

$$\hat{\rho}_0 = \Psi\Psi^\dagger, \qquad \hat{\rho}_1 = (\Psi^\dagger\Psi)^\top.$$

From the 4×4 density matrix, we cut $\hat{\rho}$ into four 2×2 blocks $\hat{\rho}^{(aa')}$ with rows $2a + b$, and

$$(\hat{\rho}_0)_{aa'} = \mathrm{Tr}\,\hat{\rho}^{(aa')}, \qquad \hat{\rho}_1 = \hat{\rho}^{(00)} + \hat{\rho}^{(11)}.$$

The block form holds for mixed registers as well. In code, `einsum("ab,Ab->aA", psi, psi.conj())`.



For the Bell state the block traces are $\frac{1}{2}, 0; 0, \frac{1}{2}$, so $\hat{\rho}_0 = \hat{I}/2$, and the sum of the diagonal blocks gives $\hat{\rho}_1 = \hat{I}/2$.

Three Reduced States

Product state $|0\rangle|+\rangle$

$\psi_{00} = \psi_{01} = \frac{1}{\sqrt{2}}$, $\psi_{10} = \psi_{11} = 0$, so $(\hat{\rho}_0)_{00} = \frac{1}{2} + \frac{1}{2} = 1$ and $\hat{\rho}_0 = |0\rangle\langle 0|$, a pure state.

Bell state

$\Psi = I/\sqrt{2}$ gives $\hat{\rho}_0 = \Psi\Psi^\dagger = \hat{I}/2$. The register is in a known pure state, while each qubit alone is a fair coin in every basis.

Family $\cos\theta|00\rangle + \sin\theta|11\rangle$

$\Psi = \mathrm{diag}(\cos\theta, \sin\theta)$ gives $\hat{\rho}_0 = \mathrm{diag}(\cos^2\theta, \sin^2\theta)$, which is $\mathrm{diag}(0.854,\ 0.146)$ at $\theta = \pi/8$.

The information of the Bell state sits in joint correlations. For Bell-diagonal states $\langle \hat{X}\hat{X}\rangle$, $\langle \hat{Y}\hat{Y}\rangle$, $\langle \hat{Z}\hat{Z}\rangle$ fix the register.

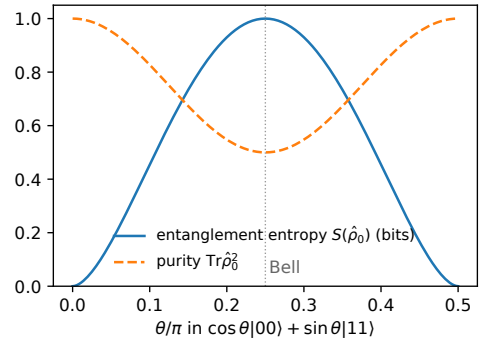
Purity and Entanglement Entropy

The **purity** $\mathrm{Tr}\hat{\rho}_0^2$ is 1 for the product, $\frac{1}{2}$ for the Bell state and $\cos^4\theta + \sin^4\theta$ for the family, 0.75 at $\theta = \pi/8$.

The **entanglement entropy** is the von Neumann entropy, in bits, of the eigenvalues λ_k of $\hat{\rho}_0$,

$$S(\hat{\rho}_0) = -\mathrm{Tr}\hat{\rho}_0 \log_2 \hat{\rho}_0 = -\sum_k \lambda_k \log_2 \lambda_k.$$

It is 0 for the product, 1 bit for the Bell state and $S_{\mathrm{bin}}(\cos^2\theta)$ for the family, 0.60 bits at $\theta = \pi/8$, with
$S_{\mathrm{bin}}(p) = -p\log_2 p - (1-p)\log_2(1-p)$.



The entropy rises to one bit at the Bell state while the purity falls to $\frac{1}{2}$.

Entanglement Criteria for a Pure Register

The SVD of the reshaped state, $\Psi = U\,\mathrm{diag}(s_k)\,V^\dagger$, gives

$$\hat{\rho}_0 = \Psi\Psi^\dagger = U\,\mathrm{diag}(s_k^2)\,U^\dagger \quad \Longrightarrow \quad \lambda_k = s_k^2.$$

$\Psi^\dagger\Psi$ has the same non-zero eigenvalues, so $S(\hat{\rho}_0) = S(\hat{\rho}_1)$ for a pure register, 0.36 bits on both sides for a random state.

pure register	Schmidt rank	$s_1,\ s_2$	$\mathrm{Tr}\hat{\rho}_0^2 = \sum_k s_k^4$	$S = -\sum_k s_k^2 \log_2 s_k^2$
product	1	$1,\ 0$	1	0
$\cos\theta\|00\rangle + \sin\theta\|11\rangle$	2 if $0 < \theta < \frac{\pi}{2}$	$\|\cos\theta\|,\ \|\sin\theta\|$	$\cos^4\theta + \sin^4\theta$	$S_{\mathrm{bin}}(\cos^2\theta)$
Bell	2	$\frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}}$	$\frac{1}{2}$	1

For a pure register, entangled $\Leftrightarrow$ Schmidt rank > 1 $\Leftrightarrow$ $\mathrm{Tr}\hat{\rho}_0^2 < 1$ $\Leftrightarrow$ $S(\hat{\rho}_0) > 0$.

Pure and Mixed, Global and Reduced

state	description	rank	purity	entropy (bits)
$\|0\rangle\|+\rangle$	pure product register	1	1	0
$\|\Phi^+\rangle$	pure entangled register	1	1	0
$\hat{\rho}_0$ of $\|\Phi^+\rangle$	reduced state of one qubit	2	1/2	1
$\hat{\rho}_{\text{coin}}$	classical mixture of $\|00\rangle, \|11\rangle$	2	1/2	1
$\hat{\rho}_0$ of $\hat{\rho}_{\text{coin}}$	reduced state of one qubit	2	1/2	1

Reduced entropy and mixed registers

The reduced entropy measures entanglement only when the global state is pure. Qubit 0 of the coin is the same $\hat{I}/2$ as qubit 0 of the Bell state, with one bit of entropy. For the coin, that bit is ignorance of which of $|00\rangle, |11\rangle$ was emitted. The coin contains no entanglement, and the two registers differ in $\langle \hat{X}\hat{X} \rangle$, $\langle \hat{Y}\hat{Y} \rangle$ and the global purity.

Measurement and Back-Action



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Projective Measurement

A projective measurement has one projector per outcome. The projectors are Hermitian, $\hat{P}_k = \hat{P}_k^\dagger$, orthogonal, $\hat{P}_k \hat{P}_l = \delta_{kl} \hat{P}_k$, so that outcomes exclude each other, and complete, $\sum_k \hat{P}_k = \hat{I}$, so that some outcome always occurs. The Born rule and the Lüders rule give

$$p_k = \mathrm{Tr}(\hat{P}_k \hat{\rho}), \qquad \hat{\rho} \longrightarrow \hat{\rho}_k = \frac{\hat{P}_k \, \hat{\rho} \, \hat{P}_k}{p_k}.$$

Measuring qubit 0 of the register

In the $\hat{Z}$ basis $\hat{P}_0 = |0\rangle\langle 0| \otimes \hat{I}$ and $\hat{P}_1 = |1\rangle\langle 1| \otimes \hat{I}$. In the $\hat{X}$ basis the projectors are $|\pm\rangle\langle\pm| \otimes \hat{I}$, with outcome bit 0 for $+$.

A pure state goes to $|\psi\rangle \to \hat{P}_k|\psi\rangle / \sqrt{p_k}$, the compatible branch renormalised.

G. Lüders, Ann. Phys. (Leipzig) **443**, 322 (1950), doi:10.1002/andp.19504430510

Back-Action on the Partner Qubit

For the Bell state, the outcome 0 of a $\hat{Z}$ measurement on qubit 0 gives

$$(|0\rangle\langle 0| \otimes \hat{I})\,\frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|00\rangle}{\sqrt{2}} \xrightarrow{\text{renormalise}} |0\rangle|0\rangle, \qquad p_0 = \tfrac{1}{2}.$$

In $\hat{X}$ the outcome $+$ leaves qubit 1 in $|+\rangle$ and the outcome $-$ in $|-\rangle$, since $|\Phi^+\rangle = (|{+}{+}\rangle + |{-}{-}\rangle)/\sqrt{2}$.

Qubit 0 is measured first and qubit 1 afterwards in the same basis, 200 shots each,

fraction of shots where qubit 1 repeats qubit 0	$\hat{Z}$	$\hat{X}$	$\hat{Y}$
Bell	1.00	1.00	0.00
classical coin	1.00	0.48	0.51

The partner of a Bell qubit is left in a definite eigenstate of the measured basis. It is the same one in $\hat{Z}$ and $\hat{X}$ and the opposite one in $\hat{Y}$, as $\langle \hat{Y}\hat{Y}\rangle = -1$. The coin agrees only in $\hat{Z}$.

No Signalling

Before the measurement qubit 1 of the Bell state is $\hat{I}/2$, a fair coin in every basis. After the measurement on qubit 0 it is in a definite pure state for an observer who knows the outcome.

At qubit 1 the outcome is unknown, and the post-measurement states are averaged,

$$\hat{\rho}_1^{\text{after}} = \sum_k p_k\, \hat{\rho}_{1,k} = \tfrac{1}{2}|0\rangle\langle 0| + \tfrac{1}{2}|1\rangle\langle 1| = \frac{\hat{I}}{2} \quad (\hat{Z}), \qquad \tfrac{1}{2}|+\rangle\langle +| + \tfrac{1}{2}|-\rangle\langle -| = \frac{\hat{I}}{2} \quad (\hat{X}).$$

- For any basis chosen at qubit 0, qubit 1 alone remains $\hat{I}/2$, as the simulation confirms for $\hat{Z}$, $\hat{X}$ and $\hat{Y}$.
- The correlations appear only when the two records are brought together.
- The basis chosen at qubit 0 cannot be read off at qubit 1 (**no signalling**).

Unread Measurement and Dephasing

A measurement whose result is not recorded maps

$$\hat{\rho} \;\longrightarrow\; \sum_k p_k \hat{\rho}_k = \sum_k \hat{P}_k \,\hat{\rho}\, \hat{P}_k .$$

It keeps the blocks of $\hat{\rho}$ inside each outcome sector and removes the coherences between sectors. For a measurement in a basis it keeps the diagonal and removes all coherences.

One qubit

An unread $\hat{Z}$ measurement on $|+\rangle$ gives $\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \to \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, with purity $1 \to 0.5$.

Register

An unread $\hat{Z}$ measurement of qubit 0 turns $\hat{\rho}_{\mathrm{Bell}}$ into $\hat{\rho}_{\mathrm{coin}}$, with $\langle \hat{X}\hat{X} \rangle \; 1 \to 0$ and $\langle \hat{Z}\hat{Z} \rangle \; 1 \to 1$.

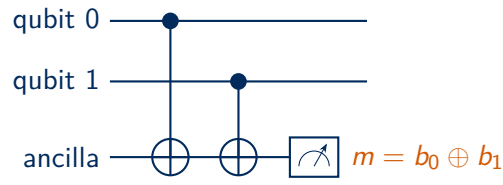
The loss of coherence in a fixed basis is called **dephasing**, and an environment that records the qubit has the same effect.

Measuring a Pauli String as a Parity

$\hat{Z}_0\hat{Z}_1$ has the eigenvalue $(-1)^{b_0 \oplus b_1}$, and measuring it reveals only the *parity* $b_0 \oplus b_1$, with the projectors $\hat{P}_{\text{even}} = |00\rangle\langle 00| + |11\rangle\langle 11|$ and $\hat{P}_{\text{odd}} = |01\rangle\langle 01| + |10\rangle\langle 10|$.

A readout of both bits, after rotating $\hat{X}$ or $\hat{Y}$ factors to $\hat{Z}$, and multiplying the ± 1 values gives the parity. It also reveals the individual bits and destroys superpositions inside each parity sector.

CNOT gates from qubits 0 and 1 onto an extra qubit (ancilla) in $|0\rangle$ write $b_0 \oplus b_1$ into it. Measuring the ancilla alone is the projective measurement $\{\hat{P}_{\text{even}}, \hat{P}_{\text{odd}}\}$. A Bell state gives even parity with certainty and is left unchanged.



The CNOT gates map $|b_0 b_1\rangle|0\rangle \to |b_0 b_1\rangle|b_0 \oplus b_1\rangle$. For $\hat{X}_0\hat{X}_1$, $\hat{H}$ acts on both data qubits before and after. The same circuit measures stabilisers in quantum error correction.

A Noisy Detector

A $\hat{Z}$ detector that reports the wrong bit with probability ϵ gives the outcome probabilities

$$p(0) = \mathrm{Tr}(\hat{\pi}_0 \hat{\rho}), \qquad \hat{\pi}_0 = (1-\epsilon)|0\rangle\langle 0| + \epsilon |1\rangle\langle 1|, \qquad \hat{\pi}_1 = \hat{I} - \hat{\pi}_0.$$

The operators $\hat{\pi}_0$ and $\hat{\pi}_1$ are positive and sum to the identity. Since $\hat{\pi}_0^2 \neq \hat{\pi}_0$ for $0 < \epsilon < 1$, they are not projectors.

With $\epsilon = 0.1$ the state $|0\rangle$ gives $p(0) = 0.9$, and $|+\rangle$ gives $p(0) = 0.5$. The same numbers form the confusion matrix of a readout error, which maps the true probabilities of the bits to the recorded ones.

General Measurements

A **positive operator-valued measure** (POVM) is a set $\{\hat{\pi}_i\}$ with $\hat{\pi}_i \geq 0$, so that probabilities are non-negative, and $\sum_i \hat{\pi}_i = \hat{I}$, so that they sum to one. The Born rule reads $p(i) = \mathrm{Tr}(\hat{\pi}_i \hat{\rho})$, and the $\hat{\pi}_i$ need be neither projectors nor orthogonal.

The POVM fixes only the statistics. The state after the outcome i follows from an **instrument**, Kraus operators $\hat{M}_{i,\alpha}$ with

$$\hat{\pi}_i = \sum_\alpha \hat{M}_{i,\alpha}^\dagger \hat{M}_{i,\alpha}, \qquad \hat{\rho} \to \hat{\rho}^{(i)} = \frac{1}{p(i)} \sum_\alpha \hat{M}_{i,\alpha}\, \hat{\rho}\, \hat{M}_{i,\alpha}^\dagger.$$

Different instruments can realise the same POVM and leave different states. One example is the **square-root instrument** $\hat{M}_i = \sqrt{\hat{\pi}_i}$, with $\hat{\rho}^{(i)} = \sqrt{\hat{\pi}_i}\hat{\rho}\sqrt{\hat{\pi}_i}/p(i)$, and a projective measurement has $\hat{\pi}_i = \hat{M}_i = \hat{P}_i$.

Informationally Complete Measurements

A measurement is **informationally complete** if its outcome probabilities fix $\hat{\rho}$. Its POVM elements must span all Hermitian $d \times d$ matrices, $d = 2^N$, so an informationally complete POVM has at least d^2 outcomes.

Pauli settings

Measuring every qubit in $\hat{X}$, $\hat{Y}$ or $\hat{Z}$ gives 3^N settings. Their parities give all 4^N Pauli-string expectation values, and

$$\hat{\rho} = \frac{1}{2^N} \sum_P \langle \hat{P} \rangle \, \hat{P}.$$

When the setting is chosen at random, it becomes part of the outcome label, and each element carries the probability of its setting.

State tomography is limited by the number of settings, which grows exponentially with N.

A Symmetric Informationally Complete POVM

A symmetric informationally complete POVM (SIC-POVM) of one qubit uses the four corners of a regular tetrahedron in the Bloch ball,

$$\mathbf{s}_1 = (0, 0, 1), \quad \mathbf{s}_2 = (\tfrac{2\sqrt{2}}{3}, 0, -\tfrac{1}{3}), \quad \mathbf{s}_{3,4} = (-\tfrac{\sqrt{2}}{3}, \pm\sqrt{\tfrac{2}{3}}, -\tfrac{1}{3}),$$

with the elements

$$\hat{\pi}_a = \tfrac{1}{4}(\hat{I} + \mathbf{s}_a \cdot \hat{\boldsymbol{\sigma}}), \qquad a = 1, \ldots, 4.$$

- $\sum_a \hat{\pi}_a = \hat{I}$, and all pairs of elements overlap equally, $\mathrm{Tr}\hat{\pi}_a\hat{\pi}_b = \frac{1}{12}$ for $a \neq b$.
- One setting returns four outcomes, and their probabilities determine the Bloch vector.
- For N qubits the products $\hat{\pi}_{a_0} \otimes \cdots \otimes \hat{\pi}_{a_{N-1}}$ give 4^N outcomes.

J. M. Renes, R. Blume-Kohout, A. J. Scott and C. M. Caves, J. Math. Phys. **45**, 2171 (2004), doi:10.1063/1.1737053

Qubit Errors, Noise Channels and Distances



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An Environment That Records the Qubit

Noise is a coupling to an unobserved environment E. In the smallest example one environment qubit starts in $|0\rangle_E$, and the coupling $\hat{\mathcal{H}} = \frac{\Omega}{2}\,|1\rangle\langle 1| \otimes \hat{Y}$ rotates it about y only when the system is $|1\rangle$,

$$\hat{U} = e^{-i\hat{\mathcal{H}}t} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_y(g), \qquad g = \Omega t, \quad 0 \le g \le \pi,$$
$$|0\rangle|0\rangle_E \to |0\rangle|0\rangle_E, \qquad |1\rangle|0\rangle_E \to |1\rangle\big(\cos\tfrac{g}{2}|0\rangle_E + \sin\tfrac{g}{2}|1\rangle_E\big).$$

Collecting the terms with $|0\rangle_E$ and with $|1\rangle_E$ leaves a 2×2 matrix on the system for each,

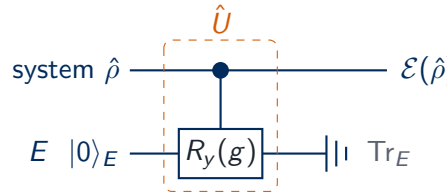
$$\hat{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & \cos\frac{g}{2} \end{pmatrix}, \qquad \hat{K}_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sin\frac{g}{2} \end{pmatrix}, \qquad \mathcal{E}(\hat{\rho}) = \begin{pmatrix} \rho_{00} & \cos\frac{g}{2}\,\rho_{01} \\ \cos\frac{g}{2}\,\rho_{10} & \rho_{11} \end{pmatrix}.$$

The populations are unchanged, and the coherence is multiplied by $\cos\frac{g}{2}$, since the environment has partly recorded whether the system is in $|1\rangle$. The channel is *phase damping* with $\kappa = \sin^2\frac{g}{2}$, and $g = \pi$ gives a full unread measurement. For $g > \pi$ the overlap $\cos\frac{g}{2}$ changes sign and adds a phase flip.

Kraus Operators

For any unitary coupling $\hat{U}$ we trace out E with the partial trace Tr_E,

$$\begin{aligned}\mathcal{E}(\hat{\rho}) &= \mathrm{Tr}_E\big[\hat{U}(\hat{\rho}\otimes|0\rangle\langle 0|_E)\hat{U}^\dagger\big] \\ &= \sum_k {}_E\langle k|\hat{U}|0\rangle_E\,\hat{\rho}\,{}_E\langle 0|\hat{U}^\dagger|k\rangle_E \\ &= \sum_k \hat{K}_k\hat{\rho}\hat{K}_k^\dagger .\end{aligned}$$



The **Kraus operator** $\hat{K}_k = {}_E\langle k|\hat{U}|0\rangle_E$ takes the matrix element of $\hat{U}$ in the environment alone, from the input $|0\rangle_E$ to the output $|k\rangle_E$, and leaves a system matrix. Inserting $\sum_k |k\rangle\langle k|_E = \hat{I}_E$ gives

$$\sum_k \hat{K}_k^\dagger\hat{K}_k = {}_E\langle 0|\hat{U}^\dagger\hat{U}|0\rangle_E = \hat{I} \quad\Longrightarrow\quad \mathrm{Tr}\,\mathcal{E}(\hat{\rho}) = \mathrm{Tr}\hat{\rho} = 1.$$

K. Kraus, *States, Effects, and Operations*, Lect. Notes Phys. **190** (Springer, 1983), doi:10.1007/3-540-12732-1

Pauli Errors and Depolarising Noise

With probability p a Pauli error $\hat{\sigma}$ hits the qubit, $\hat{K}_0 = \sqrt{1-p}\,\hat{I}$, $\hat{K}_1 = \sqrt{p}\,\hat{\sigma}$, and $\mathcal{E}(\hat{\rho}) = (1-p)\,\hat{\rho} + p\,\hat{\sigma}\hat{\rho}\hat{\sigma}$.

error	Kraus operators	Bloch vector $(r_x, r_y, r_z) \to$
bit flip	$\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{X}$	$\big(r_x,\ (1-2p)r_y,\ (1-2p)r_z\big)$
phase flip (dephasing)	$\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{Z}$	$\big((1-2p)r_x,\ (1-2p)r_y,\ r_z\big)$
bit-phase flip	$\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{Y}$	$\big((1-2p)r_x,\ r_y,\ (1-2p)r_z\big)$
depolarising	$\sqrt{1-p}\,\hat{I},\ \sqrt{p/3}\,\hat{X},\ \sqrt{p/3}\,\hat{Y},\ \sqrt{p/3}\,\hat{Z}$	$\big(1-\frac{4}{3}p\big)\,\mathbf{r}$

Anticommutation gives $\hat{\sigma}\hat{\sigma}_\alpha\hat{\sigma} = +\hat{\sigma}_\alpha$ for $\hat{\sigma}_\alpha = \hat{\sigma}$ and $-\hat{\sigma}_\alpha$ otherwise, so the component along $\hat{\sigma}$ survives and the other two shrink by $1-2p$. Depolarising noise is $\mathcal{E}(\hat{\rho}) = (1-\frac{4}{3}p)\hat{\rho} + \frac{4}{3}p\,\frac{\hat{I}}{2}$, a uniform shrink, and at $p = \frac{3}{4}$ every state becomes $\hat{I}/2$.

Amplitude Damping and Phase Damping

Amplitude damping, energy loss

$$\hat{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad \hat{K}_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}$$

$\hat{K}_1 = \sqrt{\gamma}\,\hat{S}^+$ takes $|1\rangle \to |0\rangle$ with probability γ,

$$\mathcal{E}(\hat{\rho}) = \begin{pmatrix} \rho_{00} + \gamma\rho_{11} & \sqrt{1-\gamma}\,\rho_{01} \\ \sqrt{1-\gamma}\,\rho_{10} & (1-\gamma)\rho_{11} \end{pmatrix}.$$

$r_{x,y} \to \sqrt{1-\gamma}\, r_{x,y}$, $r_z \to \gamma + (1-\gamma) r_z$, towards the fixed state $|0\rangle$.

Phase damping, no energy loss

$$\hat{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\kappa} \end{pmatrix}, \quad \hat{K}_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\kappa} \end{pmatrix}$$

The environment learns about $|1\rangle$ and takes no energy,

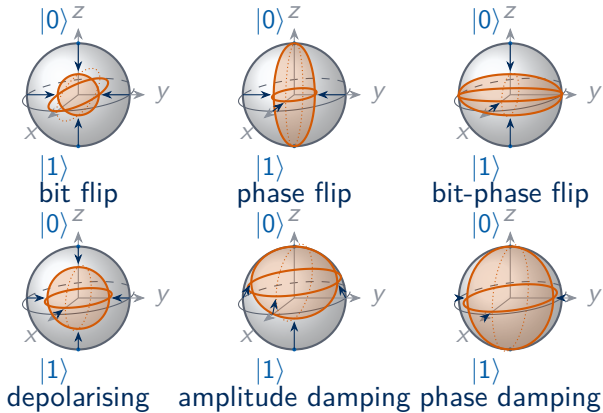
$$\mathcal{E}(\hat{\rho}) = \begin{pmatrix} \rho_{00} & \sqrt{1-\kappa}\,\rho_{01} \\ \sqrt{1-\kappa}\,\rho_{10} & \rho_{11} \end{pmatrix}.$$

$r_{x,y} \to \sqrt{1-\kappa}\, r_{x,y}$, r_z unchanged.

Phase damping equals the phase flip with $1 - 2p = \sqrt{1-\kappa}$.

Six Error Models on the Bloch Ball

Each panel shows the image of the Bloch sphere (grey) under one channel at strength 0.3. The orange curves are the images of great circles of the sphere, and arrows join pure states to their images.



Pauli errors squeeze the ball towards the axis of the error, and depolarising noise shrinks it uniformly by $1 - \frac{4}{3}p = 0.6$. Only amplitude damping moves the centre, towards $|0\rangle$. Phase damping keeps the poles and shrinks x and y by $\sqrt{1-\kappa} \approx 0.84$.

Quantum Channels

Every Kraus map $\mathcal{E}(\hat{\rho}) = \sum_k \hat{K}_k \hat{\rho} \hat{K}_k^\dagger$ with $\sum_k \hat{K}_k^\dagger \hat{K}_k = \hat{I}$ is **completely positive**, since it keeps $\hat{\rho} \geq 0$ even when it acts on one qubit of a larger register ($\mathcal{E} \otimes \mathrm{id}$). It is also **trace preserving**, $\mathrm{Tr}\, \mathcal{E}(\hat{\rho}) = 1$. Such a completely positive trace-preserving (CPTP) map, a quantum channel, is the most general physical evolution, and a unitary is the case of one Kraus operator.

Pauli channels

$\mathcal{E}(\hat{\rho}) = p_0 \hat{\rho} + p_x \hat{X} \hat{\rho} \hat{X} + p_y \hat{Y} \hat{\rho} \hat{Y} + p_z \hat{Z} \hat{\rho} \hat{Z}$ with $\sum p = 1$. Bit, phase and bit-phase flip and depolarising noise are special cases.

Unital and non-unital channels

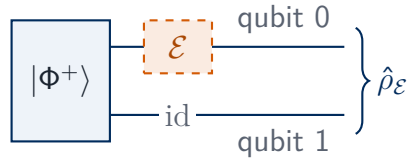
A unital channel has $\mathcal{E}(\hat{I}) = \hat{I}$ and keeps the centre of the Bloch ball fixed, as Pauli channels and phase damping do. Amplitude damping is non-unital and drives every state towards $|0\rangle$.

The Choi State of a Channel

A channel becomes a state when it acts on qubit 0 of the Bell state, with the identity on qubit 1. Measuring this state measures the channel, so process tomography is state tomography (Lecture 13),

$$\hat{\rho}_{\mathcal{E}} = (\mathcal{E} \otimes \mathrm{id})(|\Phi^+\rangle\langle\Phi^+|) = \tfrac{1}{2}\sum_{ij} \mathcal{E}(|i\rangle\langle j|) \otimes |i\rangle\langle j|, \qquad \mathcal{E}(\hat{\rho}) = 2\,\mathrm{Tr}_1[\hat{\rho}_{\mathcal{E}}\,(\hat{I} \otimes \hat{\rho}^T)].$$

- completely positive $\iff \hat{\rho}_{\mathcal{E}} \geq 0$
- trace preserving $\iff \mathrm{Tr}_0 \hat{\rho}_{\mathcal{E}} = \hat{I}/2$
- eigenvectors $\sqrt{2\lambda_k}|u_k\rangle$, reshaped like Ψ with rows labelled by qubit 0, are Kraus operators



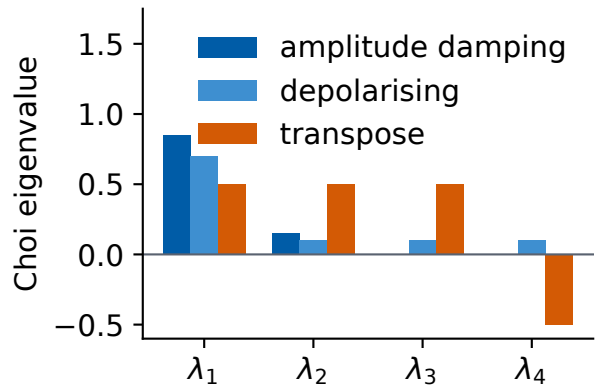
M.-D. Choi, Linear Algebra Appl. **10**, 285 (1975), doi:10.1016/0024-3795(75)90075-0; A. Jamiołkowski, Rep. Math. Phys. **3**, 275 (1972), doi:10.1016/0034-4877(72)90011-0

Choi States of Three Maps

Amplitude damping, eigenvalues $1-\gamma/2$, $\gamma/2$, 0, 0, two Kraus operators.

Depolarising, $\hat{\rho}_{\mathcal{E}} = (1-\frac{4}{3}p)|\Phi^+\rangle\langle\Phi^+| + \frac{4}{3}p\,\frac{\hat{I}}{4}$, eigenvalues $1-p$ and $p/3$ three times.

Transpose, $\hat{\rho}_{\mathcal{E}} = \widehat{\mathrm{SWAP}}/2$ with eigenvalue $-1/2$ on the singlet, positive but not completely positive.



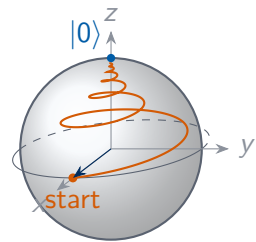
Choi-state eigenvalues, $\gamma = p = 0.3$. Lab 066a

The Lindblad Master Equation

For a qubit weakly coupled to a memoryless environment the density matrix obeys

$$\frac{d\hat{\rho}}{dt} = \underbrace{-i[\hat{\mathcal{H}}, \hat{\rho}]}_{\text{unitary part}} + \sum_j \underbrace{\left(\hat{L}_j \hat{\rho} \hat{L}_j^\dagger - \tfrac{1}{2}\{\hat{L}_j^\dagger \hat{L}_j, \hat{\rho}\}\right)}_{\text{jump operator } \hat{L}_j}.$$

The solution over a time t is a channel. Energy decay with $\hat{L} = \sqrt{\Gamma}\,\hat{S}^+$, $\hat{S}^+ = |0\rangle\langle 1|$, and $T_1 = 1/\Gamma$ gives amplitude damping, $\gamma = 1 - e^{-t/T_1}$. Pure dephasing with $\hat{L} = \sqrt{\gamma_\varphi/2}\,\hat{Z}$ gives phase damping, $\sqrt{1-\kappa} = e^{-\gamma_\varphi t}$. With both, $\rho_{11} \propto e^{-t/T_1}$ and $\rho_{01} \propto e^{-t/T_2}$, $\frac{1}{T_2} = \frac{1}{2T_1} + \gamma_\varphi$.

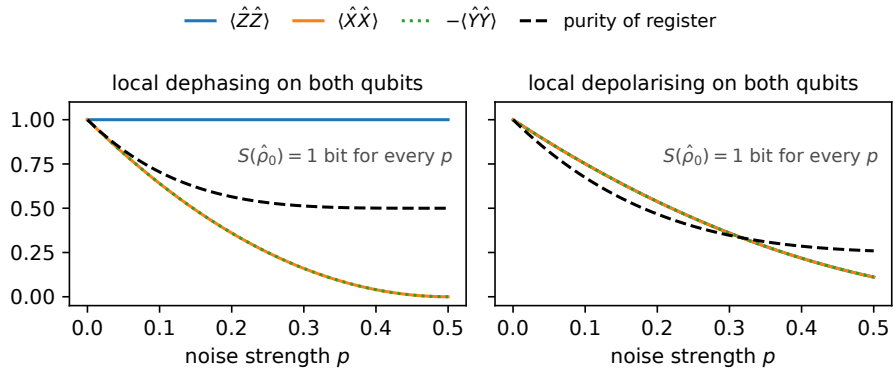


With $\hat{\mathcal{H}} = \frac{\omega}{2}\hat{Z}$ the Bloch vector precesses while it relaxes to $|0\rangle$.

G. Lindblad, Commun. Math. Phys. **48**, 119 (1976), doi:10.1007/BF01608499

The Register under Local Noise

The same channel acts on each qubit of the Bell state, $\hat{\rho} \to \sum_{k,l} (\hat{K}_k \otimes \hat{K}_l)\, \hat{\rho}\, (\hat{K}_k \otimes \hat{K}_l)^\dagger$.



Dephasing keeps $\langle \hat{Z}\hat{Z} \rangle = 1$ and shrinks $\langle \hat{X}\hat{X} \rangle$ as $(1-2p)^2$, so at $p = \frac{1}{2}$ the register becomes the classical coin. Depolarising noise shrinks all correlators by $(1 - \frac{4}{3}p)^2$.

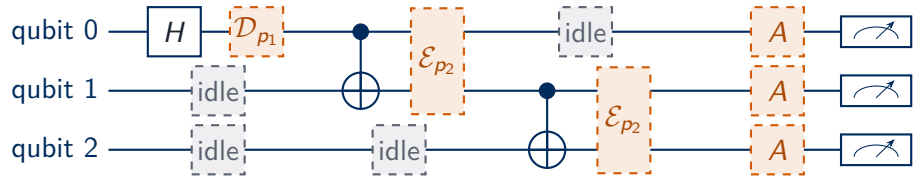
Entropy of One Qubit under Local Noise

Under local dephasing and depolarising noise the reduced entropy $S(\hat{\rho}_0)$ (grey dots) stays at 1 bit for every p, since $\hat{\rho}_0 = \hat{I}/2$ throughout. The register purity at $p = 0.3$ under dephasing is 0.51.

The loss of entanglement therefore escapes the reduced entropy. It shows in the correlators $\langle \hat{X}\hat{X} \rangle$ and $\langle \hat{Y}\hat{Y} \rangle$ and in the global purity, and each of these requires joint measurements.

The Gate-Noise Model of a Circuit

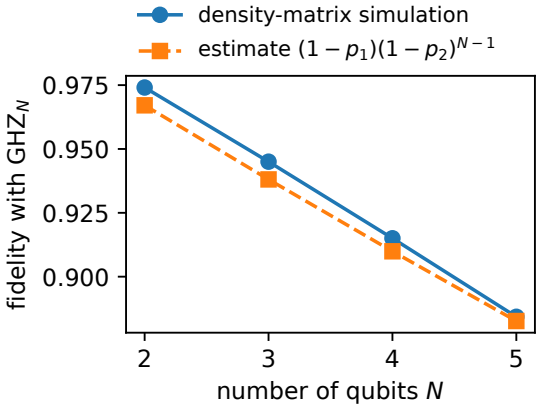
Each ideal gate is followed by a channel on the qubits it touched, $\hat{\rho} \to \mathcal{E}(\hat{U}\hat{\rho}\hat{U}^\dagger)$.



- Orange, depolarising noise with p_1 after one-qubit gates and p_2 after each CNOT, $\hat{\rho} \to (1-p_2)\hat{\rho} + \frac{p_2}{15}\sum_{\hat{P}\neq\hat{I}\hat{I}} \hat{P}\hat{\rho}\hat{P}$, typically $p_2 \approx 10\,p_1$.
- Grey, waiting qubits relax by amplitude and phase damping, $\gamma = 1 - e^{-t/T_1}$. A gate lasts θ/Ω or θ/J, so large angles cost coherence.
- A, readout error as a confusion matrix $A = A_0 \otimes A_1 \otimes A_2$ on the bit-string probabilities, with flip probabilities $\epsilon_{0\to1}$, $\epsilon_{1\to0}$ in each A_q.

A simulation applies $\hat{U}\hat{\rho}\hat{U}^\dagger$ and then the Kraus maps, at a memory cost of 4^N numbers.

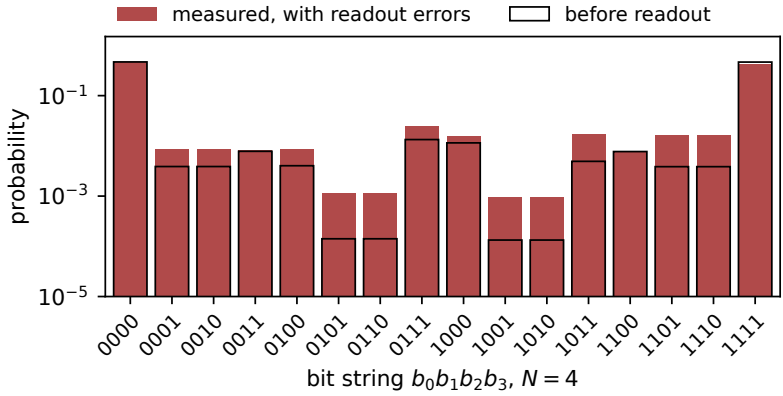
A Noisy GHZ Circuit



The target $|\mathrm{GHZ}_N\rangle$ is prepared by $\hat{H}$ and a chain of $N-1$ CNOT gates. Its **fidelity** $F = \langle \mathrm{GHZ}_N|\hat{\rho}|\mathrm{GHZ}_N\rangle$ is the probability that the noisy register passes a test for the target.

With $p_2 = 0.03$ per CNOT and ten times less per one-qubit gate, each added CNOT lowers the fidelity, which follows the estimate $(1-p_1)(1-p_2)^{N-1}$ closely.

Bit Strings of the Noisy GHZ Circuit



The ideal state gives only 0000 and 1111. Gate noise, relaxation and readout errors fill in the other fourteen strings, and the correlator $\langle \hat{Z}_0 \hat{Z}_3 \rangle$ falls below one.

Coherent and Stochastic Errors

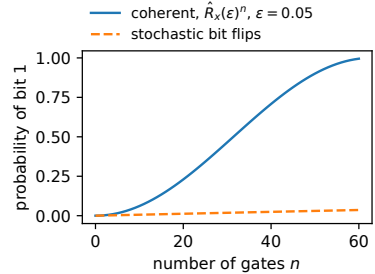
Two errors with equal infidelity per gate act k times on $|0\rangle$. Randomised compiling turns the first kind into the second.

Coherent over-rotation $\hat{R}_x(\epsilon)$

Angles add, $\hat{R}_x(\epsilon)^k = \hat{R}_x(k\epsilon)$, and $1 - F = \sin^2 \frac{k\epsilon}{2} \approx k^2 \frac{\epsilon^2}{4}$. The infidelity grows as k^2 because amplitudes add.

Stochastic bit flip with $p = \sin^2 \frac{\epsilon}{2}$

An odd number of flips gives $1 - F = \frac{1}{2}\left[1 - (1-2p)^k\right] \approx k\,p$. Probabilities add, and the infidelity grows as k.



After 60 gates the qubit is flipped with probability 0.995 by the coherent error and 0.036 by the stochastic one.

J. J. Wallman and J. Emerson, Phys. Rev. A **94**, 052325 (2016), doi:10.1103/PhysRevA.94.052325

Fidelity and Trace Distance

Fidelity

For pure states $F = |\langle\varphi|\psi\rangle|^2$, the overlap probability.

For a pure target $|\psi\rangle$ and a prepared state $\hat{\sigma}$, possibly mixed,

$$F = \langle\psi|\hat{\sigma}|\psi\rangle.$$

Two mixed states need the Uhlmann fidelity, built from matrix square roots.

Trace distance

$$D(\hat{\rho}, \hat{\sigma}) = \tfrac{1}{2} \sum_k |\mu_k|,$$

where μ_k are the eigenvalues of the Hermitian matrix $\hat{\rho} - \hat{\sigma}$.

Given one copy of $\hat{\rho}$ or $\hat{\sigma}$ with equal probability, the best measurement identifies it with probability $\frac{1}{2}(1 + D)$. For one qubit, $D = \frac{1}{2}|\mathbf{r}_\rho - \mathbf{r}_\sigma|$.

A. Uhlmann, Rep. Math. Phys. **9**, 273 (1976), doi:10.1016/0034-4877(76)90060-4

Three Pairs of States

pair	fidelity F	trace distance D	best success $\frac{1}{2}(1+D)$
$\lvert+\rangle$ and the 50/50 mixture	0.50	0.50	0.75
Bell and the classical coin	0.50	0.50	0.75
Bell and dephased Bell, $p = 0.3$	0.58	0.42	0.71

- One copy of the Bell state or the coin is identified correctly with probability 3/4 at best, for example by measuring $\hat{X}\hat{X}$.
- After local dephasing the fidelity with $|\Phi^+\rangle$ is

$$F = \langle\Phi^+|\hat{\rho}|\Phi^+\rangle = \tfrac{1}{2}\big(1 + \underbrace{(1-2p)^2}_{\text{surviving coherence}}\big) = 0.58.$$

What Quantum Information Allows and Forbids



JAGIELLONIAN UNIVERSITY
IN KRAKOW

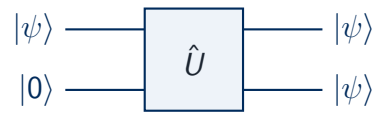
The No-Cloning Theorem

A copier $\hat{U}|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ that copies $|\psi\rangle$ and $|\varphi\rangle$ obeys

$$\underbrace{\langle\varphi|\psi\rangle\,\langle 0|0\rangle}_{\text{inputs}} = \underbrace{\langle\varphi|\psi\rangle^2}_{\text{outputs}},$$

as a unitary preserves inner products, so $|\langle\varphi|\psi\rangle| = 0$ or 1. Only orthogonal states, or equal ones up to a global phase, are copied.

Learning a state needs many fresh copies (tomography, shadows, Lecture 13).



$|0\rangle|0\rangle \mapsto |00\rangle, \quad |1\rangle|0\rangle \mapsto |11\rangle$

$|+\rangle|0\rangle \mapsto \frac{|00\rangle+|11\rangle}{\sqrt{2}} \neq |+\rangle|+\rangle$

W. K. Wootters and W. H. Zurek, Nature **299**, 802 (1982), doi:10.1038/299802a0; D. Dieks, Phys. Lett. A **92**, 271 (1982), doi:10.1016/0375-9601(82)90084-6

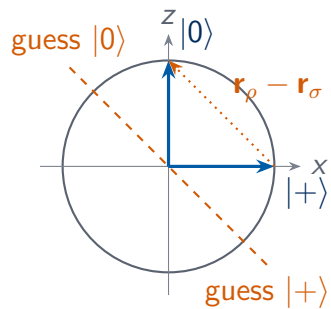
Telling Two States Apart with One Copy

One copy of $\hat{\rho}$ or $\hat{\sigma}$, each with probability $\frac{1}{2}$. We guess $\hat{\rho}$ on the outcome $\hat{\Pi}$ of $\{\hat{\Pi}, \hat{I} - \hat{\Pi}\}$,

$$P_{\text{succ}} = \tfrac{1}{2} + \tfrac{1}{2}\mathrm{Tr}\big[\hat{\Pi} \underbrace{(\hat{\rho} - \hat{\sigma})}_{\sum_k \mu_k |k\rangle\langle k|}\big]$$

$$\leq \tfrac{1}{2}\big[1 + \underbrace{D(\hat{\rho}, \hat{\sigma})}_{\text{sum of } \mu_k > 0}\big],$$

reached by the projector on the positive part of $\hat{\rho} - \hat{\sigma}$, for one qubit along $\mathbf{r}_\rho - \mathbf{r}_\sigma$.



Dashed, the measured spin axis along $\mathbf{r}_\rho - \mathbf{r}_\sigma$. Each end is read as a guess.

C. W. Helstrom, J. Stat. Phys. **1**, 231 (1969), doi:10.1007/BF01007479

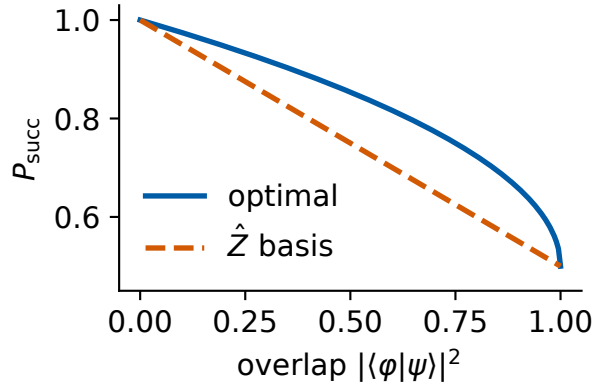
Optimal and Fixed-Basis Discrimination

For pure states

$$P_{\text{succ}}^{\max} = \tfrac{1}{2}\Big(1 + \sqrt{1 - |\langle\varphi|\psi\rangle|^2}\Big),$$

0.85 for $|0\rangle$, $|+\rangle$, against 0.75 in the $\hat{Z}$ basis. Only orthogonal states are told apart with certainty.

A classifier that labels one copy of a quantum state, as in Lecture 14, cannot exceed this bound.



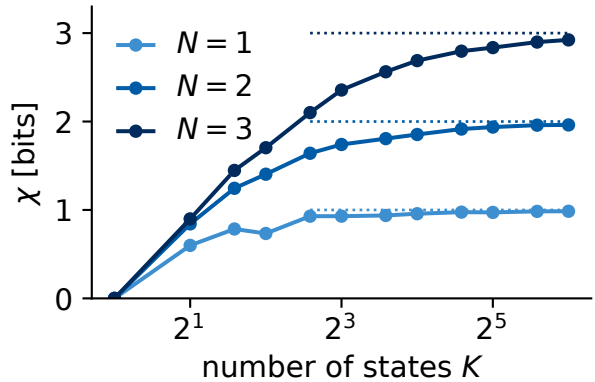
P_{succ} for $|0\rangle$ against a pure state with overlap $|\langle\varphi|\psi\rangle|^2$. Lab 066a

The Holevo Bound

A message x with probability p_x is sent as $\hat{\rho}_x$, and the receiver holds $\hat{\bar{\rho}} = \sum_x p_x \hat{\rho}_x$. The mutual information between x and any measurement outcome obeys

$$I \leq \chi = S(\hat{\bar{\rho}}) - \sum_x p_x S(\hat{\rho}_x) \leq N.$$

2^N amplitudes fix the state, yet a measurement reads at most N bits.



χ of K random pure states of N qubits with equal probabilities, averaged over five draws. Dotted, N bits. Lab 066a

A. S. Holevo, Probl. Peredachi Inf. **9**(3), 3 (1973), mathnet.ru/eng/ppi903

Controlled Gates from CNOT and Rotations

Single-qubit rotations and CNOT are **universal**, so every unitary decomposes into them, with controlled gates as the key step. Any single-qubit gate has the Euler form

$$\hat{U} = e^{i\eta}\hat{R}_z(\theta_1)\hat{R}_y(\theta_2)\hat{R}_z(\theta_3).$$

From its angles we build three rotations,

$$\hat{A} = \hat{R}_z(\theta_1)\hat{R}_y(\tfrac{\theta_2}{2}), \qquad \hat{B} = \hat{R}_y(-\tfrac{\theta_2}{2})\hat{R}_z(-\tfrac{\theta_3+\theta_1}{2}), \qquad \hat{C} = \hat{R}_z(\tfrac{\theta_3-\theta_1}{2}).$$

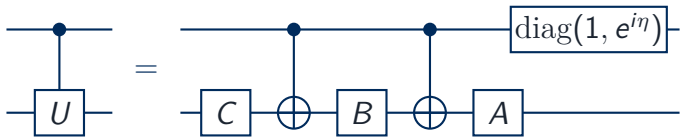
Their product collapses, and $\hat{X}\hat{R}_{y,z}(\alpha)\hat{X} = \hat{R}_{y,z}(-\alpha)$ reverses the angles of $\hat{B}$, so

$$\underbrace{\hat{A}\hat{B}\hat{C} = \hat{I}}_{\text{all angles cancel}}, \qquad \underbrace{\hat{A}\,\hat{X}\hat{B}\hat{X}\,\hat{C} = e^{-i\eta}\hat{U}}_{\text{angles of } \hat{B} \text{ reversed}}.$$

A. Barenco *et al.*, Phys. Rev. A **52**, 3457 (1995), doi:10.1103/PhysRevA.52.3457

The Controlled-$\hat{U}$ Circuit

Two CNOT gates insert $\hat{X}$ around $\hat{B}$ only when the control is $|1\rangle$.



- Control $|0\rangle$, the target sees $\hat{A}\hat{B}\hat{C} = \hat{I}$.
- Control $|1\rangle$, the target sees $\hat{A}\hat{X}\hat{B}\hat{X}\hat{C} = e^{-i\eta}\hat{U}$.
- The phase gate $\mathrm{diag}(1, e^{i\eta}) = e^{i\eta/2}\hat{R}_z(\eta)$ on the control restores the missing phase $e^{i\eta}$ on the branch $|1\rangle$.

A controlled-$\hat{U}$ therefore costs two CNOT gates and four single-qubit gates.

Cost of a General Unitary

Counting parameters

A unitary on N qubits has $4^N - 1$ real parameters, and a two-qubit gate at most $4^2 - 1 = 15$, with single-qubit gates absorbed. A circuit of G two-qubit gates reaches a generic unitary only if

$$15\,G \;\geq\; 4^N - 1, \qquad \text{at least } 69905 \text{ gates for } N = 10.$$

Finite gate sets

Error-correcting codes protect only a finite set such as $\hat{H}$, $\hat{T}$, CNOT. Any single-qubit gate is approximated to accuracy ε by a sequence whose length grows as a power of $\log(1/\varepsilon)$, the Solovay–Kitaev theorem.

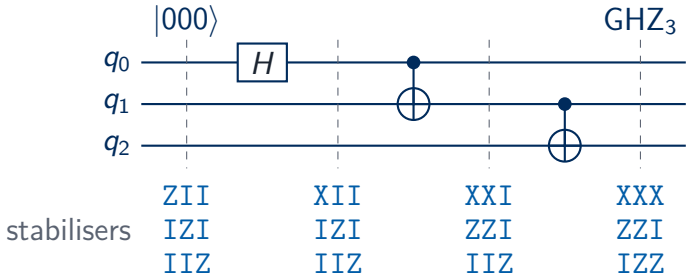
A. Yu. Kitaev, Russ. Math. Surv. **52**, 1191 (1997), doi:10.1070/RM1997v052n06ABEH002155; C. M. Dawson and M. A. Nielsen, Quantum Inf. Comput. **6**, 81 (2006), doi:10.26421/QIC6.1-6

Clifford Gates and Pauli Strings

A Clifford gate maps Pauli strings to Pauli strings, $\hat{U}\hat{P}\hat{U}^\dagger = \pm\hat{Q}$, and $\hat{H}$, $\hat{S}$, CNOT generate all of them,

$$\hat{H}:\ \hat{X} \leftrightarrow \hat{Z}, \qquad \hat{S}:\ \hat{X} \to \hat{Y},\ \hat{Y} \to -\hat{X}, \qquad \widehat{\mathrm{CNOT}}:\ \hat{X}_0 \to \hat{X}_0\hat{X}_1,\ \hat{Z}_1 \to \hat{Z}_0\hat{Z}_1.$$

Bits $(x_q, z_q) = (0,0), (1,0), (0,1), (1,1)$ store $\hat{I}, \hat{X}, \hat{Z}, \hat{Y}$, and each gate updates a few bits. $\hat{P}$ **stabilises** $|\psi\rangle$ if $\hat{P}|\psi\rangle = |\psi\rangle$, as $\hat{Z}_q$ do for $|000\rangle$.



D. Gottesman, arXiv:quant-ph/9807006 (1998)

Stabiliser States

Each $\hat{Z}_q$ stabilises $|0\ldots0\rangle$. For a Clifford circuit $\hat{V}$ it follows that

$$(\hat{V}\hat{Z}_q\hat{V}^\dagger)\,\hat{V}|0\ldots0\rangle = \hat{V}\hat{Z}_q|0\ldots0\rangle = \hat{V}|0\ldots0\rangle,$$

so the N Pauli strings $\hat{V}\hat{Z}_q\hat{V}^\dagger$ stabilise the **stabiliser state** $\hat{V}|0\ldots0\rangle$ and fix it up to a phase.

- The Bell state is stabilised by $\hat{X}\hat{X}$ and $\hat{Z}\hat{Z}$.
- The three-qubit GHZ state is stabilised by $\hat{X}\hat{X}\hat{X}$, $\hat{Z}_0\hat{Z}_1$ and $\hat{Z}_1\hat{Z}_2$.

The state is stored as N Pauli strings, $2N^2$ bits and N signs, instead of 2^N amplitudes.

The Gottesman–Knill Theorem

Pauli expectation values from bits

$$\langle \hat{P} \rangle = \langle 0 \dots 0 | \underbrace{\hat{V}^\dagger \hat{P} \hat{V}}_{\pm \hat{Q},\ \text{bit update}} | 0 \dots 0 \rangle = \begin{cases} \pm 1, & \hat{Q} \text{ only } \hat{Z}, \hat{I} \\ 0, & \text{otherwise} \end{cases}$$

Clifford circuits on $|0 \dots 0\rangle$ with Pauli measurements are simulated efficiently, even for a thousand entangled qubits. $\hat{T}$ maps $\hat{X} \to (\hat{X} + \hat{Y})/\sqrt{2}$ and makes the set universal.

S. Aaronson and D. Gottesman, Phys. Rev. A **70**, 052328 (2004), doi:10.1103/PhysRevA.70.052328; P. O. Boykin *et al.*, Inf. Process. Lett. **75**, 101 (2000), doi:10.1016/S0020-0190(00)00084-3

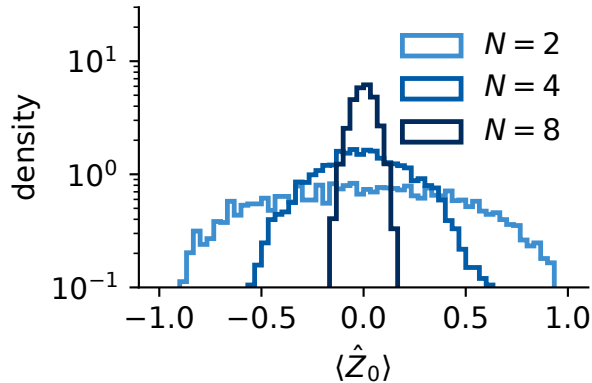
Haar-Random States

A Haar-random state is unchanged in distribution by every unitary. An anticommuting Pauli string flips the sign of $\langle \hat{P} \rangle$, so $\mathbb{E}\langle \hat{P} \rangle = 0$. With $\hat{\rho} = 2^{-N} \sum_{\hat{P}} \langle \hat{P} \rangle \hat{P}$,

$$1 = \mathrm{Tr}\hat{\rho}^2 = \frac{1}{2^N}\Big(\underbrace{1}_{\hat{I}} + \sum_{\hat{P} \neq \hat{I}} \langle \hat{P} \rangle^2 \Big),$$

and Clifford gates make the $4^N - 1$ strings equivalent,

$$\mathrm{Var}\, \langle \hat{P} \rangle = \frac{2^N - 1}{4^N - 1} = \frac{1}{2^N + 1}.$$



Distribution of $\langle \hat{Z}_0 \rangle$ over 4000 Haar-random states, $N = 2, 4, 8$, logarithmic density axis. Lab 066a

Concentration of Random States

The Haar variance $\mathrm{Var}\langle \hat{P} \rangle = 1/(2^N + 1)$ gives the spread of every Pauli expectation value,

$$\mathrm{std}\,\langle \hat{P} \rangle = \frac{1}{\sqrt{2^N + 1}} \approx 2^{-N/2}.$$

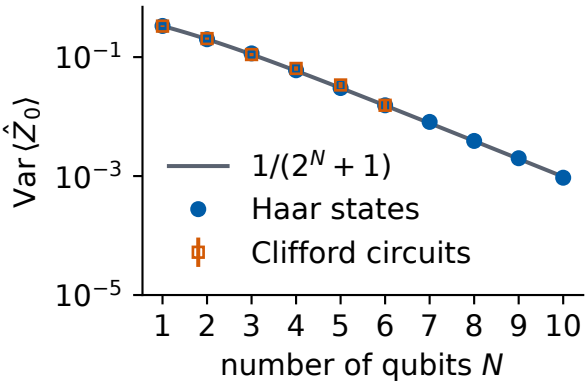
- A random state of many qubits looks maximally mixed to every Pauli measurement.
- Resolving $\langle \hat{P} \rangle$ against this spread needs a number of shots that grows as 2^N.
- Gradients of deep random circuits concentrate in the same way, the barren plateaus of variational algorithms (Lecture 12).

J. R. McClean *et al.*, Nat. Commun. **9**, 4812 (2018), doi:10.1038/s41467-018-07090-4

Unitary 2-Designs

A **2-design** is a set of unitaries that reproduces the Haar averages of all quantities quadratic in $\hat{U}$ and in $\hat{U}^\dagger$, such as $\mathrm{Var}\langle\hat{P}\rangle$.

Uniformly random Clifford circuits form one. Their $\langle\hat{P}\rangle \in \{0, \pm 1\}$, yet the variance equals the Haar value. Classical shadows rest on this (Lecture 13).



Variance of $\langle\hat{Z}_0\rangle$, Haar states and random Clifford circuits. Lab 066a

C. Dankert *et al.*, PRA **80**, 012304 (2009), doi:10.1103/PhysRevA.80.012304; H.-Y. Huang *et al.*, Nat. Phys. **16**, 1050 (2020), doi:10.1038/s41567-020-0932-7

Entanglement of Mixed States

A state is **separable** if it is a mixture of product states, $\hat{\rho} = \sum_k p_k \, \hat{\rho}_0^{(k)} \otimes \hat{\rho}_1^{(k)}$, and **entangled** otherwise. The **partial transpose** transposes the indices of qubit 1, $(\hat{\rho}^{T_1})_{ab,a'b'} = \rho_{ab',a'b}$. A separable state stays a density matrix,

$$\hat{\rho}^{T_1} = \sum_k p_k \, \hat{\rho}_0^{(k)} \otimes \underbrace{\left(\hat{\rho}_1^{(k)}\right)^T}_{\text{density matrix}} \geq 0.$$

A negative eigenvalue of $\hat{\rho}^{T_1}$ proves entanglement (Peres), and for two qubits $\hat{\rho}^{T_1} \geq 0$ proves separability (Horodecki). The **negativity** $\mathcal{N} = \sum_{\lambda<0} |\lambda|$ sums the moduli of the negative eigenvalues λ of $\hat{\rho}^{T_1}$.

A. Peres, PRL **77**, 1413 (1996), doi:10.1103/PhysRevLett.77.1413; M. Horodecki *et al.*, Phys. Lett. A **223**, 1 (1996), doi:10.1016/S0375-9601(96)00706-2; G. Vidal, R. F. Werner, PRA **65**, 032314 (2002), doi:10.1103/PhysRevA.65.032314

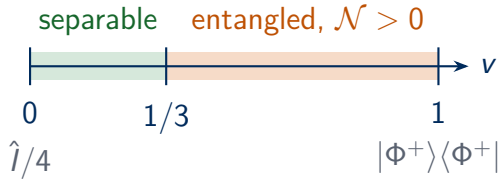
The Werner State

$$\hat{\rho}_W = v\,|\Phi^+\rangle\langle\Phi^+| + (1-v)\,\frac{\hat{I}}{4}, \qquad \text{visibility } 0 \le v \le 1.$$

The partial transpose of $|\Phi^+\rangle\langle\Phi^+|$ is $\widehat{\mathrm{SWAP}}/2$, with SWAP eigenvalues $+1$ on the three symmetric states and -1 on the singlet, so

$$\hat{\rho}_W^{T_1} = \tfrac{v}{2}\,\widehat{\mathrm{SWAP}} + \tfrac{1-v}{4}\,\hat{I} \;\text{ has }\; \underbrace{\tfrac{1+v}{4}}_{\text{symmetric},\,\times 3},\ \underbrace{\tfrac{1-3v}{4}}_{\text{singlet}},$$

so it is entangled if and only if $v > 1/3$, with $\mathcal{N} = \max\left(0, \tfrac{3v-1}{4}\right)$.

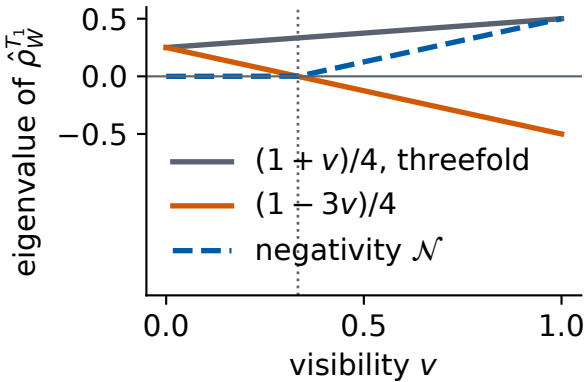


R. F. Werner, Phys. Rev. A **40**, 4277 (1989), doi:10.1103/PhysRevA.40.4277

Negativity of the Werner State and of Noisy Bell States

The lowest eigenvalue of $\hat{\rho}_W^{T_1}$ crosses zero at $v = 1/3$, where the fidelity with $|\Phi^+\rangle$ is $1/2$.

Local depolarising noise turns the Bell state into $\hat{\rho}_W$ with $v = (1 - \frac{4}{3}p)^2$, separable from $p \approx 0.32$. Local dephasing keeps $\mathcal{N} = \frac{1}{2}(1 - 2p)^2 > 0$ up to $p = 1/2$.



Eigenvalues of $\hat{\rho}_W^{T_1}$ and the negativity against v. Lab 066a

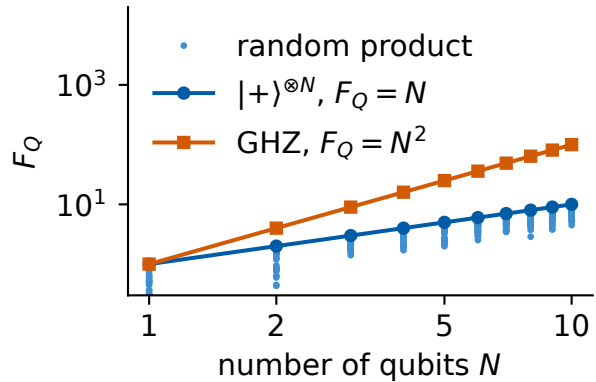
Quantum Fisher Information

$e^{-i\theta\hat{J}_z}$, $\hat{J}_z = \frac{1}{2}\sum_j \hat{Z}_j$, acts on M copies of a pure state, and $\Delta\theta$ is the spread of an unbiased estimate,

$$\langle\psi|e^{-id\theta\hat{J}_z}|\psi\rangle = 1 - id\theta\langle\hat{J}_z\rangle - \tfrac{1}{2}d\theta^2\langle\hat{J}_z^2\rangle,$$

$$|\langle\psi_\theta|\psi_{\theta+d\theta}\rangle|^2 = 1 - \tfrac{1}{4}\underbrace{4\,\mathrm{Var}(\hat{J}_z)}_{F_Q}\,d\theta^2.$$

F_Q, the quantum Fisher information, measures how fast the state moves, and $\Delta\theta \geq 1/\sqrt{MF_Q}$.



F_Q against N, random product states as dots. Lab 066a

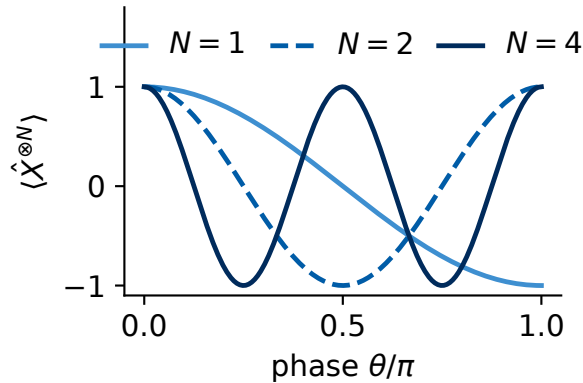
S. L. Braunstein and C. M. Caves, PRL **72**, 3439 (1994), doi:10.1103/PhysRevLett.72.3439

Heisenberg Scaling and Entanglement Certification

$$F_Q^{\text{product}} = \sum_j \underbrace{\mathrm{Var}(\hat{Z}_j)}_{\leq 1} \leq N,$$

$$F_Q^{\text{GHZ}} = N^2.$$

At fixed M the error falls as $1/\sqrt{N}$, or as $1/N$ for GHZ, the **Heisenberg scaling**. Separable states have $F_Q \leq N$, so $F_Q/N > 1$ certifies entanglement (Lecture 13).



GHZ parity $\langle \hat{X}^{\otimes N} \rangle = \cos N\theta$. Lab 066a

L. Pezzé and A. Smerzi, PRL **102**, 100401 (2009), doi:10.1103/PhysRevLett.102.100401

Methods Built on These Foundations

The limits and resources of Part I reappear in the methods of the following lectures.

- Classical shadows rest on the Clifford 2-design, barren plateaus on the concentration of random states.
- Classifiers of quantum data meet the Helstrom bound, and entanglement certification uses the quantum Fisher information.
- Variational algorithms prepare states with rotations and entangling gates and estimate energies from bit strings.
- Tomography reconstructs density matrices from records in many settings, and Hamiltonian learning infers the generator from time traces.
- Tensor networks turn the reshape-and-SVD test into matrix product states.

Part II turns to physical qubits, the devices whose drives and couplings provide the generators that Part I used abstractly.

Part II

Physical Realisations of Qubits

How atoms, superconducting circuits, spins and photons provide qubits, and how their interactions produce two-qubit gates.



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Single Physical Qubits and Their Drives



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Requirements for a Physical Qubit

A physical system provides a qubit when two of its states can be separated from all others, driven into each other by a controllable field and read out. Three systems meet these conditions for different reasons.

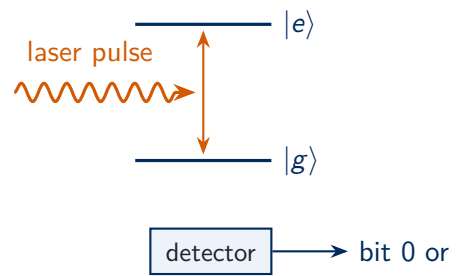
- **Atom.** Two levels are selected from an infinite spectrum by resonance and selection rules.
- **Transmon.** The level spacing of an electrical oscillator is engineered until two levels can be addressed alone.
- **Spin-1/2.** The system has only two levels from the start.

In all three the drive leads to the same 2×2 Hamiltonian. Photon polarisation provides a qubit without energy levels.

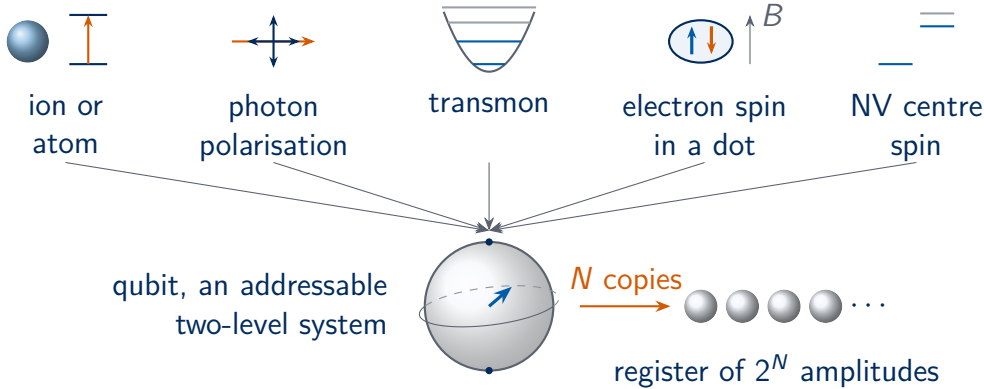
Driving and Measuring an Atom

A laser tuned to the energy gap drives the atom between $|g\rangle$ and $|e\rangle$. The pulse duration sets how far the state rotates, and a short pulse leaves the atom in a superposition of both levels.

A second laser makes the atom fluoresce in only one of the two levels. A detector records light or no light, one bit per run.



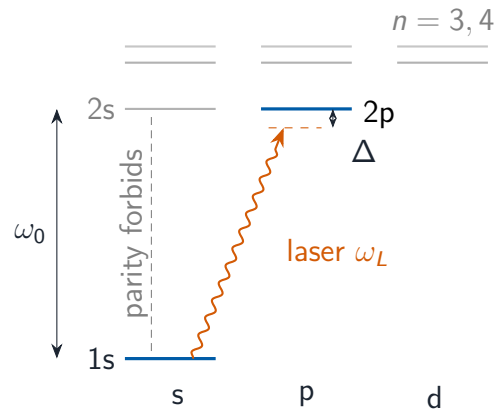
Platforms for Qubits



Ions, atoms, photons, superconducting circuits and spins, in quantum dots or in NV centres (defects in diamond), all provide addressable two-level systems. A register of N qubits is N such systems, driven and measured individually and made to interact.

An Atom in Laser Light

- Atoms have discrete levels. Ions such as Ca^+ with one valence electron serve as qubits.
- A laser with $\hbar\omega_L$ near a level difference drives that transition.
- **Selection rule**. Light couples only levels of opposite parity, so 1s and 2s stay uncoupled.
- Hydrogen shows the numbers but is no qubit, as 2p decays in nanoseconds. The appendix derives all of this.



The Qubit Hamiltonian of a Driven Atom

With $|g\rangle = |0\rangle$, $|e\rangle = |1\rangle$, $\hbar = 1$ and the zero of energy halfway between the levels, a laser field $E_0 \cos(\omega_L t - \phi_L)$ gives

$$\hat{\mathcal{H}}(t) = \underbrace{-\frac{\omega_0}{2}\hat{Z}}_{\hat{\mathcal{H}}_0,\ \text{free atom}} + \underbrace{\Omega \cos(\omega_L t - \phi_L)\,\hat{X}}_{\text{laser coupling}}, \qquad \Omega = -\frac{d_{eg} E_0}{\hbar}\ \text{(Rabi frequency)}.$$

- $\hat{\mathcal{H}}_0$ puts $|g\rangle$ at $-\omega_0/2$ and $|e\rangle$ at $+\omega_0/2$.
- Ω grows with the field amplitude E_0 and the dipole matrix element d_{eg}.
- Atomic-physics texts write $\hat{\sigma}_z^{\text{at}} = |e\rangle\langle e| - |g\rangle\langle g| = -\hat{Z}$ and $\hat{\sigma}_{\text{at}}^{+} = |e\rangle\langle g|$, which excites. Our $\hat{\sigma}^{+} = \hat{X} + i\hat{Y} = 2|g\rangle\langle e|$ lowers the energy.

C. J. Foot, *Atomic Physics* (Oxford University Press, 2005), doi:10.1093/oso/9780198506959.001.0001

The Rotating Frame

$\hat{\mathcal{H}}(t)$ oscillates at an optical frequency, but $e^{-it\hat{\mathcal{H}}}$ needs a constant Hamiltonian. We follow the qubit in axes that turn about z with the laser.

- The drive $\Omega\cos(\omega_L t)\hat{X}$ is two couplings of strength $\Omega/2$ rotating about z in opposite senses at ω_L.
- $-\frac{\omega_0}{2}\hat{Z}$ turns the Bloch vector clockwise at ω_0, as $\hat{R}_z(-\omega_0 t)$.
- Axes turning clockwise at ω_L see one drive component at rest and the precession slowed to $\omega_0 - \omega_L$.

Turning the axes clockwise means turning every state counterclockwise,

$$|\psi_{\mathrm{RF}}(t)\rangle = \hat{R}_z(\omega_L t)\,|\psi(t)\rangle,$$

which changes only the phases of the amplitudes, so all readout probabilities are the same in both frames.

Rotating-Wave Approximation

Differentiating $|\psi_{\mathrm{RF}}\rangle$ and using $\hat{R}_z(a)\hat{X}\hat{R}_z(-a) = \cos a\,\hat{X} + \sin a\,\hat{Y}$,

$$\hat{\mathcal{H}}_{\mathrm{RF}}(t) = \frac{\omega_L - \omega_0}{2}\hat{Z} + \underbrace{\frac{\Omega}{2}\big(\cos\phi_L\hat{X} + \sin\phi_L\hat{Y}\big)}_{\text{co-rotating, constant}}$$

$$+ \underbrace{\frac{\Omega}{2}\big[\cos(2\omega_L t - \phi_L)\hat{X} + \sin(2\omega_L t - \phi_L)\hat{Y}\big]}_{\text{counter-rotating, turns at } 2\omega_L \text{ and averages out}}.$$

Dropping the counter-rotating term, valid for $|\Omega|, |\Delta| \ll \omega_L$, leaves

$$\hat{\mathcal{H}}_{\mathrm{RF}} = \frac{\Delta}{2}\hat{Z} + \frac{\Omega}{2}\big(\cos\phi_L\hat{X} + \sin\phi_L\hat{Y}\big), \qquad \Delta = \omega_L - \omega_0 \text{ (detuning)}.$$

- The sign follows from the remaining clockwise precession at $\omega_0 - \omega_L = -\Delta$, generated by $-\frac{\omega_0-\omega_L}{2}\hat{Z} = +\frac{\Delta}{2}\hat{Z}$.

Limits of the Two-Level Description

- Far-detuned levels are only nearly empty. They shift the qubit levels in proportion to the laser intensity, the **light shift**, and the counter-rotating term adds the smaller Bloch–Siegert shift. Both are calibrated into the detuning.
- The excited level decays by spontaneous emission. Hydrogen 2p lives about 1.60 ns, an energy relaxation far too fast for a computation.
- Atomic and ionic qubits therefore use long-lived pairs of levels. **Hyperfine levels** of the ground state differ in the relative orientation of the electron and nuclear spins, split by their magnetic interaction at a microwave frequency. A **metastable** level cannot decay by a dipole transition.
- Microwaves, laser pairs or narrow optical lines drive these levels with the same $\hat{\mathcal{H}}_{\mathrm{RF}}$.

From the Qubit Hamiltonian to Gates

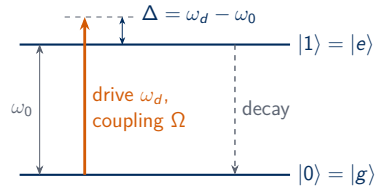
The same Hamiltonian also describes other qubits, with a drive frequency ω_d ($= \omega_L$ for a laser) and a drive phase ϕ. Results built on it hold for any qubit.

$$\hat{\mathcal{H}}_{\mathrm{RF}} = \frac{\Delta}{2}\hat{Z} + \frac{\Omega}{2}(\cos\phi\,\hat{X} + \sin\phi\,\hat{Y}), \qquad \Delta = \omega_d - \omega_0.$$

On resonance and for $\phi = 0$,
$\hat{U}(t) = e^{-i\Omega t\hat{X}/2} = \hat{R}_x(\Omega t)$ and

$$P_e(t) = |\langle e|\hat{U}(t)|g\rangle|^2 = \sin^2\frac{\Omega t}{2},$$

the **Rabi oscillation**. $\Omega t = \pi$ takes $|g\rangle$ to $|e\rangle$,
and $\Omega t = \pi/2$ creates an equal superposition.



I. I. Rabi, Phys. Rev. **51**, 652 (1937), doi:10.1103/PhysRev.51.652

Detuning and Drive Phase

- Off resonance $\hat{\mathcal{H}}_{\mathrm{RF}} = \frac{W}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ with $W = \sqrt{\Omega^2+\Delta^2}$ and the tilted axis $\mathbf{n} = (\Omega, 0, \Delta)/W$. The rotation $\hat{R}_{\mathbf{n}}(Wt)$ gives

$$P_e(t) = \frac{\Omega^2}{\Omega^2+\Delta^2}\sin^2\frac{Wt}{2},$$

  faster and incomplete, since the Bloch vector circles a cone that misses the south pole.

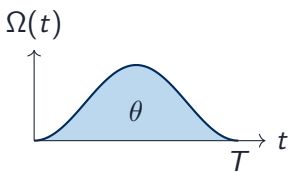
- A detuning alone, $\Omega = 0$, gives the precession $\hat{R}_z(\Delta t)$ about z in the rotating frame.

- The drive phase ϕ turns the rotation axis to $(\cos\phi, \sin\phi, 0)$ in the x–y plane.

Rotation Angle from Pulse Area

[holds for any qubit] On resonance a drive with phase ϕ and a slowly varying amplitude gives $\hat{\mathcal{H}}(t) = \frac{\Omega(t)}{2}(\cos\phi\,\hat{X} + \sin\phi\,\hat{Y})$, which commutes with itself at all times, so

$$\hat{U} = \hat{R}_\phi(\theta) = e^{-i\theta(\cos\phi\,\hat{X}+\sin\phi\,\hat{Y})/2}, \qquad \theta = \underbrace{\int_0^T \Omega(t')\,dt'}_{\text{pulse area}}.$$



- The area sets the angle, and ϕ sets the axis.
- A smooth envelope of equal area gives the same gate.
- With a detuning $\hat{\mathcal{H}}$ contains $\hat{Z}$, and the shape matters too.

Phase Updates as $\hat{Z}$ Rotations

[holds for any qubit] A rotation about z turns $\hat{X}$ into $\hat{R}_z(\phi)\hat{X}\hat{R}_z(-\phi) = \cos\phi\,\hat{X} + \sin\phi\,\hat{Y}$, so

$$\hat{R}_\phi(\theta) = \hat{R}_z(\phi)\,\hat{R}_x(\theta)\,\hat{R}_z(-\phi) \quad \Longrightarrow \quad \hat{R}_\phi(\theta)\,\hat{R}_z(\beta) = \hat{R}_z(\beta)\,\hat{R}_{\phi-\beta}(\theta).$$

- A rotation $\hat{R}_z(\beta)$ followed by pulses with phases ϕ_k equals the same pulses with phases $\phi_k - \beta$, then a final $\hat{R}_z(\beta)$.
- The final z rotation leaves the probabilities of a readout in the computational basis unchanged.
- The control electronics apply $\hat{R}_z(\beta)$ by shifting the phase of every later pulse, a **virtual $\hat{Z}$ gate**. It takes no time, so the qubit does not decohere during it.

Single-Qubit Gates in the Laboratory

gate	action on basis states	Bloch sphere	implementation
$\hat{X}$	$\|0\rangle \leftrightarrow \|1\rangle$, $\|\pm\rangle \to \pm\|\pm\rangle$	half turn about x	pulse of area π, $\phi = 0$
$\hat{Z}$	$\|1\rangle \to -\|1\rangle$, $\|+\rangle \leftrightarrow \|-\rangle$	half turn about z	virtual $\hat{Z}$
$\hat{H}$	$\|0\rangle \to \|+\rangle$, $\|1\rangle \to \|-\rangle$	half turn about $(x+z)/\sqrt{2}$	virtual $\hat{Z}$, then area $\pi/2$ at $\phi = \pi/2$
$\hat{S}$, $\hat{T}$	$\|1\rangle \to i\|1\rangle$, $\|1\rangle \to e^{i\pi/4}\|1\rangle$	quarter, eighth turn about z	virtual $\hat{Z}$
$\hat{R}_\phi(\theta)$	mixes $\|0\rangle, \|1\rangle$ with $\cos\frac{\theta}{2}$, $\sin\frac{\theta}{2}$	turn by θ about $(\cos\phi, \sin\phi, 0)$	area θ, phase ϕ
$\hat{R}_z(\theta)$	$\|1\rangle \to e^{i\theta}\|1\rangle$ up to a phase	turn by θ about z	detuning for $t = \theta/\Delta$, or virtual $\hat{Z}$

[holds for any qubit] $\hat{H} = \hat{R}_y(\pi/2)\,\hat{Z}$ is a virtual $\hat{Z}$ gate and then one pulse.

Superconducting Circuits as Qubits

- In a superconductor electrons bind into Cooper pairs that flow without resistance, so a circuit loses almost no energy and keeps the phase of its oscillations.
- Its levels act as a quantum system only if thermal energy cannot excite them, $k_B T \ll h f_{01}$. For $f_{01} = 5$ GHz, $h f_{01}/k_B \approx 0.24$ K, so chips run in dilution refrigerators near 10 mK.
- The spectrum is set by the design. An LC circuit is a harmonic oscillator, and a Josephson junction makes it anharmonic.

P. Krantz *et al.*, Appl. Phys. Rev. **6**, 021318 (2019), doi:10.1063/1.5089550

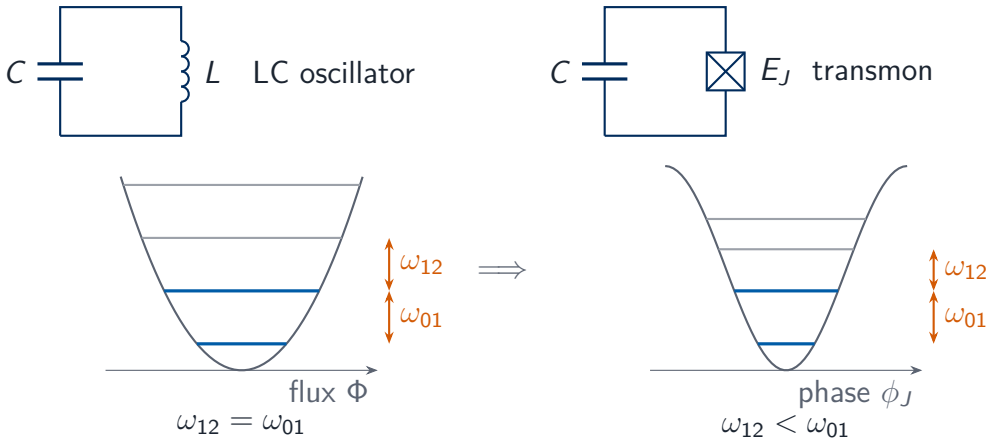
The Quantum Harmonic Oscillator

A mass m in the potential $\frac{1}{2}m\omega^2 x^2$, with $[\hat{x}, \hat{p}] = i$, has equally spaced levels,

$$\hat{\mathcal{H}} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2} = \omega\Big(\hat{b}^\dagger\hat{b} + \frac{1}{2}\Big), \qquad E_k = \omega\Big(k + \frac{1}{2}\Big), \quad k = 0, 1, 2, \ldots$$

- The lowering and raising operators act as $\hat{b}|k\rangle = \sqrt{k}\,|k-1\rangle$, $\hat{b}^\dagger|k\rangle = \sqrt{k+1}\,|k+1\rangle$, $\hat{b}^\dagger\hat{b}|k\rangle = k|k\rangle$.
- Position $\hat{x} = x_0(\hat{b} + \hat{b}^\dagger)$, $x_0^2 = 1/(2m\omega)$, so a coupling $\propto \hat{x}$ connects only $k \leftrightarrow k \pm 1$.
- The LC circuit is this oscillator with $\hat{x} = \hat{\Phi}$ (flux), $\hat{p} = \hat{Q}$ (charge), $m = C$, $\omega_r = 1/\sqrt{LC}$. The motion of trapped ions is another, with $\hat{a}$ in place of $\hat{b}$.

LC Oscillator and Transmon



A drive at ω_{01} climbs the whole LC ladder. A Josephson junction (boxed cross) makes the spacing shrink, and the two lowest levels (blue) form the qubit.

The Transmon Hamiltonian

All Cooper pairs share one wave function, whose complex phase is the superconducting phase. A Josephson junction, an insulating barrier between two superconductors, has the energy $-E_J \cos\phi_J$, with the phase difference ϕ_J. With a large shunt capacitance,

$$\hat{\mathcal{H}}_{\mathrm{tr}} = \underbrace{4E_C(\hat{n}-n_g)^2}_{\text{charging energy}} \underbrace{-E_J\cos\hat{\phi}_J}_{\text{junction}}, \qquad [\hat{\phi}_J,\hat{n}] = i, \qquad E_C = \frac{e_0^2}{2C}.$$

- $\hat{n} = \hat{Q}/2e_0$ counts Cooper pairs across the junction, $\hat{\phi}_J = 2e_0\hat{\Phi}$, and n_g is an offset charge. In the basis $|n\rangle$, $\cos\hat{\phi}_J = \frac{1}{2}\sum_n(|n+1\rangle\langle n| + \text{h.c.})$ is tridiagonal.
- $E_J/E_C \approx 50$ is a compromise. Charge noise falls exponentially with $\sqrt{8E_J/E_C}$, the relative anharmonicity only as $(8E_J/E_C)^{-1/2}$.

J. Koch *et al.*, Phys. Rev. A **76**, 042319 (2007), doi:10.1103/PhysRevA.76.042319; B. D. Josephson, Phys. Lett. **1**, 251 (1962), doi:10.1016/0031-9163(62)91369-0

The Transmon Near the Bottom of Its Well

For $E_J \gg E_C$ the phase stays near $\phi_J = 0$, n_g drops out, and $\cos\phi_J \approx 1 - \frac{\phi_J^2}{2} + \frac{\phi_J^4}{24}$ gives, up to a constant,

$$\hat{\mathcal{H}}_{\mathrm{tr}} \approx \underbrace{4E_C\hat{n}^2 + \tfrac{E_J}{2}\hat{\phi}_J^2}_{\text{oscillator}} \quad \underbrace{-\tfrac{E_J}{24}\hat{\phi}_J^4}_{\text{small correction}} \ .$$

- The oscillator has $\hat{x} = \hat{\phi}_J$, $\hat{p} = \hat{n}$, $m = 1/8E_C$ and $\omega_p = \sqrt{8E_JE_C}$.
- Its lowering operator gives $\hat{\phi}_J = \phi_0(\hat{b} + \hat{b}^\dagger)$ with

  $$\phi_0^2 = \frac{1}{2m\omega_p} = \frac{4E_C}{\sqrt{8E_JE_C}} = \sqrt{\frac{2E_C}{E_J}} \ll 1,$$

  the spread of the phase in the ground state.

Anharmonicity of the Transmon

First-order perturbation theory adds the mean of the quartic term in $|k\rangle$. Of the 16 products in $(\hat{b}+\hat{b}^\dagger)^4$ only the 6 with two $\hat{b}$ and two $\hat{b}^\dagger$ return to $|k\rangle$,

$$\langle k|(\hat{b}+\hat{b}^\dagger)^4|k\rangle = (k+1)(k+2)+(k+1)^2+2k(k+1)+k^2+k(k-1) = 6k^2+6k+3.$$

With $E_J\phi_0^4 = 2E_C$,

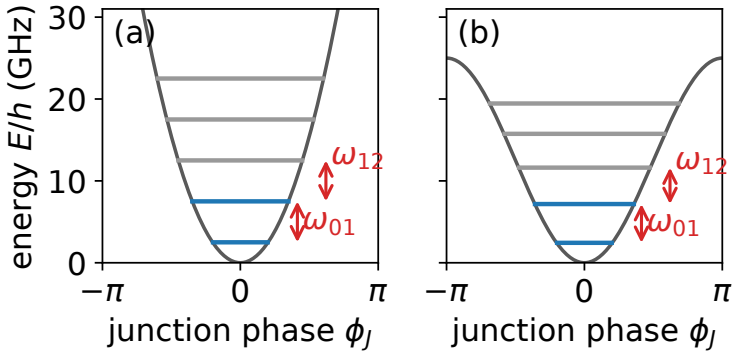
$$E_k \approx \omega_p\Big(k+\tfrac{1}{2}\Big) - \underbrace{\tfrac{E_C}{12}(6k^2+6k+3)}_{\text{quartic shift}},$$

$$\omega_{01} = E_1 - E_0 \approx \sqrt{8E_JE_C} - E_C, \qquad \alpha = \omega_{12}-\omega_{01} \approx -E_C.$$

Typical transmons have $\omega_{01}/2\pi$ of 3–6 GHz and $|\alpha|/2\pi$ of 100–300 MHz.

P. Krantz *et al.*, Appl. Phys. Rev. **6**, 021318 (2019), doi:10.1063/1.5089550

Transmon Spectrum



$E_C/h = 0.25$ GHz, $E_J = 50\,E_C$. (a) Quadratic approximation. (b) Cosine well, $\omega_{01}/2\pi = 4.74$ GHz (first order 4.75), $\alpha/2\pi = -287$ MHz, with $|\alpha|$ fifteen per cent above E_C/h.

Lab 067

Microwave Drive of the Transmon

A microwave voltage $V_0 \cos(\omega_d t - \phi)$ through a capacitance C_d adds $2e_0 \frac{C_d}{C} V(t)\, \hat{n}$. At $n_g = 0$ the eigenstates have definite parity, $\langle k|\hat{n}|k\rangle = 0$, and $\hat{n} \to n_{01}\hat{X}$ on the two lowest levels,

$$\hat{\mathcal{H}}(t) = \underbrace{-\frac{\omega_{01}}{2}\hat{Z}}_{\text{qubit}} + \underbrace{\Omega \cos(\omega_d t - \phi)\, \hat{X}}_{\text{drive}}, \qquad \Omega = 2e_0 \frac{C_d}{C} V_0\, n_{01}.$$

This is the driven atom with ω_{01} in place of ω_0. The rotating frame and the rotating-wave approximation give $\hat{\mathcal{H}}_{\mathrm{RF}}$ with $\Delta = \omega_d - \omega_{01}$, and a resonant pulse rotates the qubit by its area.

P. Krantz *et al.*, Appl. Phys. Rev. **6**, 021318 (2019), doi:10.1063/1.5089550

Leakage Out of the Qubit

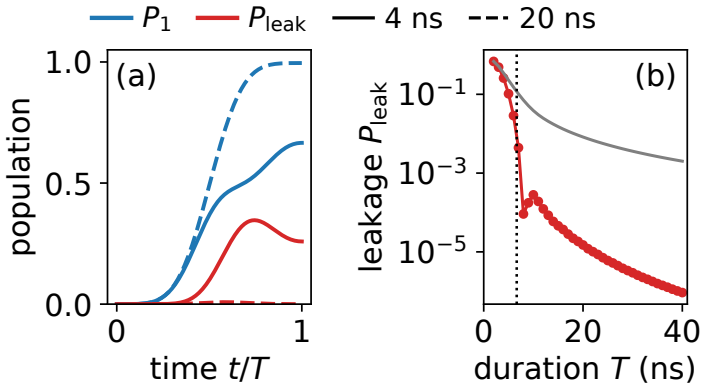
- The drive also couples $|1\rangle \to |2\rangle$, with $n_{12} = 1.37\, n_{01}$, detuned only by α. The rotating frame $|\psi\rangle \to e^{i\omega_d t \hat{N}}|\psi\rangle$, $\hat{N} = \sum_k k|k\rangle\langle k|$, gives $|k\rangle$ the energy $\delta_k = E_k - E_0 - k\omega_d$.
- **Leakage** P_{leak} is the population left in $|2\rangle$ and above after a resonant π pulse, simulated on five levels.
- A Gaussian envelope with $\sigma = T/4$ spans angular frequencies $\sim 1/\sigma$. The transition $1 \to 2$ stays unexcited once $\sigma|\alpha| \gtrsim 3$,

$$T \gtrsim \frac{12}{|\alpha|} \approx 6.6 \text{ ns}.$$

- DRAG pulses add a *quadrature* component, shifted by 90° in phase and thus along $\hat{Y}$, proportional to the envelope derivative.

F. Motzoi *et al.*, Phys. Rev. Lett. **103**, 110501 (2009), doi:10.1103/PhysRevLett.103.110501

Leakage During a π Pulse



(a) Pulses of 4 ns (below the border) and 20 ns. (b) Final (dots) and largest (grey) leakage. Dotted line $12/|\alpha|$. Below 10^{-3} from about 8 ns.

Lab 067

Dispersive Readout of the Transmon

- A microwave resonator, a circuit with one oscillation mode, is coupled to the transmon and detuned from its transition.
- The resonator frequency shifts by an amount that depends on whether the transmon is in $|0\rangle$ or $|1\rangle$.
- A weak tone sent through the resonator returns with a phase that reveals the state, a measurement of $\hat{Z}$.

A. Blais *et al.*, Phys. Rev. A **69**, 062320 (2004), doi:10.1103/PhysRevA.69.062320

A Spin-1/2 in a Magnetic Field

An electron or a proton has the spin $\hat{\mathbf{S}} = \hat{\boldsymbol{\sigma}}/2$, only two levels, and the magnetic moment $\gamma\hat{\mathbf{S}}$, with the gyromagnetic ratio γ,

$$\hat{\mathcal{H}} = -\gamma\,\mathbf{B}\cdot\hat{\mathbf{S}} \xrightarrow{\mathbf{B}=B_z\mathbf{e}_z} -\frac{\omega_0}{2}\hat{Z}, \qquad \omega_0 = \gamma B_z \text{ (Larmor frequency).}$$

- We orient z so that $\omega_0 > 0$. Then $|0\rangle = |\uparrow\rangle$ is the ground state, and the levels are split by ω_0 (Zeeman splitting). The electron has $\gamma < 0$, so its field then points along $-z$.
- The Bloch vector is $2\langle\hat{\mathbf{S}}\rangle$, the spin direction itself, and by the next frame $\frac{d}{dt}\langle\hat{\mathbf{S}}\rangle = \gamma\,\langle\hat{\mathbf{S}}\rangle \times \mathbf{B}$. A static field turns it clockwise about z, $\hat{R}_z(-\omega_0 t)$.
- Electron $|\gamma|/2\pi \approx 28$ GHz/T (microwaves), proton 42.6 MHz/T (radio waves).

P. J. Mohr *et al.* (CODATA 2022), Rev. Mod. Phys. **97**, 025002 (2025), doi:10.1103/RevModPhys.97.025002

Equation of Motion of a Mean Value

[holds for any qubit] From $i\frac{d}{dt}|\psi\rangle = \hat{\mathcal{H}}|\psi\rangle$ and its adjoint $-i\frac{d}{dt}\langle\psi| = \langle\psi|\hat{\mathcal{H}}$,

$$\frac{d}{dt}\langle\hat{O}\rangle = \Big(\frac{d}{dt}\langle\psi|\Big)\hat{O}|\psi\rangle + \langle\psi|\hat{O}\Big(\frac{d}{dt}|\psi\rangle\Big) = i\langle\psi|\hat{\mathcal{H}}\hat{O}|\psi\rangle - i\langle\psi|\hat{O}\hat{\mathcal{H}}|\psi\rangle = i\langle[\hat{\mathcal{H}},\hat{O}]\rangle.$$

For the spin, $[\hat{S}_a,\hat{S}_b] = i\varepsilon_{abc}\hat{S}_c$ and $\hat{\mathcal{H}} = -\gamma\mathbf{B}\cdot\hat{\mathbf{S}}$ give

$$\frac{d}{dt}\langle\hat{\mathbf{S}}\rangle = \gamma\,\langle\hat{\mathbf{S}}\rangle \times \mathbf{B},$$

the motion of a magnetic moment turning about a field.

Magnetic Resonance

A transverse field $B_1 \cos(\omega_d t)\,\mathbf{e}_x$, at the drive frequency ω_d, gives the Hamiltonian of the laser-driven atom,

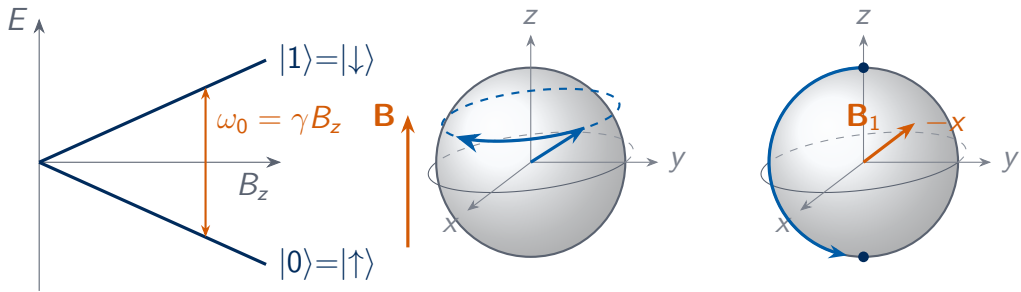
$$\hat{\mathcal{H}}(t) = -\frac{\omega_0}{2}\hat{Z} + \underbrace{\Omega \cos(\omega_d t)\,\hat{X}}_{\text{transverse field}}, \qquad \Omega = -\frac{\gamma B_1}{2} > 0 \text{ (sign of } B_1 \text{ chosen)},$$

$$\hat{\mathcal{H}}_{\mathrm{RF}} = \frac{\Delta}{2}\hat{Z} + \frac{\Omega}{2}\hat{X}, \qquad \Delta = \omega_d - \omega_0.$$

- For $\gamma > 0$, $\Omega > 0$ means $B_1 < 0$, a field along $-x$, made of two fields of amplitude $|B_1|/2$ rotating oppositely about z. In the frame rotating at $\omega_d = \omega_0$ one is static along $-x$ and the other averages out.
- The spin turns about it, $\hat{R}_x(\Omega t)$, from the north pole through $-y$, and flips with probability $\sin^2(\Omega t/2)$ (electron spin or nuclear magnetic resonance).

I. I. Rabi, J. R. Zacharias, S. Millman and P. Kusch, Phys. Rev. **53**, 318 (1938), doi:10.1103/PhysRev.53.318

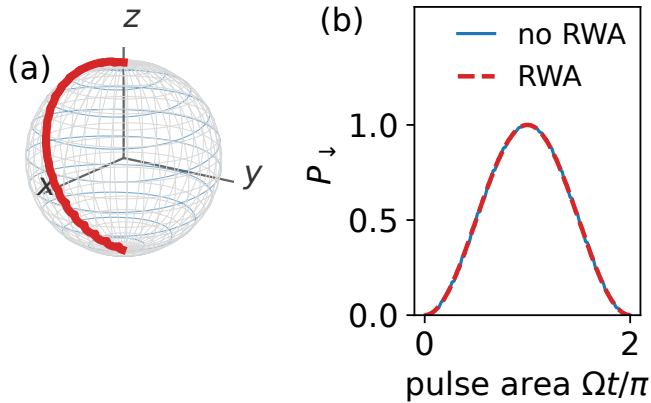
Zeeman Splitting, Precession and Resonance



(a) Zeeman splitting (b) precession, $\hat{R}_z(-\omega_0 t)$ (c) drive, $\hat{R}_x(\Omega t)$

Drawn for $\gamma > 0$, $B_z > 0$ and $\Omega > 0$. (a) Levels against B_z. (b) Clockwise Larmor precession about $\mathbf{B}$. (c) In the rotating frame $\mathbf{B}_1$ is static along $-x$, and the spin turns through $-y$, $\hat{R}_x(\Omega t)$.

Magnetic Resonance in Two Frames



$\omega_0 = 20\,\Omega > 0$, laboratory (blue) and rotating frame (red), difference at most 0.013.

Lab 067

Spin Readout

- **Quantum dot.** A single electron is held in a nanometre-sized region of a semiconductor. The levels are tuned so that only spin down can tunnel to a nearby reservoir, and a charge sensor records whether the electron left, a single-shot measurement of $\hat{Z}$.
- **Nuclear magnetic resonance.** An ensemble of nuclear spins precesses after a pulse, and its magnetisation induces a voltage in a pick-up coil, which measures the mean of $\hat{X}$ or $\hat{Y}$.

D. Loss and D. P. DiVincenzo, Phys. Rev. A **57**, 120 (1998), doi:10.1103/PhysRevA.57.120; J. M. Elzerman *et al.*, Nature **430**, 431 (2004), doi:10.1038/nature02693; F. Bloch, Phys. Rev. **70**, 460 (1946), doi:10.1103/PhysRev.70.460

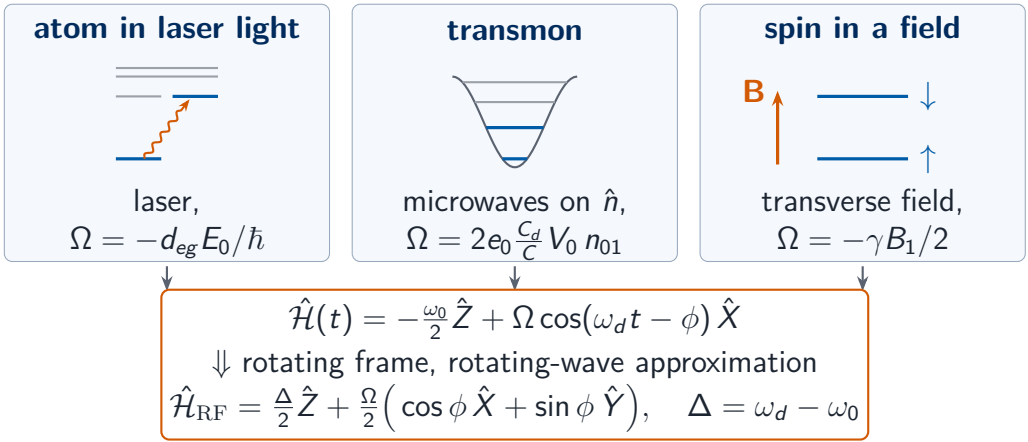
Photon Polarisation

A photon carries a qubit in its polarisation, $|0\rangle = |H\rangle$ (horizontal), $|1\rangle = |V\rangle$ (vertical).

- The two states have equal energy, so there is no splitting, no precession and no resonant drive. A photon has no drive Hamiltonian to switch on.
- A wave plate, a birefringent crystal slab, applies a fixed rotation while the photon passes. Thickness and orientation play the roles of pulse area and phase.
- A polarising beam splitter sends H and V to two detectors, a measurement in the computational basis. A wave plate in front of it chooses another basis.
- Photons travel far with little disturbance, but two photons barely interact, so a gate acting on two photon qubits at once needs extra photons and measurements.

E. Knill, R. Laflamme and G. J. Milburn, Nature **409**, 46 (2001), doi:10.1038/35051009

Three Platforms, One Qubit Hamiltonian



ω_d and ϕ are the drive frequency and phase (ω_L, ϕ_L for a laser), and ω_0 is the qubit frequency.

Platforms Compared

platform	the two levels	driven by	two-qubit coupling	read out by
trapped ions	electronic or hyperfine levels	lasers or microwaves	shared ion vibrations	fluorescence
atoms in tweezers	two atomic levels	lasers	Rydberg interactions	fluorescence imaging
transmons	two lowest oscillator levels	microwave pulses	capacitors, couplers	microwave resonator
quantum-dot spins	spin up and down	microwaves, electric fields	exchange interaction	charge sensing
photons	H and V polarisation	wave plates	interference, measurement	photon detectors

Hyperfine levels are ground-state levels split by the electron–nuclear spin interaction, tweezers are focused beams holding one atom each, Rydberg states are highly excited, and a resonator is a one-mode circuit.

Natural and Fabricated Qubits

- Atomic and ionic qubits in hyperfine or metastable levels keep superpositions for a long time, and atoms of one isotope are identical, so every qubit has the same frequency.
- Superconducting circuits are fabricated. Their gates are fast, with durations set by the anharmonicity, whereas every transmon has a slightly different frequency, is calibrated individually and decoheres sooner.
- The interaction between qubits fixes the two-qubit gates a device applies directly, its **native gates**, and the pairs of qubits it can couple, its **connectivity**.

Interacting Qubits and Two-Qubit Gates in the Laboratory



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Where Qubit Couplings Come From

platform	mechanism	coupling	native gate
neutral atoms	Rydberg interaction $V \propto 1/R^6$	$V\lvert rr\rangle\langle rr\rvert$	controlled-Z by a blockade pulse sequence
trapped ions	Coulomb force, shared vibrations	$\hat{X}\hat{X}$	$\hat{R}_{xx}(\theta)$
superconducting circuits	capacitor or resonator	$\hat{X}\hat{X} + \hat{Y}\hat{Y}$, or $\hat{Z}\hat{X}$	$\widehat{\mathrm{iSWAP}}^{\dagger}$ or $\widehat{\mathrm{iSWAP}}$, or CNOT
quantum-dot spins	exchange interaction	$J_{\mathrm{ex}}(t)\,\hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1$	$\sqrt{\widehat{\mathrm{SWAP}}}$

Saffman *et al.*, RMP **82**, 2313 (2010), doi:10.1103/RevModPhys.82.2313; Rigetti, Devoret, PRB **81**, 134507 (2010), doi:10.1103/PhysRevB.81.134507; Loss, DiVincenzo, PRA **57**, 120 (1998), doi:10.1103/PhysRevA.57.120

Rydberg Interaction in Pauli Language

Projectors become Pauli operators, $|0\rangle\langle 0| = \frac{1}{2}(\hat{I} + \hat{Z})$ and $|1\rangle\langle 1| = \frac{1}{2}(\hat{I} - \hat{Z})$. With the qubit stored in $|g\rangle = |0\rangle$ and a Rydberg state $|r\rangle = |1\rangle$ (ground–Rydberg encoding), the interaction of two atoms is

$$V|11\rangle\langle 11| = V\,\frac{\hat{I} - \hat{Z}_0}{2}\,\frac{\hat{I} - \hat{Z}_1}{2} = \underbrace{\tfrac{V}{4}\hat{Z}_0\hat{Z}_1}_{\text{Ising coupling}} \quad \underbrace{-\tfrac{V}{4}(\hat{Z}_0 + \hat{Z}_1)}_{\text{detuning of each atom}} \quad + \quad \underbrace{\tfrac{V}{4}\hat{I}}_{\text{constant}} .$$

The other couplings follow in the same way,

$$|01\rangle\langle 10| + |10\rangle\langle 01| = \tfrac{1}{2}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1), \qquad J_{\text{ex}}\hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1 = \tfrac{J_{\text{ex}}}{4}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1 + \hat{Z}_0\hat{Z}_1).$$

Two Ions and Their Shared Motion

The **Mølmer–Sørensen** gate couples two ions through their shared motion and works also for ions not cooled to the motional ground state.

- The ions repel, and their joint oscillation is a **motional mode** of frequency ν, the oscillator of the transmon slides with $\hat{a}$ in place of $\hat{b}$. Its quanta are **phonons**.
- A laser at the qubit frequency $\pm\nu$ flips a qubit and adds or removes one phonon, the blue and red **sidebands**, weakened by the **Lamb–Dicke parameter** $\eta \ll 1$.
- Two tones detuned by $\pm(\nu - \delta)$ drive both ions. To first order in η they exert a force whose sign depends on the qubits,

$$\hat{\mathcal{H}}(t) = g\,\hat{S}_x\big(\hat{a}\,e^{-i\delta t} + \hat{a}^\dagger e^{i\delta t}\big), \qquad \hat{S}_x = \tfrac{1}{2}(\hat{X}_0 + \hat{X}_1), \quad g = \eta\Omega.$$

Sørensen, Mølmer, PRL **82**, 1971 (1999), doi:10.1103/PhysRevLett.82.1971; Mølmer, Sørensen, PRL **82**, 1835 (1999), doi:10.1103/PhysRevLett.82.1835

A Closed Loop in Phase Space

For an eigenvalue $m \in \{1, 0, -1\}$ of $\hat{S}_x$ the force pushes the motion around a circle in the plane of mean position and momentum,

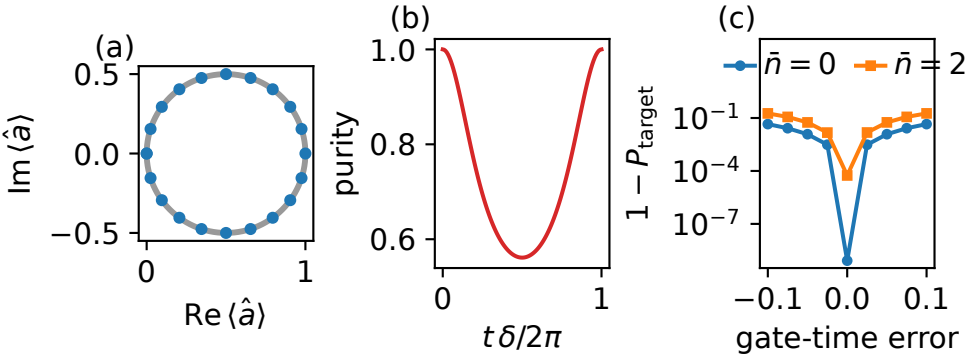
$$\langle \hat{a} \rangle(t) = \langle \hat{a} \rangle(0) + m\,\alpha(t), \qquad \alpha(t) = \frac{g}{\delta}(1 - e^{i\delta t}).$$

At $t = 2\pi K/\delta$ the circle closes for every motional state, and the qubits are free of the motion. The loop leaves a phase $\propto m^2$, the effective coupling $-\frac{J}{2}\hat{X}_0\hat{X}_1$ with $J = g^2/\delta$,

$$\hat{U}\Big(\frac{2\pi K}{\delta}\Big) = e^{i\Phi/2}\,\hat{R}_{xx}(-\Phi), \qquad \Phi = \frac{2\pi K g^2}{\delta^2} = Jt.$$

One loop with $\delta = 2g$ gives $\Phi = \pi/2$ and $|00\rangle \to (|00\rangle + i|11\rangle)/\sqrt{2}$, opposite in sense to the $\hat{R}_{xx}(+\pi/2)$ of Part I. Derivation in the appendix at the end of the deck.

A Gate Insensitive to Thermal Motion



(a) $\langle \hat{a} \rangle$ for $|{+}{+}\rangle$ (dots) runs around the circle (line) and returns. (b) Qubit purity from $|00\rangle$ drops during the entanglement with the motion and returns to one. (c) Error $1 - P_{\text{target}}$. At the closing time $P_{\text{target}} = 0.9999$ also at $\bar{n} = 2$, and a gate time off by ten per cent gives 0.045 (cold) and 0.182 ($\bar{n} = 2$).

Native Two-Qubit Gates in the Laboratory

The rotations and controlled gates of Part I are applied by switching physical couplings on for calibrated times.

- The Ising coupling $\frac{J}{2}\hat{Z}_0\hat{Z}_1$ comes from the Rydberg interaction or a tunable coupler. CZ is this coupling on for $t = \pi/(2J)$, with the two z rotations applied as virtual $\hat{Z}$ gates.
- Neutral atoms also reach CZ with a laser-pulse sequence that uses the Rydberg blockade.
- Superconducting circuits obtain CNOT from the cross-resonance coupling $\hat{Z}\hat{X}$ on for a calibrated time, with single-qubit corrections, and $\widehat{\mathrm{iSWAP}}$ or $\widehat{\mathrm{iSWAP}}^{\dagger}$ from the exchange $\frac{J}{2}(\hat{X}\hat{X} + \hat{Y}\hat{Y})$.
- Quantum-dot spins give $\sqrt{\widehat{\mathrm{SWAP}}}$ from an exchange pulse $J_{\mathrm{ex}}\hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1 = \frac{J_{\mathrm{ex}}}{4}(\hat{X}\hat{X} + \hat{Y}\hat{Y} + \hat{Z}\hat{Z})$ with $\int J_{\mathrm{ex}}\, dt = \pi/2$.

Jaksch *et al.*, PRL **85**, 2208 (2000), doi:10.1103/PhysRevLett.85.2208; Rigetti, Devoret, PRB **81**, 134507 (2010), doi:10.1103/PhysRevB.81.134507; Loss, DiVincenzo, PRA **57**, 120 (1998), doi:10.1103/PhysRevA.57.120

Gate Parameters and Their Physical Processes

The coupling $J(t)$ is set by a tunable coupler, by the ion drive or by the gate voltage of quantum-dot exchange.

gate	parameter	physical process	control setting
$\hat{R}_\phi(\theta)$	$\theta = \int \Omega\, dt$, axis angle ϕ	resonant laser or microwave pulse	duration or amplitude, phase
$\hat{R}_z(\theta)$, $\hat{S}$, $\hat{T}$	$\theta = \Delta t$ or phase shift	detuning, or virtual $\hat{Z}$	waiting time, or later phases
$\hat{R}_{zz}(\theta)$	$\theta = \int J\, dt$	Ising coupling, Rydberg or tunable coupler	interaction time
$\hat{R}_{xx}(\theta)$	$\theta = \int J\, dt$	Mølmer–Sørensen drive of ions	drive duration
$\sqrt{\widehat{\mathrm{SWAP}}}$, $\mathrm{i}\widehat{\mathrm{SWAP}}$	$\int J_{\mathrm{ex}} dt$, $\int J\, dt = \frac{\pi}{2}$	exchange of spins or circuits	exchange pulse
CZ, CNOT	$\hat{R}_{zz}(\pi/2)$ and fixed phases	Ising coupling, blockade, cross-resonance	calibrated duration

A Circuit as Timed Evolutions

Single-qubit drives and two-qubit couplings are both Hamiltonians switched on for a time. A circuit of K steps is

$$\hat{U} = e^{-it_K \hat{\mathcal{H}}_K} \cdots e^{-it_2 \hat{\mathcal{H}}_2} \, e^{-it_1 \hat{\mathcal{H}}_1}.$$

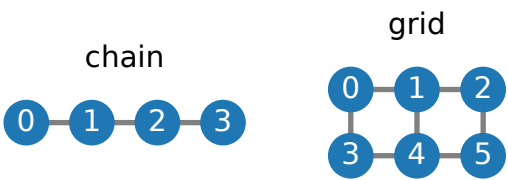
- Gates on different qubits run in parallel within a layer, and the circuit lasts the sum of its layer durations.
- Two-qubit gates take longer, as couplings between qubits are often weaker than the drives.

Each angle in a circuit diagram stands for the duration of a physical evolution.

A Register of Interacting Qubits

Each qubit of a register is driven on its own, and pairs joined by an edge of the **connectivity graph** interact,

$$\hat{\mathcal{H}}_{\mathrm{dev}} = \sum_{q} \left(\frac{\Omega_q}{2} \hat{X}_q + \frac{\Delta_q}{2} \hat{Z}_q \right) + \sum_{(q,q') \in \mathrm{edges}} \frac{J_{qq'}}{2} \hat{Z}_q \hat{Z}_{q'}.$$



- Every term is a Pauli string, and pulses switch the terms on and off.
- Distant qubits at distance d need $2(d-1)$ SWAP gates of three CNOTs each.

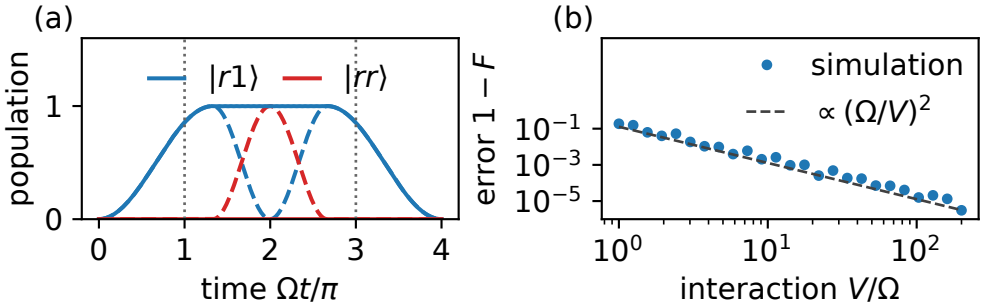
A Blockade Gate

- Rydberg states live only about $100\,\mu$s, too short to store a qubit. The **hyperfine encoding** keeps the qubit in two long-lived ground levels $|0\rangle, |1\rangle$ and visits $|r\rangle$ only during the gate.
- A laser couples $|1\rangle \leftrightarrow |r\rangle$, and two atoms in $|r\rangle$ shift by V.
- Pulses π (control), 2π (target), π (control). A 2π pulse multiplies $|1\rangle$ by -1, so $|01\rangle$ and $|10\rangle$ change sign.
- In $|11\rangle$ the excited control shifts the target out of resonance, the **blockade**, and only the control pulses act.

The result is $\mathrm{diag}(1,-1,-1,-1) = (\hat{Z}\otimes\hat{Z})\,\widehat{\mathrm{CZ}}$.

D. Jaksch *et al.*, PRL **85**, 2208 (2000), doi:10.1103/PhysRevLett.85.2208; M. Saffman *et al.*, RMP **82**, 2313 (2010), doi:10.1103/RevModPhys.82.2313

Error of the Blockade Gate



(a) Populations of $|r1\rangle$, control in $|r\rangle$, and of $|rr\rangle$ for $|11\rangle$ through the three pulses, separated by dotted lines, dashed $V = 0$, solid $V = 20\,\Omega$. (b) The light shift $\Omega^2/(4V)$ of $|r1\rangle$ gives an error $1 - F \propto (\Omega/V)^2$, and 10^{-3} needs $V \approx 11\,\Omega$.

Lab 068

Summary



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Summary of Part I

1. A qubit is a two-level system described by two complex amplitudes, and the Born rule turns them into the probabilities of a bit.
2. A gate is evolution under a chosen Hamiltonian for a chosen time. A single-qubit gate rotates the Bloch vector. A generator with $\hat{P}^2 = \hat{I}$ gives $e^{-ia\hat{P}} = \cos a\,\hat{I} - i \sin a\,\hat{P}$, and controlled gates are exponentials of projectors.
3. Other bases are measured by rotating first, and finite records give errors proportional to $1/\sqrt{M}$.
4. N qubits are 2^N amplitudes in binary order, qubit 0 most significant. Operators are sparse Kronecker products, and state-tensor contractions give the same results.

Summary of Part I, Continued

5. A pure two-qubit register with Schmidt rank 2 is entangled. The Bell state and the correlated coin give different records in the $\hat{X}\hat{X}$ and $\hat{Y}\hat{Y}$ settings.

6. The density matrix describes mixtures and parts of registers. Purity separates pure states from mixtures, and the reduced state measures the entanglement of a pure register.

7. Measurement has back-action and sends no signal. Errors are Kraus channels accumulating gate by gate. Fidelity and trace distance measure how far a state has moved.

Summary of Part I, Concluded

8. The Choi state $\hat{\rho}_{\mathcal{E}} = (\mathcal{E} \otimes \mathrm{id})(|\Phi^+\rangle\langle\Phi^+|)$ stores a channel, positive for a completely positive map, and its eigenvectors give the Kraus operators. The transpose, with $\hat{\rho}_{\mathcal{E}} = \widehat{\mathrm{SWAP}}/2$, is positive but not completely positive.

9. Unknown states cannot be copied, one copy is identified with probability at most $\frac{1}{2}(1 + D)$, and a measurement on N qubits yields at most N bits about an encoded message.

10. CNOT and rotations are universal, but a generic unitary needs $\sim 4^N$ gates. Clifford circuits are classically simulable, and random states have $\mathrm{Var}\langle\hat{P}\rangle = 1/(2^N + 1)$. The partial transpose detects mixed-state entanglement, and $F_Q/N > 1$ certifies it.

Summary of Part II

1. Atoms, transmons and spins become qubits for different reasons, but a resonant drive gives the same rotating-frame Hamiltonian $\frac{\Delta}{2}\hat{Z} + \frac{\Omega}{2}\hat{X}$ in the frame $\hat{R}_z(\omega_d t)$, and pulse area and phase set the rotation of any qubit.

2. The platforms differ in how the two levels are obtained, driven, coupled and read out, and in how long their superpositions survive.

3. Each platform couples its qubits through its own interaction, a sum of Pauli strings. The Rydberg interaction gives an Ising coupling and the blockade CZ gate, the Mølmer–Sørensen drive gives $\hat{R}_{xx}$, and exchange gives $\widehat{\mathrm{iSWAP}}$ and $\sqrt{\mathrm{SWAP}}$.

4. Every angle of a circuit is a pulse area or an integrated coupling, and the native gates and the connectivity of a device fix how a circuit is compiled.

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Labs and Reading

Labs

Part I

- Lab 064, one qubit and gates as rotations.
- Lab 065, the two-spin register and entanglement.
- Lab 066, density matrices, measurement and noise.
- Lab 066a, what quantum information allows and forbids.

Part II

- Lab 067, physical qubits, an atom, a transmon and a spin.
- Lab 068, from interacting qubits to two-qubit gates.

Textbooks and notes

- C. J. Foot, *Atomic Physics* (Oxford University Press, 2005), doi:10.1093/oso/9780198506959.001.0001.
- L. Allen and J. H. Eberly, *Optical Resonance and Two-Level Atoms* (Dover, 1987), doverpublications.com.
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2010), doi:10.1017/CBO9780511976667.
- J. Preskill, lecture notes on quantum computation, www.preskill.caltech.edu/ph229.

Appendix: Atom in Laser Light, Derivation



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The Atom and Its Hamiltonian

A nucleus of charge $Z_{\mathrm{at}} e_0$ (atomic number Z_{at}) and mass M binds N_e electrons of charge $-e_0$ and mass m_e,

$$\hat{\mathcal{H}}_{\mathrm{atom}} = \underbrace{\frac{\hat{\mathbf{P}}^2}{2M} + \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e}}_{\text{kinetic energies}} \underbrace{- \sum_i \frac{Z_{\mathrm{at}} e_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{R}}|}}_{\text{electron–nucleus attraction}} + \underbrace{\sum_{i<j} \frac{e_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|}}_{\text{electron–electron repulsion}} ,$$

with nucleus $\hat{\mathbf{R}}, \hat{\mathbf{P}}$, electrons $\hat{\mathbf{r}}_i, \hat{\mathbf{p}}_i$ and the vacuum permittivity ε_0.

- The centre of mass separates exactly. With $M/m_e \approx 1823\,A$ (mass number A) it sits at the nucleus, $\mathbf{R} = 0$, and the electrons obey $\hat{\mathcal{H}}_{\mathrm{el}}$, the same expression without $\hat{\mathbf{P}}^2/2M$. For hydrogen only the reduced mass remains, a 0.05 % shift.
- The fine structure (relativistic) and the hyperfine splitting (electron–nuclear spin coupling) are omitted.

Energy Eigenstates as a Basis

The bound eigenstates $\hat{\mathcal{H}}_{\rm el}|j\rangle = E_j|j\rangle$ are discrete and orthonormal, with j collecting all quantum numbers. Every state is a superposition,

$$|\psi(t)\rangle = \sum_j c_j(t)\,|j\rangle, \qquad c_j = \langle j|\psi\rangle, \qquad \sum_j |c_j|^2 = 1,$$

and $|c_j|^2$ is the probability of finding the atom in $|j\rangle$. Projecting the Schrödinger equation on $\langle j|$ gives its matrix form

$$i\hbar\,\dot{c}_j = \sum_k H_{jk}\,c_k, \qquad H_{jk} = \langle j|\hat{\mathcal{H}}|k\rangle,$$

with $c_j(t) = c_j(0)e^{-iE_j t/\hbar}$ for the free atom.

Coupling to Laser Light

A charge q at $\mathbf{r}$ in a uniform field $\mathbf{E}$ has the energy $-q\,\mathbf{E}\cdot\mathbf{r}$. An optical wavelength exceeds the atom by orders of magnitude, so $\mathbf{E}(t) = E_0\mathbf{e}_z\cos(\omega_L t - \phi_L)$ is uniform over it (**dipole approximation**). Summing over the electrons, $q = -e_0$,

$$\hat{\mathcal{H}}_{\mathrm{int}} = -\hat{\mathbf{d}}\cdot\mathbf{E}(t), \qquad \hat{\mathbf{d}} = -e_0\sum_i \hat{\mathbf{r}}_i, \qquad d_{jk} = \langle j|\hat{\mathbf{d}}\cdot\mathbf{e}_z|k\rangle.$$

$\hat{\mathcal{H}}_{\mathrm{el}}$ is unchanged when all $\mathbf{r}_i \to -\mathbf{r}_i$, so its eigenstates can be chosen even or odd, while $\hat{\mathbf{d}}$ changes sign. Hence $d_{jj} = 0$, and states of equal parity are not coupled, the **selection rule**.

L. Allen and J. H. Eberly, *Optical Resonance and Two-Level Atoms* (Dover, 1987), store.doverpublications.com

Interaction Picture and Slow Couplings

The interaction picture $c_j = a_j e^{-iE_j t/\hbar}$ removes the free rotation and leaves only the light,

$$i\hbar\,\dot{a}_j = -E_0\cos(\omega_L t - \phi_L)\sum_k d_{jk}\,e^{i\omega_{jk}t}\,a_k, \qquad \omega_{jk} = \frac{E_j - E_k}{\hbar}.$$

Each coupling splits into two oscillating terms,

$$e^{i\omega_{jk}t}\cos(\omega_L t - \phi_L) = \tfrac{1}{2}e^{i\phi_L}e^{i(\omega_{jk}-\omega_L)t} + \tfrac{1}{2}e^{-i\phi_L}e^{i(\omega_{jk}+\omega_L)t}.$$

A term much faster than the coupling strength $\Omega_{jk} = -d_{jk}E_0/\hbar$ averages out, so a pair stays uncoupled when

$$|\omega_{jk} - \omega_L| \gg |\Omega_{jk}| \quad \text{and} \quad |\omega_{jk} + \omega_L| \gg |\Omega_{jk}|.$$

Only the resonant pair g, e with $\omega_{eg} = \omega_0 \approx \omega_L$ violates this. Lab 067 shows that the averaged coupling matrix keeps only 1s–2p.

Two Amplitudes in the Rotating-Wave Approximation

For $j = e$, $k = g$ and for $j = g$, $k = e$ (with $\omega_{ge} = -\omega_0$),

$$e^{\pm i\omega_0 t}\cos(\omega_L t - \phi_L) = \underbrace{\tfrac{1}{2}e^{\pm i\phi_L}e^{\mp i\Delta t}}_{\text{slow, } \Delta=\omega_L-\omega_0} + \underbrace{\tfrac{1}{2}e^{\mp i\phi_L}e^{\pm i(\omega_0+\omega_L)t}}_{\text{fast, averages out}}.$$

Keeping the slow terms, with the phases of $|g\rangle$, $|e\rangle$ chosen so that d_{eg} is real,

$$i\dot{a}_g = \tfrac{\Omega}{2}e^{-i\phi_L}e^{i\Delta t}a_e, \qquad i\dot{a}_e = \tfrac{\Omega}{2}e^{i\phi_L}e^{-i\Delta t}a_g, \qquad \Omega = -\frac{d_{eg}E_0}{\hbar}.$$

The first line is the conjugate partner of the second. For $|\Omega|, |\Delta| \ll \omega_0$ all other levels stay nearly empty.

The 2×2 Qubit Matrix

With $b_g = a_g e^{-i\Delta t/2}$, $b_e = a_e e^{i\Delta t/2}$, $i\dot{b}_g = i\dot{a}_g e^{-i\Delta t/2} + \frac{\Delta}{2} b_g$ and $i\dot{b}_e = i\dot{a}_e e^{i\Delta t/2} - \frac{\Delta}{2} b_e$,

$$i\frac{d}{dt}\begin{pmatrix} b_g \\ b_e \end{pmatrix} = \begin{pmatrix} \frac{\Delta}{2} & \frac{\Omega}{2} e^{-i\phi_L} \\ \frac{\Omega}{2} e^{i\phi_L} & -\frac{\Delta}{2} \end{pmatrix} \begin{pmatrix} b_g \\ b_e \end{pmatrix} = \Big[\frac{\Delta}{2} \hat{Z} + \frac{\Omega}{2} \big(\cos\phi_L \hat{X} + \sin\phi_L \hat{Y} \big) \Big] \begin{pmatrix} b_g \\ b_e \end{pmatrix}.$$

- The off-diagonal entries $\frac{\Omega}{2}(\cos\phi_L \mp i \sin\phi_L)$ are those of $\frac{\Omega}{2}(\cos\phi_L \hat{X} + \sin\phi_L \hat{Y})$.
- With $E_{g,e} = \mp\hbar\omega_0/2$, $b_g = c_g e^{-i\omega_L t/2}$ and $b_e = c_e e^{i\omega_L t/2}$, the rotating-frame state $\hat{R}_z(\omega_L t)|\psi\rangle$ of the main slides.

Hydrogen as an Example

Hydrogen has eigenstates $|nlm\rangle$ with $E_n = -\mathrm{Ry}/n^2$, $\mathrm{Ry} \approx 13.6$ eV, and $\langle \mathbf{r}|nlm\rangle = R_{nl}(r)\,Y_{lm}(\vartheta,\varphi)$, a radial function times a spherical harmonic.

- Light polarised along z drives only $l' = l \pm 1$, $m' = m$.
- $\langle 2p_0|\hat{z}|1s\rangle = \frac{128\sqrt{2}}{243} a_0 \approx 0.745\, a_0$.
- $\hbar\omega_0 = \frac{3}{4}\mathrm{Ry} \approx 10.2$ eV (Lyman-α, 121.5 nm).
- A far level r shifts g by $\delta E_g = \Omega_r^2/(4\Delta_r)$, the exact eigenvalue of the 2×2 pair expanded to second order in Ω_r (light shift).
- Bloch–Siegert shift $\Omega^2/(4\omega_0)$.
- Decay $\Gamma = \frac{\omega_0^3 |d_{eg}|^2}{3\pi\varepsilon_0\hbar c^3}$, $1/\Gamma \approx 1.60$ ns for 2p.

Appendix: Mølmer–Sørensen Gate, Derivation



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Displacements in Phase Space

- The complex number $\langle \hat{a} \rangle$, with scaled mean position and momentum as real and imaginary parts, is a point in **phase space**.
- The displacement $\hat{D}(\beta) = e^{\beta \hat{a}^\dagger - \beta^* \hat{a}}$ shifts the motion by β, $\hat{D}^\dagger \hat{a} \hat{D} = \hat{a} + \beta$, and $\hat{D}(\beta_2)\hat{D}(\beta_1) = e^{i \operatorname{Im}(\beta_2 \beta_1^*)} \hat{D}(\beta_1 + \beta_2)$.
- For an eigenvalue m of $\hat{S}_x$, $\frac{d}{dt}\langle \hat{a} \rangle = i\langle [\hat{\mathcal{H}}, \hat{a}] \rangle = -img\, e^{i\delta t}$, so $\langle \hat{a} \rangle$ moves by $m\alpha(t)$, $\alpha(t) = \frac{g}{\delta}(1 - e^{i\delta t})$.

$$\hat{U}_m(t) = e^{i\Phi(t)m^2} \hat{D}(m\alpha(t)), \qquad \Phi(t) = \int_0^t \operatorname{Im}[\alpha^* \dot{\alpha}]\, dt' = \frac{g^2}{\delta}\Big(t - \frac{\sin \delta t}{\delta}\Big),$$

twice the enclosed area. At $t = 2\pi K/\delta$ the displacement vanishes for every motional state.

Loop Phase, Effective Coupling and Sign

With $m^2 \to \hat{S}_x^2 = \frac{1}{2}(\hat{I} + \hat{X}_0\hat{X}_1)$ the closed loop gives

$$\hat{U}\Big(\frac{2\pi K}{\delta}\Big) = e^{\,i\Phi\hat{S}_x^2} = e^{i\Phi/2}\,\hat{R}_{xx}(-\Phi), \qquad \Phi = \frac{2\pi K g^2}{\delta^2}.$$

- On average $\Phi \approx Jt$ with $J = g^2/\delta$, the effective Hamiltonian $-J\hat{S}_x^2 = -\frac{J}{2}(\hat{I} + \hat{X}_0\hat{X}_1)$.
- For $\delta > 0$ the gate is $\hat{R}_{xx}(-\Phi)$, opposite to the $\hat{R}_{xx}(+\pi/2)$ of Part I, and at $\Phi = \pi/2$ it gives $|00\rangle \to (|00\rangle + i|11\rangle)/\sqrt{2}$. The opposite sign of δ reverses it.

Sørensen, Mølmer, PRL **82**, 1971 (1999), doi:10.1103/PhysRevLett.82.1971; Mølmer, Sørensen, PRL **82**, 1835 (1999), doi:10.1103/PhysRevLett.82.1835